

Analysis Review Bundle — Full Extraction

analysis_review_20260627_220800

Complete listing of every measured result, table and stated conclusion

ChampSim trace profiling + normal prefetchers vs. standalone multi-horizon LSTM replay (v3.1 / v3.3)

Note: This document extracts and lists, item by item, every measured value and every stated conclusion from each result file in the bundle: trace profiles, the evidence report, L2C demand-event attribution, the capacity sweep, build info, and the limitation statements. All source data is English / numeric.

1 Provenance, scope and global facts

Source directories (manifest):

- A.ROI_DIR = formal_NN_training/results/trace_profiles/roi25m_trace_profile_20260625_115128
- B.FULL_DIR = formal_NN_training/results/trace_profiles/full_trace_profile_20260625_150232
- D.EVENT_ROOT = formal_NN_training/results/prefetch_explainability/event_evidence_all15_20260625_121115
- F.REPORT_DIR = formal_NN_training/reports/evidence/evidence_parse_report_20260625_142546
- G.CAPACITY_DIR = formal_NN_training/results/cache_capacity_sweep/capacity_all13_half_double_20260626_105924

Workloads (5 SPEC CPU traces, all runs): 602.gcc_s-734B, 619.lbm_s-4268B, 605.mcf_s-994B, 620.omnetpp_s-874B, 623.xalancbmk_s-700B.

Normal prefetchers swept: no_pref, sandbox, sms, ampm, spp (event-attribution adds stride, streamer, ipcp, power7).

Standalone NN variants: v3.1 = standalone_multihorizon_lstm_v3_1; v3.3 = standalone_multihorizon_lstm_v3_3_context_coverage.

Sim parameters (all runs): WARMUP = 25,000,000; SIM = 25,000,000 instructions; CHUNK_LEN = 1024; DEDUP_CAPACITY = 256; EXPORT_SUFFIX = pure.balanced_lru256.

Build info (representative binary): built_utc = 2026-06-26T16:03:23Z; champ_sim_head = 838be17e3e5eb61d5c4fb615e5f8eae1883aca28; line_bytes = 64; source binary = /home/qianruw/cache/external/ChampSim/bin/perceptron-no-multi-no-ship-1core.

1.1 Stated limitations / caveats (verbatim meaning)

- **Assembly mapping (C):** N/A. Reason — “No matching original benchmark executable with matching PC address space was available.” Trace records contain no opcode bytes, so binary mapping is best-effort and valid only when binary static addresses match the trace PC address space (assembly_available_from_trace=false, assembly_mapping_attempted=false for every trace).
- **Event analysis scope:** L2C demand-event analysis.
- **event_rows_written = False.**
- **Limit:** Frozen standalone exports contain selections only. no_earlier_selected_standalone_export is not proof of candidate-bank absence.
- **Next-notebook requirement:** Export a full decision ledger with every candidate, score/probabilities, candidate-source, and reject_reason to distinguish candidate absence from policy rejection.
- **Frozen-list caveat (capacity sweep, every config):** “Frozen lists use the baseline-capacity raw oracle. This is not a capacity-trained NN experiment.”
- **Evidence report scope:** records measured associations only; “does not claim a causal explanation beyond what these counters and event outcomes establish.” Top residual contexts are evidence for later candidate-representation analysis, not NN labels.

2 Workload / trace characterization (ROI, 25M instructions)

All five traces: 25,000,000 instructions profiled; profile window skipped the first 25M records and read records 25M–50M; record format = ChampSim input_instr, 64 bytes; full_trace_counted=false.

2.1 Trace profile summary

trace	uniq PCs	uniq load pp	branch frac	taken frac	load frac	store frac	mem frac
602.gcc_s-734B	485	4,965	0.2120	0.4730	0.2622	0.1124	0.3745
619.lbm_s-4268B	510	4,603	0.0178	0.8964	0.2603	0.1836	0.4439
605.mcf_s-994B	715	102,356	0.1919	0.5246	0.3516	0.1012	0.4186
620.omnetpp_s-874B	6,591	32,774	0.1491	0.4441	0.3381	0.1855	0.5104
623.xalancbmk_s-700B	4,156	2,203	0.2115	0.3405	0.3081	0.1128	0.4086

Compressed trace bytes: 602.gcc=368,694,692; 619.lbm=773,592,004; 605.mcf=697,584,448; 620.omnetpp=802,283,392; 623.xalancbmk=160,985,836.

2.2 602.gcc_s-734B

Counters: instructions=25,000,000; branch=5,299,375 (frac 0.211975, taken frac 0.472968); load=6,554,962 (frac 0.262198); store=2,809,016 (frac 0.112361); memory=9,363,621 (frac 0.374545); unique PCs=485; unique load pages=4,965.

Window: first PC 0x40f511, last PC 0x40f4f3; records read 25,000,000; ended at record 50,000,000; skip 25,000,000.

trace_sha256: a32be73c53946748a1393cca4999f374c5d9d3737a414437d81bb20ed6311bd5.

—load line delta— **buckets:** 0=2,598,631; 1=494,786; 2_4=787; 5_16=43; 17_64=21; 65_256=9; gt_256=3,460,684.

top load line deltas			top load pages	
line_delta	share		page	share
0	0.39644	34,358,534,190	0.33395	
1	0.07545	652,008	0.02660	
5,083,415	0.00101	652,009	0.02654	
5,631,043	0.00097	652,005	0.02649	
5,627,328	0.00093	652,007	0.02648	
5,684,124	0.00082	652,003	0.02646	
5,082,884	0.00081	652,006	0.02646	
5,631,574	0.00081	652,004	0.02641	
5,630,512	0.00078			
5,080,230	0.00072			
5,684,655	0.00072			
-5,631,043	0.00068			

top PCs		top src reg		top dst reg	
pc	instr share	reg	share	reg	share
0x40f700	0.03695	26	0.1539	25	0.2288
0x40f703	0.03695	6	0.1453	26	0.2060
0x40f705	0.02463	25	0.1282	6	0.1733
0x40f708	0.02463	15	0.0846	4	0.0537
0x40f70a	0.02463	4	0.0676	15	0.0324
0x40f70f	0.02463	8	0.0594	7	0.0324
0x40f714	0.02463	7	0.0593	9	0.0323
0x40f719	0.02463	17	0.0520	16	0.0323
0x40f511	0.01235				
0x40f514	0.01235				

2.3 619.lbm_s-4268B

Counters: instructions=25,000,000; branch=445,543 (frac 0.017822, taken frac 0.896443); load=6,507,537 (frac 0.260301); store=4,590,390 (frac 0.183616); memory=11,097,927 (frac 0.443917); unique PCs=510; unique load pages=4,603.

Window: first PC 0x402773, last PC 0x4020a7; records read 25,000,000; ended at record 50,000,000; skip 25,000,000.

trace_sha256: 4516bb1d4c8985678ada3f09706f551dfcb01b2f840fea8a59307a938b263b8f.

—load line delta— **buckets:** 0=3,024,797; 1=940,327; 2_4=1,065,994; gt_256=1,476,418.

top load line deltas			top load pages	
line_delta	share		page	share
0	0.46481		34,358,891,164	0.43958
1	0.08778		1,179	0.07090
-1	0.05672		34,190,983,063	0.00015
-2	0.04450		34,190,983,073	0.00015
3	0.04450		34,190,983,075	0.00015
2	0.03032		34,190,983,078	0.00015
-3	0.02323		34,190,983,080	0.00015
-2,198,968,959,038	0.01418		34,190,983,083	0.00015
4	0.01418			
2,198,968,959,037	0.00709			
-2,198,968,959,037	0.00709			
2,198,968,959,038	0.00709			

top PCs			top src reg		top dst reg	
pc	instr share		reg	share	reg	share
0x4027cb	0.00471		10	0.1213	101	0.1570
0x4027cf	0.00471		6	0.1118	103	0.1256
0x4027d5	0.00471		101	0.0890	100	0.1166
0x401cb0	0.00471		100	0.0722	9	0.0627
0x401cb7	0.00471		3	0.0683	105	0.0606
0x401cba	0.00471		103	0.0663	102	0.0583
0x401cbf	0.00471		106	0.0504	115	0.0538
0x401df0	0.00286		4	0.0480	107	0.0538
0x401df5	0.00286					
0x401dfa	0.00286					

2.4 605.mcf_s-994B

Counters: instructions=25,000,000; branch=4,796,849 (frac 0.191874, taken frac 0.524561); load=8,791,216 (frac 0.351649); store=2,531,012 (frac 0.101240); memory=10,465,445 (frac 0.418618); unique PCs=715; unique load pages=102,356.

Window: first PC 0x406383, last PC 0x40244a; records read 25,000,000; ended at record 50,000,000; skip 25,000,000.

trace_sha256: 3216f131809626217f0c6130c6537704b5143ad11b0b6364bdd3a787da0a9167.

—load line delta— buckets: 0=1,071,402; 1=1,855,162; 2.4=747,213; 5.16=132,130; 17.64=169,681; 65.256=231,169; gt.256=4,584,458.

top load line deltas			top load pages	
line_delta	share		page	share
-1	0.16116		34,357,359,865	0.40866
0	0.12187		34,122,636,279	0.03252
1	0.04986		34,357,359,864	0.02958
2	0.04547		34,122,166,985	0.02276
-30,034,854	0.01858		34,122,169,806	0.01851
-2	0.01855		34,122,636,209	0.00702
3	0.01433		34,122,636,242	0.00549
-29,854,322	0.01072		34,122,636,256	0.00514
15,052,163,820	0.00779			
15,052,344,359	0.00418			
-3	0.00386			
-17,041,110	0.00217			

top PCs		top src reg		top dst reg	
pc	instr share	reg	share	reg	share
0x402440	0.03247	26	0.1679	26	0.2476
0x402443	0.03247	6	0.1530	25	0.1983
0x402446	0.03247	25	0.1498	56	0.1344
0x40244a	0.03247	9	0.0686	6	0.0666
0x40244f	0.03247	5	0.0662	4	0.0663
0x402453	0.03247	3	0.0542	9	0.0541
0x40246d	0.03247	4	0.0526	3	0.0478
0x402455	0.03035	8	0.0514	8	0.0379
0x40245a	0.03035				
0x40245c	0.02537				

2.5 620.omnetpp_s-874B

Counters: instructions=25,000,000; branch=3,728,126 (frac 0.149125, taken frac 0.444056); load=8,452,555 (frac 0.338102); store=4,637,242 (frac 0.185490); memory=12,758,863 (frac 0.510355); unique PCs=6,591; unique load pages=32,774.

Window: first PC 0x55b6e3, last PC 0x54c95c; records read 25,000,000; ended at record 50,000,000; skip 25,000,000.

trace_sha256: 4034c81a92536f75121a6c5321d5448e0b2b82a8ffb367ae4d1a60bbc5d430a0.

—load line delta— **buckets:** 0=2,534,417; 1=527,169; 2_4=106,655; 5_16=72,919; 17_64=228,762; 65_256=72,868; gt_256=4,909,764.

top load line deltas		top load pages	
line_delta	share	page	share
0	0.29984	34,356,209,220	0.33694
1	0.05454	6,442	0.11150
244,559	0.02625	1,793	0.05202
244,031	0.01004	33,619	0.04126
-1	0.00783	2,621	0.04037
2	0.00558	1,794	0.01588
-244,559	0.00487	2,629	0.01437
-297,746	0.00486	1,723	0.01303
-21	0.00448		
21	0.00416		
1,739,318	0.00398		
-2,198,796,977,812	0.00381		

top PCs		top src reg		top dst reg	
pc	instr share	reg	share	reg	share
0x471325	0.00561	6	0.1833	6	0.1940
0x471328	0.00561	26	0.1416	25	0.1855
0x47132e	0.00532	25	0.0967	26	0.1848
0x471331	0.00532	10	0.0858	10	0.0760
0x471334	0.00532	3	0.0603	3	0.0368
0x471386	0.00532	7	0.0517	7	0.0324
0x471389	0.00532	9	0.0354	56	0.0282
0x47138d	0.00532	55	0.0313	9	0.0268
0x471390	0.00532				
0x471394	0.00532				

2.6 623.xalancbmk_s-700B

Counters: instructions=25,000,000; branch=5,288,514 (frac 0.211541, taken frac 0.340503); load=7,702,366 (frac 0.308095); store=2,819,867 (frac 0.112795); memory=10,215,151 (frac 0.408606); unique PCs=4,156; unique load pages=2,203.

Window: first PC 0x64a798, last PC 0x5a0fea; records read 25,000,000; ended at record 50,000,000; skip 25,000,000.

trace_sha256: 84a437aa68592584522fea1effc5a931e818d110a3d7a181979ea20b7caed4e7.

—load line delta— **buckets:** 0=3,335,073; 1=850,389; 2_4=124,915; 5_16=123,020; 17_64=66,810; 65_256=121,638; gt_256=3,080,520.

top load line deltas		top load pages	
line_delta	share	page	share
0	0.43299	34,356,208,070	0.14006
1	0.09166	34,356,208,071	0.13419
-1	0.01875	125,425	0.06458
2	0.00691	125,426	0.04756
2,242	0.00651	125,424	0.03563
-2,242	0.00592	5,867	0.03321
-6	0.00419	125,427	0.03054
-229	0.00385	94,325	0.02339
-5	0.00379		
-233,209	0.00370		
6	0.00368		
229	0.00356		

top PCs		top src reg		top dst reg	
pc	instr share	reg	share	reg	share
0x638eb4	0.01773	26	0.1605	25	0.2758
0x638eb7	0.01773	25	0.1360	26	0.2208
0x638ebd	0.01773	6	0.1182	6	0.1284
0x638ec0	0.01773	10	0.1130	10	0.0557
0x638ec3	0.01773	7	0.0857	7	0.0509
0x638ec5	0.01773	4	0.0686	54	0.0350
0x638ec8	0.01773	9	0.0424	9	0.0322
0x638ecb	0.01773	8	0.0419	4	0.0286
0x638ed1	0.01772				
0x638ed5	0.01772				

3 Trace and prefetch evidence report (core measured results)

For each benchmark: best normal prefetcher (by IPC) vs. the two standalone NN replays, plus L2C demand-event overlap. Source: `trace_prefetch_evidence_report.md` / `event_evidence` run 20260625_121115 (ROI 25M).

3.1 602.gcc_s-734B

Trace window: 25,000,000 instructions; unique PCs=485; branch frac=0.211975; load frac=0.26219848; mem frac=0.37454484.

Best normal IPC: sandbox at 0.43628; issued=3,839,474, useful=390,102, useless=32,524, late=59, selected acc=0.10184, timeliness=0.99985.

std	IPC	Δ vs best	sel.acc	ctr timeli.	evt coverage	evt timeli.	issued	useful	useless	late
v3.1	0.42863	-0.007650	0.973754	0.998591	0.899996	0.998640	187,803	182,874	4,068	258
v3.3	0.42887	-0.007410	0.980913	0.999002	0.901364	0.999002	186,723	183,158	3,056	183

Demand-event overlap vs. best normal prefetcher:

std	both timely	normal-only	std-only	both late	neither	sel. but late	no earlier sel.
v3.1	180,554	15,340	2,263	1	4,933	248	14,961
v3.3	180,405	15,489	2,690	1	4,508	182	15,153

3.2 619.lbm_s-4268B

Trace window: 25,000,000 instructions; unique PCs=510; branch frac=0.01782172; load frac=0.26030148; mem frac=0.44391708.

Best normal IPC: sms at 0.38105; issued=568,017, useful=535,130, useless=84, late=26,099, selected acc=0.94210, timeliness=0.95350.

std	IPC	Δ vs best	sel.acc	ctr timeli.	evt coverage	evt timeli.	issued	useful	useless	late
v3.1	0.38492	0.003870	0.978263	0.984176	0.972339	0.984191	293,605	287,223	1,732	4,618
v3.3	0.38434	0.003290	0.985636	0.992914	0.978893	0.992919	293,442	289,227	2,115	2,064

Demand-event overlap vs. best normal prefetcher:

std	both timely	normal-only	std-only	both late	neither	sel. but late	no earlier sel.
v3.1	250,052	5,218	36,437	729	1,457	3,545	974
v3.3	252,030	3,240	36,390	384	2,023	1,517	961

3.3 605.mcf_s-994B

Trace window: 25,000,000 instructions; unique PCs=715; branch frac=0.19187396; load frac=0.35164864; mem frac=0.4186178.

Best normal IPC: ampm at 0.18874; issued=509,840, useful=90,495, useless=58,586, late=1,305, selected acc=0.17787, timeliness=0.98578.

std	IPC	Δ vs best	sel.acc	ctr timeli.	evt coverage	evt timeli.	issued	useful	useless	late
v3.1	0.18862	-0.000120	0.526061	0.801153	0.073720	0.800813	104,121	54,774	5,304	13,595
v3.3	0.18729	-0.001450	0.431481	0.738338	0.069089	0.737923	118,930	51,316	5,380	18,186

Demand-event overlap vs. best normal prefetcher:

std	both timely	normal-only	std-only	both late	neither	sel. but late	no earlier sel.
v3.1	40,159	17,513	14,410	80	653,507	65	17,080
v3.3	28,390	29,282	22,751	230	640,924	239	27,960

3.4 620.omnetpp_s-874B

Trace window: 25,000,000 instructions; unique PCs=6591; branch frac=0.14912504; load frac=0.3381022; mem frac=0.51035452.

Best normal IPC: sms at 0.24695; issued=808,086, useful=197,737, useless=394,038, late=3,655, selected acc=0.24470, timeliness=0.98185.

std	IPC	Δ vs best	sel.acc	ctr timeli.	evt coverage	evt timeli.	issued	useful	useless	late
v3.1	0.24503	-0.001920	0.454822	0.990821	0.236165	0.990658	169,940	77,286	63,543	716
v3.3	0.24559	-0.001360	0.444369	0.992572	0.255163	0.992529	187,366	83,252	74,440	623

Demand-event overlap vs. best normal prefetcher:

std	both timely	normal-only	std-only	both late	neither	sel. but late	no earlier sel.
v3.1	38,558	47,449	36,201	48	192,359	209	46,252
v3.3	40,520	45,487	40,253	51	188,378	180	44,089

3.5 623.xalancbmk_s-700B

Trace window: 25,000,000 instructions; unique PCs=4156; branch frac=0.21154056; load frac=0.30809464; mem frac=0.40860604.

Best normal IPC: spp at 0.35391; issued=1,670,201, useful=7,816, useless=23,108, late=1,870, selected acc=0.004846, timeliness=0.80694.

std	IPC	Δ vs best	sel.acc	ctr timeli.	evt coverage	evt timeli.	issued	useful	useless	late
v3.1	0.36407	0.010160	0.683155	0.987180	0.454340	0.987595	317,571	216,910	21,844	2,817
v3.3	0.37893	0.025020	0.751036	0.997001	0.899447	0.997356	570,297	428,219	25,092	1,288

Demand-event overlap vs. best normal prefetcher:

std	both timely	normal-only	std-only	both late	neither	sel. but late	no earlier sel.
v3.1	880	252	203,732	39	242,859	12	225
v3.3	1,012	120	404,054	48	43,997	7	101

4 L2C demand-event attribution (D/E)

Run `event_evidence_all5.20260625.121115`. “no_pref_miss_events” = demand L2C misses under no_pref baseline (the universe of events). Coverage = unique demand-event coverage; timeliness = timeliness over covered events.

4.1 Per-(trace,family,variant) event outcomes

trace	variant	evt rows	unmatched	no_pref miss	timely	late	residual / cov / timeli.
602.gcc.s-734B	no_pref	404,219	0	203,131	0	0	203,131 / 0.0000 / 0.0000
602.gcc.s-734B	stride	404,219	3,370	203,131	34,854	164	168,110 / 0.1716 / 0.9953
602.gcc.s-734B	streamer	404,219	40,527	203,131	187,580	47	15,414 / 0.9234 / 0.9997
602.gcc.s-734B	ampm	404,219	40,569	203,131	186,005	600	16,396 / 0.9157 / 0.9968
602.gcc.s-734B	spp	404,219	30,824	203,131	132,205	384	70,471 / 0.6508 / 0.9971
602.gcc.s-734B	ipcp	404,219	6	203,131	0	1	203,130 / 0.0000 / 0.0000
602.gcc.s-734B	sms	404,219	35,180	203,131	161,912	173	40,963 / 0.7971 / 0.9989
602.gcc.s-734B	sandbox	404,219	42,180	203,131	195,894	43	7,064 / 0.9644 / 0.9998
602.gcc.s-734B	power7	404,219	40,166	203,131	185,747	125	17,169 / 0.9144 / 0.9993
602.gcc.s-734B	v3.1	404,219	39,849	203,131	182,817	249	19,982 / 0.9000 / 0.9986
602.gcc.s-734B	v3.3	404,219	39,723	203,131	183,095	183	19,772 / 0.9014 / 0.9990
619.lbm.s-4268B	no_pref	295,037	0	294,639	0	0	294,639 / 0.0000 / 0.0000
619.lbm.s-4268B	stride	295,026	385	294,639	67,229	15,664	211,245 / 0.2282 / 0.8110
619.lbm.s-4268B	streamer	295,032	345	294,639	142,164	112,238	39,792 / 0.4825 / 0.5588
619.lbm.s-4268B	ampm	295,027	672	294,639	235,435	24,838	33,578 / 0.7991 / 0.9046
619.lbm.s-4268B	spp	295,024	480	294,639	68,702	63,167	162,156 / 0.2332 / 0.5210
619.lbm.s-4268B	ipcp	295,039	86	294,639	0	0	294,491 / 0.0000 / 0.0000
619.lbm.s-4268B	sms	295,545	1,260	294,639	255,270	25,203	13,298 / 0.8664 / 0.9101
619.lbm.s-4268B	sandbox	295,024	448	294,639	209,025	76,845	8,199 / 0.7094 / 0.7312
619.lbm.s-4268B	power7	295,027	393	294,639	160,327	76,585	57,217 / 0.5441 / 0.6767
619.lbm.s-4268B	v3.1	295,033	761	294,639	286,489	4,602	2,674 / 0.9723 / 0.9842
619.lbm.s-4268B	v3.3	295,044	837	294,639	288,420	2,057	3,227 / 0.9789 / 0.9929
605.mcf.s-994B	no_pref	985,276	0	740,223	0	0	740,223 / 0.0000 / 0.0000
605.mcf.s-994B	stride	985,274	646	740,223	4,858	0	734,820 / 0.0066 / 1.0000
605.mcf.s-994B	streamer	985,268	15,434	740,223	85,808	15,919	626,627 / 0.1159 / 0.8435
605.mcf.s-994B	ampm	985,273	15,289	740,223	57,672	1,232	668,468 / 0.0779 / 0.9791
605.mcf.s-994B	spp	985,273	12,741	740,223	5,156	1,089	722,964 / 0.0070 / 0.8256
605.mcf.s-994B	ipcp	985,276	0	740,223	0	0	740,223 / 0.0000 / 0.0000
605.mcf.s-994B	sms	985,312	15,608	740,223	91,604	54,251	582,137 / 0.1238 / 0.6280
605.mcf.s-994B	sandbox	1,130,709	161,820	740,223	157,849	109,635	460,390 / 0.2132 / 0.5901
605.mcf.s-994B	power7	985,411	12,533	740,223	71,010	15,522	644,718 / 0.0959 / 0.8206
605.mcf.s-994B	v3.1	985,274	9,696	740,223	54,569	13,573	663,793 / 0.0737 / 0.8008
605.mcf.s-994B	v3.3	985,273	3,388	740,223	51,141	18,163	668,021 / 0.0691 / 0.7379
620.omnetpp.s-874B	no_pref	451,398	0	316,554	0	0	316,554 / 0.0000 / 0.0000
620.omnetpp.s-874B	stride	451,396	37	316,554	2,990	0	313,502 / 0.0094 / 1.0000
620.omnetpp.s-874B	streamer	451,360	1,470	316,554	51,399	1,932	261,401 / 0.1624 / 0.9638
620.omnetpp.s-874B	ampm	451,359	767	316,554	21,888	878	292,579 / 0.0691 / 0.9614
620.omnetpp.s-874B	spp	451,389	762	316,554	49	13	315,658 / 0.0002 / 0.7903
620.omnetpp.s-874B	ipcp	451,406	87	316,554	2,139	292	313,839 / 0.0068 / 0.8799
620.omnetpp.s-874B	sms	451,284	3,317	316,554	86,007	1,731	225,755 / 0.2717 / 0.9803
620.omnetpp.s-874B	sandbox	466,225	19,396	316,554	134,047	8,004	171,771 / 0.4235 / 0.9437
620.omnetpp.s-874B	power7	451,341	1,294	316,554	51,322	1,997	261,512 / 0.1621 / 0.9625
620.omnetpp.s-874B	v3.1	451,356	331	316,554	74,759	705	239,934 / 0.2362 / 0.9907
620.omnetpp.s-874B	v3.3	451,317	318	316,554	80,773	608	233,979 / 0.2552 / 0.9925
623.xalancbmk.s-700B	no_pref	1,218,494	0	450,350	0	0	450,350 / 0.0000 / 0.0000
623.xalancbmk.s-700B	stride	1,218,495	21	450,350	2,115	7,469	440,312 / 0.0047 / 0.2207
623.xalancbmk.s-700B	streamer	1,218,352	2,424	450,350	25,610	3,711	415,300 / 0.0569 / 0.8734
623.xalancbmk.s-700B	ampm	1,218,334	872	450,350	18,842	2,673	423,874 / 0.0418 / 0.8758
623.xalancbmk.s-700B	spp	1,218,483	222	450,350	1,132	1,837	434,440 / 0.0025 / 0.3813
623.xalancbmk.s-700B	ipcp	1,218,487	349	450,350	2,817	79	446,603 / 0.0063 / 0.9727
623.xalancbmk.s-700B	sms	1,218,827	4,800	450,350	68,566	4,908	368,045 / 0.1523 / 0.9332
623.xalancbmk.s-700B	sandbox	1,218,513	2,343	450,350	164,738	9,037	274,702 / 0.3658 / 0.9480
623.xalancbmk.s-700B	power7	1,218,439	1,477	450,350	17,247	4,522	422,903 / 0.0383 / 0.7923
623.xalancbmk.s-700B	v3.1	1,219,357	2,354	450,350	204,612	2,570	240,492 / 0.4543 / 0.9876
623.xalancbmk.s-700B	v3.3	1,219,263	2,394	450,350	405,066	1,074	38,696 / 0.8994 / 0.9974

4.2 Normal-vs-standalone target attribution

Each row: a normal prefetcher's demand-event timeliness compared against a standalone variant on the same trace. Columns are event counts.

trace	std	normal	both timely	normal-only	std-only	neither	no-earlier-sel	sel-but-late
602.gcc.s-734B	v3.1	no_pref			182,817	20,065		
602.gcc.s-734B	v3.1	stride	33,372	1,482	149,445	18,500	1,418	48
602.gcc.s-734B	v3.1	streamer	175,444	12,136	7,373	8,077	11,839	181
602.gcc.s-734B	v3.1	ampm	174,921	11,084	7,896	8,870	10,830	139
602.gcc.s-734B	v3.1	spp	125,048	7,157	57,769	12,917	7,026	52
602.gcc.s-734B	v3.1	ipcp			182,817	20,065		
602.gcc.s-734B	v3.1	sms	150,699	11,213	32,118	8,930	10,966	130
602.gcc.s-734B	v3.1	sandbox	180,554	15,340	2,263	4,933	14,961	248
602.gcc.s-734B	v3.1	power7	173,862	11,885	8,955	8,264	11,595	177
602.gcc.s-734B	v3.3	no_pref			183,095	19,853		
602.gcc.s-734B	v3.3	stride	33,134	1,720	149,961	18,138	1,672	36
602.gcc.s-734B	v3.3	streamer	174,753	12,827	8,342	7,112	12,567	117
602.gcc.s-734B	v3.3	ampm	174,295	11,710	8,800	8,042	11,489	83
602.gcc.s-734B	v3.3	spp	125,087	7,118	58,008	12,720	7,003	13
602.gcc.s-734B	v3.3	ipcp			183,095	19,853		
602.gcc.s-734B	v3.3	sms	150,583	11,329	32,512	8,551	11,101	75
602.gcc.s-734B	v3.3	sandbox	180,405	15,489	2,690	4,508	15,153	182
602.gcc.s-734B	v3.3	power7	173,171	12,576	9,924	7,340	12,320	113
619.lbm.s-4268B	v3.1	no_pref			286,489	3,548		
619.lbm.s-4268B	v3.1	stride	65,873	1,356	220,616	2,973	117	868
619.lbm.s-4268B	v3.1	streamer	139,622	2,542	146,867	2,062	417	1,695
619.lbm.s-4268B	v3.1	ampm	231,113	4,322	55,376	2,065	593	3,045
619.lbm.s-4268B	v3.1	spp	67,616	1,086	218,873	2,608	246	691
619.lbm.s-4268B	v3.1	ipcp			286,489	3,548		
619.lbm.s-4268B	v3.1	sms	250,052	5,218	36,437	1,457	974	3,545
619.lbm.s-4268B	v3.1	sandbox	205,131	3,894	81,358	1,437	855	2,437
619.lbm.s-4268B	v3.1	power7	157,195	3,132	129,294	2,104	461	2,127
619.lbm.s-4268B	v3.3	no_pref			288,420	4,162		
619.lbm.s-4268B	v3.3	stride	66,397	832	222,023	3,587	96	327
619.lbm.s-4268B	v3.3	streamer	140,507	1,657	147,913	2,649	453	737
619.lbm.s-4268B	v3.3	ampm	232,867	2,568	55,553	2,667	558	1,284
619.lbm.s-4268B	v3.3	spp	67,903	799	220,517	3,096	293	320
619.lbm.s-4268B	v3.3	ipcp			288,420	4,162		
619.lbm.s-4268B	v3.3	sms	252,030	3,240	36,390	2,023	961	1,517
619.lbm.s-4268B	v3.3	sandbox	206,484	2,541	81,936	2,022	880	1,051
619.lbm.s-4268B	v3.3	power7	158,424	1,903	129,996	2,727	428	911
605.mcf.s-994B	v3.1	no_pref			54,569	672,081		
605.mcf.s-994B	v3.1	stride	3,992	866	50,577	671,215	813	
605.mcf.s-994B	v3.1	streamer	41,015	44,793	13,554	613,373	43,517	436
605.mcf.s-994B	v3.1	ampm	40,159	17,513	14,410	653,507	17,080	65
605.mcf.s-994B	v3.1	spp	4,228	928	50,341	670,236	840	13
605.mcf.s-994B	v3.1	ipcp			54,569	672,081		
605.mcf.s-994B	v3.1	sms	44,637	46,967	9,932	583,970	45,050	741
605.mcf.s-994B	v3.1	sandbox	44,791	113,058	9,778	470,537	109,956	1,360
605.mcf.s-994B	v3.1	power7	34,030	36,980	20,539	621,599	35,698	429
605.mcf.s-994B	v3.3	no_pref			51,141	670,919		
605.mcf.s-994B	v3.3	stride	3,144	1,714	47,997	669,205	1,625	
605.mcf.s-994B	v3.3	streamer	31,459	54,349	19,682	606,654	51,539	1,221
605.mcf.s-994B	v3.3	ampm	28,390	29,282	22,751	640,924	27,960	239
605.mcf.s-994B	v3.3	spp	4,129	1,027	47,012	669,108	892	40
605.mcf.s-994B	v3.3	ipcp			51,141	670,919		
605.mcf.s-994B	v3.3	sms	36,872	54,732	14,269	575,501	51,878	1,241
605.mcf.s-994B	v3.3	sandbox	35,772	122,077	15,369	469,130	115,765	3,518
605.mcf.s-994B	v3.3	power7	26,642	44,368	24,499	617,055	41,789	1,159
620.omnetpp.s-874B	v3.1	no_pref			74,759	241,090		
620.omnetpp.s-874B	v3.1	stride	1,627	1,363	73,132	239,729	1,322	2
620.omnetpp.s-874B	v3.1	streamer	36,221	15,178	38,538	225,008	14,691	108
620.omnetpp.s-874B	v3.1	ampm	16,308	5,580	58,451	235,202	5,332	40
620.omnetpp.s-874B	v3.1	spp	29	20	74,730	241,070	20	
620.omnetpp.s-874B	v3.1	ipcp	1,551	588	73,208	240,427	521	6
620.omnetpp.s-874B	v3.1	sms	38,558	47,449	36,201	192,359	46,252	209
620.omnetpp.s-874B	v3.1	sandbox	62,511	71,536	12,248	164,142	69,886	325

trace	std	normal	both timely	normal-only	std-only	neither	no-earlier-sel	sel-but-late
620.omnetpp_s-874B	v3.1	power7	35,681	15,641	39,078	224,478	15,142	106
620.omnetpp_s-874B	v3.3	no_pref			80,773	235,173		
620.omnetpp_s-874B	v3.3	stride	1,897	1,093	78,876	234,084	1,037	4
620.omnetpp_s-874B	v3.3	streamer	37,399	14,000	43,374	220,227	13,489	82
620.omnetpp_s-874B	v3.3	ampm	17,011	4,877	63,762	229,977	4,563	28
620.omnetpp_s-874B	v3.3	spp	37	12	80,736	235,161	12	
620.omnetpp_s-874B	v3.3	ipcp	1,504	635	79,269	234,413	576	4
620.omnetpp_s-874B	v3.3	sms	40,520	45,487	40,253	188,378	44,089	180
620.omnetpp_s-874B	v3.3	sandbox	65,651	68,396	15,122	161,300	66,289	292
620.omnetpp_s-874B	v3.3	power7	36,965	14,357	43,808	219,803	13,839	73
623.xalancbm_s-700B	v3.1	no_pref			204,612	243,168		
623.xalancbm_s-700B	v3.1	stride	1,875	240	202,737	242,595	108	23
623.xalancbm_s-700B	v3.1	streamer	18,783	6,827	185,829	235,547	5,948	64
623.xalancbm_s-700B	v3.1	ampm	15,972	2,870	188,640	239,715	2,291	86
623.xalancbm_s-700B	v3.1	spp	880	252	203,732	242,859	225	12
623.xalancbm_s-700B	v3.1	ipcp	2,493	324	202,119	242,803	294	1
623.xalancbm_s-700B	v3.1	sms	40,631	27,935	163,981	213,285	25,275	491
623.xalancbm_s-700B	v3.1	sandbox	82,491	82,247	122,121	157,284	78,329	454
623.xalancbm_s-700B	v3.1	power7	12,690	4,557	191,922	237,851	3,935	35
623.xalancbm_s-700B	v3.3	no_pref			405,066	44,210		
623.xalancbm_s-700B	v3.3	stride	1,841	274	403,225	43,469	142	13
623.xalancbm_s-700B	v3.3	streamer	23,184	2,426	381,882	41,440	1,655	63
623.xalancbm_s-700B	v3.3	ampm	17,273	1,569	387,793	42,255	980	92
623.xalancbm_s-700B	v3.3	spp	1,012	120	404,054	43,997	101	7
623.xalancbm_s-700B	v3.3	ipcp	2,427	390	402,639	43,778	366	
623.xalancbm_s-700B	v3.3	sms	62,407	6,159	342,659	37,714	5,124	160
623.xalancbm_s-700B	v3.3	sandbox	149,467	15,271	255,599	28,222	13,435	119
623.xalancbm_s-700B	v3.3	power7	15,579	1,668	389,487	42,125	1,152	31

4.3 Reasons normal-only-timely events were missed by the standalone export

Decomposes “normal-only timely” events by why the standalone list did not cover them in time. (See limitation: absence of an earlier selected export is NOT proof of candidate-bank absence.)

trace	std	normal	reason	events
602.gcc_s-734B	v3.1	ampm	no earlier selected standalone export	10,830
602.gcc_s-734B	v3.1	ampm	standalone selected but late	139
602.gcc_s-734B	v3.1	ampm	standalone selected earlier but not timely	115
602.gcc_s-734B	v3.1	power7	no earlier selected standalone export	11,595
602.gcc_s-734B	v3.1	power7	standalone selected but late	177
602.gcc_s-734B	v3.1	power7	standalone selected earlier but not timely	113
602.gcc_s-734B	v3.1	sandbox	no earlier selected standalone export	14,961
602.gcc_s-734B	v3.1	sandbox	standalone selected but late	248
602.gcc_s-734B	v3.1	sandbox	standalone selected earlier but not timely	131
602.gcc_s-734B	v3.1	sms	no earlier selected standalone export	10,966
602.gcc_s-734B	v3.1	sms	standalone selected but late	130
602.gcc_s-734B	v3.1	sms	standalone selected earlier but not timely	117
602.gcc_s-734B	v3.1	spp	no earlier selected standalone export	7,026
602.gcc_s-734B	v3.1	spp	standalone selected but late	52
602.gcc_s-734B	v3.1	spp	standalone selected earlier but not timely	79
602.gcc_s-734B	v3.1	streamer	no earlier selected standalone export	11,839
602.gcc_s-734B	v3.1	streamer	standalone selected but late	181
602.gcc_s-734B	v3.1	streamer	standalone selected earlier but not timely	116
602.gcc_s-734B	v3.1	stride	no earlier selected standalone export	1,418
602.gcc_s-734B	v3.1	stride	standalone selected but late	48
602.gcc_s-734B	v3.1	stride	standalone selected earlier but not timely	16
602.gcc_s-734B	v3.3	ampm	no earlier selected standalone export	11,489
602.gcc_s-734B	v3.3	ampm	standalone selected but late	83
602.gcc_s-734B	v3.3	ampm	standalone selected earlier but not timely	138
602.gcc_s-734B	v3.3	power7	no earlier selected standalone export	12,320
602.gcc_s-734B	v3.3	power7	standalone selected but late	113
602.gcc_s-734B	v3.3	power7	standalone selected earlier but not timely	143
602.gcc_s-734B	v3.3	sandbox	no earlier selected standalone export	15,153

trace	std	normal	reason	events
602.gcc_s-734B	v3.3	sandbox	standalone selected but late	182
602.gcc_s-734B	v3.3	sandbox	standalone selected earlier but not timely	154
602.gcc_s-734B	v3.3	sms	no earlier selected standalone export	11,101
602.gcc_s-734B	v3.3	sms	standalone selected but late	75
602.gcc_s-734B	v3.3	sms	standalone selected earlier but not timely	153
602.gcc_s-734B	v3.3	spp	no earlier selected standalone export	7,003
602.gcc_s-734B	v3.3	spp	standalone selected but late	13
602.gcc_s-734B	v3.3	spp	standalone selected earlier but not timely	102
602.gcc_s-734B	v3.3	streamer	no earlier selected standalone export	12,567
602.gcc_s-734B	v3.3	streamer	standalone selected but late	117
602.gcc_s-734B	v3.3	streamer	standalone selected earlier but not timely	143
602.gcc_s-734B	v3.3	stride	no earlier selected standalone export	1,672
602.gcc_s-734B	v3.3	stride	standalone selected but late	36
602.gcc_s-734B	v3.3	stride	standalone selected earlier but not timely	12
605.mcf_s-994B	v3.1	ampm	no earlier selected standalone export	17,080
605.mcf_s-994B	v3.1	ampm	standalone selected but late	65
605.mcf_s-994B	v3.1	ampm	standalone selected earlier but not timely	368
605.mcf_s-994B	v3.1	power7	no earlier selected standalone export	35,698
605.mcf_s-994B	v3.1	power7	standalone selected but late	429
605.mcf_s-994B	v3.1	power7	standalone selected earlier but not timely	853
605.mcf_s-994B	v3.1	sandbox	no earlier selected standalone export	109,956
605.mcf_s-994B	v3.1	sandbox	standalone selected but late	1,360
605.mcf_s-994B	v3.1	sandbox	standalone selected earlier but not timely	1,742
605.mcf_s-994B	v3.1	sms	no earlier selected standalone export	45,050
605.mcf_s-994B	v3.1	sms	standalone selected but late	741
605.mcf_s-994B	v3.1	sms	standalone selected earlier but not timely	1,176
605.mcf_s-994B	v3.1	spp	no earlier selected standalone export	840
605.mcf_s-994B	v3.1	spp	standalone selected but late	13
605.mcf_s-994B	v3.1	spp	standalone selected earlier but not timely	75
605.mcf_s-994B	v3.1	streamer	no earlier selected standalone export	43,517
605.mcf_s-994B	v3.1	streamer	standalone selected but late	436
605.mcf_s-994B	v3.1	streamer	standalone selected earlier but not timely	840
605.mcf_s-994B	v3.1	stride	no earlier selected standalone export	813
605.mcf_s-994B	v3.1	stride	standalone selected earlier but not timely	53
605.mcf_s-994B	v3.3	ampm	no earlier selected standalone export	27,960
605.mcf_s-994B	v3.3	ampm	standalone selected but late	239
605.mcf_s-994B	v3.3	ampm	standalone selected earlier but not timely	1,083
605.mcf_s-994B	v3.3	power7	no earlier selected standalone export	41,789
605.mcf_s-994B	v3.3	power7	standalone selected but late	1,159
605.mcf_s-994B	v3.3	power7	standalone selected earlier but not timely	1,420
605.mcf_s-994B	v3.3	sandbox	no earlier selected standalone export	115,765
605.mcf_s-994B	v3.3	sandbox	standalone selected but late	3,518
605.mcf_s-994B	v3.3	sandbox	standalone selected earlier but not timely	2,794
605.mcf_s-994B	v3.3	sms	no earlier selected standalone export	51,878
605.mcf_s-994B	v3.3	sms	standalone selected but late	1,241
605.mcf_s-994B	v3.3	sms	standalone selected earlier but not timely	1,613
605.mcf_s-994B	v3.3	spp	no earlier selected standalone export	892
605.mcf_s-994B	v3.3	spp	standalone selected but late	40
605.mcf_s-994B	v3.3	spp	standalone selected earlier but not timely	95
605.mcf_s-994B	v3.3	streamer	no earlier selected standalone export	51,539
605.mcf_s-994B	v3.3	streamer	standalone selected but late	1,221
605.mcf_s-994B	v3.3	streamer	standalone selected earlier but not timely	1,589
605.mcf_s-994B	v3.3	stride	no earlier selected standalone export	1,625
605.mcf_s-994B	v3.3	stride	standalone selected earlier but not timely	89
619.lbm_s-4268B	v3.1	ampm	no earlier selected standalone export	593
619.lbm_s-4268B	v3.1	ampm	standalone selected but late	3,045
619.lbm_s-4268B	v3.1	ampm	standalone selected earlier but not timely	684
619.lbm_s-4268B	v3.1	power7	no earlier selected standalone export	461
619.lbm_s-4268B	v3.1	power7	standalone selected but late	2,127
619.lbm_s-4268B	v3.1	power7	standalone selected earlier but not timely	544
619.lbm_s-4268B	v3.1	sandbox	no earlier selected standalone export	855
619.lbm_s-4268B	v3.1	sandbox	standalone selected but late	2,437
619.lbm_s-4268B	v3.1	sandbox	standalone selected earlier but not timely	602
619.lbm_s-4268B	v3.1	sms	no earlier selected standalone export	974

trace	std	normal	reason	events
619.lbm_s-4268B	v3.1	sms	standalone selected but late	3,545
619.lbm_s-4268B	v3.1	sms	standalone selected earlier but not timely	699
619.lbm_s-4268B	v3.1	spp	no earlier selected standalone export	246
619.lbm_s-4268B	v3.1	spp	standalone selected but late	691
619.lbm_s-4268B	v3.1	spp	standalone selected earlier but not timely	149
619.lbm_s-4268B	v3.1	streamer	no earlier selected standalone export	417
619.lbm_s-4268B	v3.1	streamer	standalone selected but late	1,695
619.lbm_s-4268B	v3.1	streamer	standalone selected earlier but not timely	430
619.lbm_s-4268B	v3.1	stride	no earlier selected standalone export	117
619.lbm_s-4268B	v3.1	stride	standalone selected but late	868
619.lbm_s-4268B	v3.1	stride	standalone selected earlier but not timely	371
619.lbm_s-4268B	v3.3	ampm	no earlier selected standalone export	558
619.lbm_s-4268B	v3.3	ampm	standalone selected but late	1,284
619.lbm_s-4268B	v3.3	ampm	standalone selected earlier but not timely	726
619.lbm_s-4268B	v3.3	power7	no earlier selected standalone export	428
619.lbm_s-4268B	v3.3	power7	standalone selected but late	911
619.lbm_s-4268B	v3.3	power7	standalone selected earlier but not timely	564
619.lbm_s-4268B	v3.3	sandbox	no earlier selected standalone export	880
619.lbm_s-4268B	v3.3	sandbox	standalone selected but late	1,051
619.lbm_s-4268B	v3.3	sandbox	standalone selected earlier but not timely	610
619.lbm_s-4268B	v3.3	sms	no earlier selected standalone export	961
619.lbm_s-4268B	v3.3	sms	standalone selected but late	1,517
619.lbm_s-4268B	v3.3	sms	standalone selected earlier but not timely	762
619.lbm_s-4268B	v3.3	spp	no earlier selected standalone export	293
619.lbm_s-4268B	v3.3	spp	standalone selected but late	320
619.lbm_s-4268B	v3.3	spp	standalone selected earlier but not timely	186
619.lbm_s-4268B	v3.3	streamer	no earlier selected standalone export	453
619.lbm_s-4268B	v3.3	streamer	standalone selected but late	737
619.lbm_s-4268B	v3.3	streamer	standalone selected earlier but not timely	467
619.lbm_s-4268B	v3.3	stride	no earlier selected standalone export	96
619.lbm_s-4268B	v3.3	stride	standalone selected but late	327
619.lbm_s-4268B	v3.3	stride	standalone selected earlier but not timely	409
620.omnetpp_s-874B	v3.1	ampm	no earlier selected standalone export	5,332
620.omnetpp_s-874B	v3.1	ampm	standalone selected but late	40
620.omnetpp_s-874B	v3.1	ampm	standalone selected earlier but not timely	208
620.omnetpp_s-874B	v3.1	ipcp	no earlier selected standalone export	521
620.omnetpp_s-874B	v3.1	ipcp	standalone selected but late	6
620.omnetpp_s-874B	v3.1	ipcp	standalone selected earlier but not timely	61
620.omnetpp_s-874B	v3.1	power7	no earlier selected standalone export	15,142
620.omnetpp_s-874B	v3.1	power7	standalone selected but late	106
620.omnetpp_s-874B	v3.1	power7	standalone selected earlier but not timely	393
620.omnetpp_s-874B	v3.1	sandbox	no earlier selected standalone export	69,886
620.omnetpp_s-874B	v3.1	sandbox	standalone selected but late	325
620.omnetpp_s-874B	v3.1	sandbox	standalone selected earlier but not timely	1,325
620.omnetpp_s-874B	v3.1	sms	no earlier selected standalone export	46,252
620.omnetpp_s-874B	v3.1	sms	standalone selected but late	209
620.omnetpp_s-874B	v3.1	sms	standalone selected earlier but not timely	988
620.omnetpp_s-874B	v3.1	spp	no earlier selected standalone export	20
620.omnetpp_s-874B	v3.1	streamer	no earlier selected standalone export	14,691
620.omnetpp_s-874B	v3.1	streamer	standalone selected but late	108
620.omnetpp_s-874B	v3.1	streamer	standalone selected earlier but not timely	379
620.omnetpp_s-874B	v3.1	stride	no earlier selected standalone export	1,322
620.omnetpp_s-874B	v3.1	stride	standalone selected but late	2
620.omnetpp_s-874B	v3.1	stride	standalone selected earlier but not timely	39
620.omnetpp_s-874B	v3.3	ampm	no earlier selected standalone export	4,563
620.omnetpp_s-874B	v3.3	ampm	standalone selected but late	28
620.omnetpp_s-874B	v3.3	ampm	standalone selected earlier but not timely	286
620.omnetpp_s-874B	v3.3	ipcp	no earlier selected standalone export	576
620.omnetpp_s-874B	v3.3	ipcp	standalone selected but late	4
620.omnetpp_s-874B	v3.3	ipcp	standalone selected earlier but not timely	55
620.omnetpp_s-874B	v3.3	power7	no earlier selected standalone export	13,839
620.omnetpp_s-874B	v3.3	power7	standalone selected but late	73
620.omnetpp_s-874B	v3.3	power7	standalone selected earlier but not timely	445
620.omnetpp_s-874B	v3.3	sandbox	no earlier selected standalone export	66,289

trace	std	normal	reason	events
620.omnetpp_s-874B	v3.3	sandbox	standalone selected but late	292
620.omnetpp_s-874B	v3.3	sandbox	standalone selected earlier but not timely	1,815
620.omnetpp_s-874B	v3.3	sms	no earlier selected standalone export	44,089
620.omnetpp_s-874B	v3.3	sms	standalone selected but late	180
620.omnetpp_s-874B	v3.3	sms	standalone selected earlier but not timely	1,218
620.omnetpp_s-874B	v3.3	spp	no earlier selected standalone export	12
620.omnetpp_s-874B	v3.3	streamer	no earlier selected standalone export	13,489
620.omnetpp_s-874B	v3.3	streamer	standalone selected but late	82
620.omnetpp_s-874B	v3.3	streamer	standalone selected earlier but not timely	429
620.omnetpp_s-874B	v3.3	stride	no earlier selected standalone export	1,037
620.omnetpp_s-874B	v3.3	stride	standalone selected but late	4
620.omnetpp_s-874B	v3.3	stride	standalone selected earlier but not timely	52
623.xalancbmk_s-700B	v3.1	ampm	no earlier selected standalone export	2,291
623.xalancbmk_s-700B	v3.1	ampm	standalone selected but late	86
623.xalancbmk_s-700B	v3.1	ampm	standalone selected earlier but not timely	493
623.xalancbmk_s-700B	v3.1	ipcp	no earlier selected standalone export	294
623.xalancbmk_s-700B	v3.1	ipcp	standalone selected but late	1
623.xalancbmk_s-700B	v3.1	ipcp	standalone selected earlier but not timely	29
623.xalancbmk_s-700B	v3.1	power7	no earlier selected standalone export	3,935
623.xalancbmk_s-700B	v3.1	power7	standalone selected but late	35
623.xalancbmk_s-700B	v3.1	power7	standalone selected earlier but not timely	587
623.xalancbmk_s-700B	v3.1	sandbox	no earlier selected standalone export	78,329
623.xalancbmk_s-700B	v3.1	sandbox	standalone selected but late	454
623.xalancbmk_s-700B	v3.1	sandbox	standalone selected earlier but not timely	3,464
623.xalancbmk_s-700B	v3.1	sms	no earlier selected standalone export	25,275
623.xalancbmk_s-700B	v3.1	sms	standalone selected but late	491
623.xalancbmk_s-700B	v3.1	sms	standalone selected earlier but not timely	2,169
623.xalancbmk_s-700B	v3.1	spp	no earlier selected standalone export	225
623.xalancbmk_s-700B	v3.1	spp	standalone selected but late	12
623.xalancbmk_s-700B	v3.1	spp	standalone selected earlier but not timely	15
623.xalancbmk_s-700B	v3.1	streamer	no earlier selected standalone export	5,948
623.xalancbmk_s-700B	v3.1	streamer	standalone selected but late	64
623.xalancbmk_s-700B	v3.1	streamer	standalone selected earlier but not timely	815
623.xalancbmk_s-700B	v3.1	stride	no earlier selected standalone export	108
623.xalancbmk_s-700B	v3.1	stride	standalone selected but late	23
623.xalancbmk_s-700B	v3.1	stride	standalone selected earlier but not timely	109
623.xalancbmk_s-700B	v3.3	ampm	no earlier selected standalone export	980
623.xalancbmk_s-700B	v3.3	ampm	standalone selected but late	92
623.xalancbmk_s-700B	v3.3	ampm	standalone selected earlier but not timely	497
623.xalancbmk_s-700B	v3.3	ipcp	no earlier selected standalone export	366
623.xalancbmk_s-700B	v3.3	ipcp	standalone selected earlier but not timely	24
623.xalancbmk_s-700B	v3.3	power7	no earlier selected standalone export	1,152
623.xalancbmk_s-700B	v3.3	power7	standalone selected but late	31
623.xalancbmk_s-700B	v3.3	power7	standalone selected earlier but not timely	485
623.xalancbmk_s-700B	v3.3	sandbox	no earlier selected standalone export	13,435
623.xalancbmk_s-700B	v3.3	sandbox	standalone selected but late	119
623.xalancbmk_s-700B	v3.3	sandbox	standalone selected earlier but not timely	1,717
623.xalancbmk_s-700B	v3.3	sms	no earlier selected standalone export	5,124
623.xalancbmk_s-700B	v3.3	sms	standalone selected but late	160
623.xalancbmk_s-700B	v3.3	sms	standalone selected earlier but not timely	875
623.xalancbmk_s-700B	v3.3	spp	no earlier selected standalone export	101
623.xalancbmk_s-700B	v3.3	spp	standalone selected but late	7
623.xalancbmk_s-700B	v3.3	spp	standalone selected earlier but not timely	12
623.xalancbmk_s-700B	v3.3	streamer	no earlier selected standalone export	1,655
623.xalancbmk_s-700B	v3.3	streamer	standalone selected but late	63
623.xalancbmk_s-700B	v3.3	streamer	standalone selected earlier but not timely	708
623.xalancbmk_s-700B	v3.3	stride	no earlier selected standalone export	142
623.xalancbmk_s-700B	v3.3	stride	standalone selected but late	13
623.xalancbmk_s-700B	v3.3	stride	standalone selected earlier but not timely	119

5 Cache-capacity sensitivity sweep (G)

Run `capacity_all13_half_double_20260626_105924`: frozen L2C capacity-sensitivity control. Each config re-runs the 5 traces under all normal prefetchers and replays the frozen v3.1 / v3.3 lists. **Caveat (all configs)**: frozen lists use the baseline-capacity raw oracle — this is NOT a capacity-trained NN experiment.

5.1 l1d_32set_8way_16KiB (L1D, half, 32 sets $\times$ 8 ways, 16 KiB)

Normal prefetcher baselines:

trace	pref	IPC	spd/no_pref	L2 miss rate	sel.acc	timeli.	cov. vs no_pref
602.gcc.s-734B	no_pref	0.36800	1.0000	0.5025	0.0000	0.0000	0.0000
602.gcc.s-734B	sandbox	0.43628	1.1855	0.0177	0.1018	0.9999	1.9204
602.gcc.s-734B	sms	0.42212	1.1471	0.1019	0.4587	0.9994	1.5732
602.gcc.s-734B	ampm	0.43069	1.1704	0.0421	0.1312	0.9984	1.8367
602.gcc.s-734B	spp	0.42842	1.1642	0.1749	0.0592	0.9985	1.3061
619.lbm.s-4268B	no_pref	0.32192	1.0000	0.9721	0.0000	0.0000	0.0000
619.lbm.s-4268B	sandbox	0.37443	1.1631	0.2788	0.2020	0.8643	1.6598
619.lbm.s-4268B	sms	0.38103	1.1836	0.1262	0.9428	0.9540	1.8183
619.lbm.s-4268B	ampm	0.37483	1.1644	0.1929	0.3249	0.9517	1.6859
619.lbm.s-4268B	spp	0.37927	1.1781	0.7460	0.0817	0.7455	0.6460
605.mcf.s-994B	no_pref	0.18276	1.0000	0.6902	0.0000	0.0000	0.0000
605.mcf.s-994B	sandbox	0.16722	0.9150	0.6724	0.0308	0.7690	0.6241
605.mcf.s-994B	sms	0.18713	1.0239	0.6380	0.1858	0.8074	0.3488
605.mcf.s-994B	ampm	0.18817	1.0296	0.6410	0.1552	0.9858	0.1226
605.mcf.s-994B	spp	0.18808	1.0291	0.6850	0.0172	0.9210	0.0198
620.omnetpp.s-874B	no_pref	0.23617	1.0000	0.5525	0.0000	0.0000	0.0000
620.omnetpp.s-874B	sandbox	0.24380	1.0323	0.4143	0.0509	0.9737	1.1803
620.omnetpp.s-874B	sms	0.24562	1.0400	0.4284	0.2263	0.9823	0.6278
620.omnetpp.s-874B	ampm	0.23823	1.0087	0.5244	0.1184	0.9566	0.1428
620.omnetpp.s-874B	spp	0.23716	1.0042	0.5522	0.0003	0.6667	0.0002
623.xalancbmk.s-700B	no_pref	0.34960	1.0000	0.3040	0.0000	0.0000	0.0000
623.xalancbmk.s-700B	sandbox	0.33544	0.9595	0.4853	0.0341	0.9596	0.8307
623.xalancbmk.s-700B	sms	0.33861	0.9686	0.4388	0.1334	0.9605	0.6602
623.xalancbmk.s-700B	ampm	0.34537	0.9879	0.3452	0.0280	0.9411	0.1214
623.xalancbmk.s-700B	spp	0.35004	1.0013	0.2983	0.0043	0.8119	0.0195

Standalone v3.1 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42863	0.9825	1.1648	0.9737	0.9986	0.0501	sandbox@0.4363
619.lbm.s-4268B	0.38472	1.0097	1.1951	0.9785	0.9844	0.0237	sms@0.3810
605.mcf.s-994B	0.18818	1.0001	1.0297	0.5254	0.8011	0.6408	ampm@0.1882
620.omnetpp.s-874B	0.24363	0.9919	1.0316	0.4462	0.9908	0.4351	sms@0.2456
623.xalancbmk.s-700B	0.35953	1.0271	1.0284	0.5848	0.9869	0.1977	spp@0.3500

Standalone v3.3 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42887	0.9830	1.1654	0.9809	0.9990	0.0494	sandbox@0.4363
619.lbm.s-4268B	0.38419	1.0083	1.1934	0.9858	0.9931	0.0170	sms@0.3810
605.mcf.s-994B	0.18686	0.9930	1.0224	0.4307	0.7382	0.6440	ampm@0.1882
620.omnetpp.s-874B	0.24415	0.9940	1.0338	0.4345	0.9930	0.4264	sms@0.2456
623.xalancbmk.s-700B	0.37369	1.0676	1.0689	0.6926	0.9971	0.0566	spp@0.3500

5.2 l1d_128set_8way_64KiB (L1D, double, 128 sets $\times$ 8 ways, 64 KiB)

Normal prefetcher baselines:

trace	pref	IPC	spd/no_pref	L2 miss rate	sel.acc	timeli.	cov. vs no_pref
602.gcc.s-734B	no_pref	0.36799	1.0000	0.6001	0.0000	0.0000	0.0000
602.gcc.s-734B	sandbox	0.43656	1.1863	0.0216	0.1182	0.9998	1.9096
602.gcc.s-734B	sms	0.42347	1.1508	0.1101	0.5284	0.9993	1.6176
602.gcc.s-734B	ampm	0.43087	1.1709	0.0511	0.1530	0.9984	1.8247
602.gcc.s-734B	spp	0.42849	1.1644	0.2104	0.0692	0.9985	1.3031
619.lbm.s-4268B	no_pref	0.32193	1.0000	1.0000	0.0000	0.0000	0.0000
619.lbm.s-4268B	sandbox	0.37446	1.1632	0.2887	0.2096	0.8634	1.6577
619.lbm.s-4268B	sms	0.38112	1.1839	0.1301	0.9427	0.9538	1.8181
619.lbm.s-4268B	ampm	0.37479	1.1642	0.1989	0.3274	0.9514	1.6856
619.lbm.s-4268B	spp	0.37949	1.1788	0.7661	0.0826	0.7461	0.6479
605.mcf.s-994B	no_pref	0.18358	1.0000	0.8190	0.0000	0.0000	0.0000
605.mcf.s-994B	sandbox	0.16806	0.9155	0.7626	0.0352	0.7582	0.5723
605.mcf.s-994B	sms	0.18838	1.0261	0.7521	0.2151	0.8066	0.3438
605.mcf.s-994B	ampm	0.18919	1.0306	0.7598	0.2012	0.9857	0.1219
605.mcf.s-994B	spp	0.18923	1.0308	0.8138	0.0238	0.9161	0.0199
620.omnetpp.s-874B	no_pref	0.23810	1.0000	0.8028	0.0000	0.0000	0.0000
620.omnetpp.s-874B	sandbox	0.24613	1.0337	0.5733	0.0649	0.9716	1.0956
620.omnetpp.s-874B	sms	0.24782	1.0408	0.6228	0.2735	0.9814	0.6159
620.omnetpp.s-874B	ampm	0.24031	1.0093	0.7620	0.1805	0.9614	0.1388
620.omnetpp.s-874B	spp	0.23960	1.0063	0.8024	0.0011	0.6512	0.0006
623.xalancbm.s-700B	no_pref	0.35519	1.0000	0.4404	0.0000	0.0000	0.0000
623.xalancbm.s-700B	sandbox	0.34789	0.9794	0.5694	0.0566	0.9594	1.1566
623.xalancbm.s-700B	sms	0.34412	0.9688	0.6211	0.1713	0.9658	0.6536
623.xalancbm.s-700B	ampm	0.35146	0.9895	0.4880	0.0987	0.9418	0.1224
623.xalancbm.s-700B	spp	0.35244	0.9923	0.4849	0.0156	0.8143	0.0486

Standalone v3.1 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42855	0.9817	1.1646	0.9742	0.9987	0.0656	sandbox@0.4366
619.lbm.s-4268B	0.38501	1.0102	1.1959	0.9782	0.9841	0.0245	sms@0.3811
605.mcf.s-994B	0.18909	0.9993	1.0300	0.5310	0.8008	0.7603	spp@0.1892
620.omnetpp.s-874B	0.24554	0.9908	1.0312	0.4790	0.9911	0.6354	sms@0.2478
623.xalancbm.s-700B	0.36617	1.0390	1.0309	0.6984	0.9865	0.2731	spp@0.3524

Standalone v3.3 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42880	0.9822	1.1652	0.9813	0.9991	0.0647	sandbox@0.4366
619.lbm.s-4268B	0.38433	1.0084	1.1938	0.9858	0.9931	0.0175	sms@0.3811
605.mcf.s-994B	0.18771	0.9920	1.0225	0.4310	0.7372	0.7643	spp@0.1892
620.omnetpp.s-874B	0.24604	0.9928	1.0333	0.4640	0.9929	0.6240	sms@0.2478
623.xalancbm.s-700B	0.38029	1.0790	1.0707	0.7563	0.9970	0.0821	spp@0.3524

5.3 l2c_256set_8way_128KiB (L2C, half, 256 sets × 8 ways, 128 KiB)**Normal prefetcher baselines:**

trace	pref	IPC	spd/no_pref	L2 miss rate	sel.acc	timeli.	cov. vs no_pref
602.gcc.s-734B	no_pref	0.36787	1.0000	0.5036	0.0000	0.0000	0.0000
602.gcc.s-734B	sandbox	0.43634	1.1861	0.0187	0.1042	0.9999	1.9600
602.gcc.s-734B	sms	0.42198	1.1471	0.1036	0.4609	0.9994	1.5774
602.gcc.s-734B	ampm	0.43059	1.1705	0.0434	0.1317	0.9984	1.8400
602.gcc.s-734B	spp	0.42840	1.1645	0.1758	0.0594	0.9985	1.3058
619.lbm.s-4268B	no_pref	0.32184	1.0000	0.9992	0.0000	0.0000	0.0000
619.lbm.s-4268B	sandbox	0.37374	1.1613	0.2836	0.2104	0.8659	1.6621
619.lbm.s-4268B	sms	0.37942	1.1789	0.1307	0.9426	0.9542	1.8162
619.lbm.s-4268B	ampm	0.37368	1.1611	0.1983	0.3285	0.9522	1.6849
619.lbm.s-4268B	spp	0.37632	1.1693	0.7804	0.0807	0.7284	0.6304
605.mcf.s-994B	no_pref	0.18162	1.0000	0.8331	0.0000	0.0000	0.0000
605.mcf.s-994B	sandbox	0.16829	0.9266	0.7597	0.0364	0.7712	0.5873
605.mcf.s-994B	sms	0.18662	1.0275	0.7588	0.2324	0.8212	0.3645
605.mcf.s-994B	ampm	0.18774	1.0337	0.7741	0.2004	0.9864	0.1242
605.mcf.s-994B	spp	0.18771	1.0335	0.8264	0.0233	0.9221	0.0209
620.omnetpp.s-874B	no_pref	0.23615	1.0000	0.7868	0.0000	0.0000	0.0000
620.omnetpp.s-874B	sandbox	0.24266	1.0276	0.5864	0.0569	0.9715	1.0682
620.omnetpp.s-874B	sms	0.24545	1.0394	0.6212	0.2617	0.9823	0.5955
620.omnetpp.s-874B	ampm	0.23844	1.0097	0.7495	0.1607	0.9605	0.1311
620.omnetpp.s-874B	spp	0.23740	1.0053	0.7864	0.0005	0.5695	0.0002
623.xalancbmk.s-700B	no_pref	0.32794	1.0000	0.7721	0.0000	0.0000	0.0000
623.xalancbmk.s-700B	sandbox	0.34172	1.0420	0.5540	0.0468	0.9535	0.5352
623.xalancbmk.s-700B	sms	0.33408	1.0187	0.6594	0.1897	0.9778	0.3896
623.xalancbmk.s-700B	ampm	0.32905	1.0034	0.7337	0.1454	0.9600	0.1298
623.xalancbmk.s-700B	spp	0.32819	1.0008	0.7664	0.0118	0.8348	0.0208

Standalone v3.1 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42851	0.9821	1.1648	0.9736	0.9986	0.0515	sandbox@0.4363
619.lbm.s-4268B	0.38340	1.0105	1.1913	0.9794	0.9853	0.0246	sms@0.3794
605.mcf.s-994B	0.18747	0.9986	1.0322	0.5648	0.8087	0.7748	ampm@0.1877
620.omnetpp.s-874B	0.24361	0.9925	1.0316	0.4828	0.9910	0.6259	sms@0.2455
623.xalancbmk.s-700B	0.34140	0.9991	1.0410	0.8685	0.9893	0.5482	sandbox@0.3417

Standalone v3.3 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42874	0.9826	1.1655	0.9808	0.9990	0.0508	sandbox@0.4363
619.lbm.s-4268B	0.38302	1.0095	1.1901	0.9865	0.9938	0.0179	sms@0.3794
605.mcf.s-994B	0.18599	0.9907	1.0241	0.4696	0.7489	0.7774	ampm@0.1877
620.omnetpp.s-874B	0.24408	0.9944	1.0336	0.4687	0.9928	0.6152	sms@0.2455
623.xalancbmk.s-700B	0.35571	1.0409	1.0847	0.9176	0.9954	0.3462	sandbox@0.3417

5.4 l2c_1024set_8way_512KiB (L2C, double, 1024 sets × 8 ways, 512 KiB)**Normal prefetcher baselines:**

trace	pref	IPC	spd/no_pref	L2 miss rate	sel.acc	timeli.	cov. vs no_pref
602.gcc.s-734B	no_pref	0.36804	1.0000	0.5024	0.0000	0.0000	0.0000
602.gcc.s-734B	sandbox	0.43629	1.1854	0.0175	0.1018	0.9998	1.9207
602.gcc.s-734B	sms	0.42222	1.1472	0.1016	0.4588	0.9994	1.5741
602.gcc.s-734B	ampm	0.43070	1.1703	0.0418	0.1312	0.9984	1.8378
602.gcc.s-734B	spp	0.42839	1.1640	0.1748	0.0592	0.9986	1.3061
619.lbm.s-4268B	no_pref	0.33043	1.0000	0.9983	0.0000	0.0000	0.0000
619.lbm.s-4268B	sandbox	0.38675	1.1704	0.2992	0.2061	0.8572	1.6457
619.lbm.s-4268B	sms	0.39463	1.1943	0.1331	0.9409	0.9506	1.8147
619.lbm.s-4268B	ampm	0.38717	1.1717	0.2020	0.3245	0.9498	1.6819
619.lbm.s-4268B	spp	0.39163	1.1852	0.7522	0.0837	0.7518	0.6609
605.mcf.s-994B	no_pref	0.18376	1.0000	0.6714	0.0000	0.0000	0.0000
605.mcf.s-994B	sandbox	0.16952	0.9225	0.6715	0.0296	0.7467	0.5930
605.mcf.s-994B	sms	0.18909	1.0290	0.6293	0.1651	0.7896	0.3212
605.mcf.s-994B	ampm	0.19001	1.0340	0.6229	0.1552	0.9850	0.1194
605.mcf.s-994B	spp	0.19026	1.0354	0.6670	0.0171	0.9224	0.0186
620.omnetpp.s-874B	no_pref	0.23908	1.0000	0.6036	0.0000	0.0000	0.0000
620.omnetpp.s-874B	sandbox	0.24630	1.0302	0.4565	0.0526	0.9710	1.2841
620.omnetpp.s-874B	sms	0.24865	1.0400	0.4714	0.2229	0.9814	0.6610
620.omnetpp.s-874B	ampm	0.24142	1.0098	0.5742	0.1427	0.9584	0.1517
620.omnetpp.s-874B	spp	0.24028	1.0050	0.6034	0.0004	0.5672	0.0003
623.xalancbmk.s-700B	no_pref	0.38070	1.0000	0.0220	0.0000	0.0000	0.0000
623.xalancbmk.s-700B	sandbox	0.35066	0.9211	0.4132	0.0420	0.9531	16.8495
623.xalancbmk.s-700B	sms	0.37488	0.9847	0.0869	0.0240	0.8965	1.7303
623.xalancbmk.s-700B	ampm	0.37949	0.9968	0.0344	0.0089	0.9813	0.2778
623.xalancbmk.s-700B	spp	0.38065	0.9999	0.0214	0.0007	0.9274	0.0448

Standalone v3.1 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42868	0.9826	1.1648	0.9738	0.9986	0.0500	sandbox@0.4363
619.lbm.s-4268B	0.39783	1.0081	1.2040	0.9765	0.9824	0.0267	sms@0.3946
605.mcf.s-994B	0.18933	0.9951	1.0303	0.4586	0.7853	0.6248	spp@0.1903
620.omnetpp.s-874B	0.24693	0.9931	1.0328	0.4121	0.9902	0.4718	sms@0.2487
623.xalancbmk.s-700B	0.38128	1.0017	1.0015	0.0416	0.9793	0.0151	spp@0.3806

Standalone v3.3 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42891	0.9831	1.1654	0.9810	0.9990	0.0493	sandbox@0.4363
619.lbm.s-4268B	0.39741	1.0070	1.2027	0.9847	0.9920	0.0189	sms@0.3946
605.mcf.s-994B	0.18808	0.9885	1.0235	0.3634	0.7148	0.6296	spp@0.1903
620.omnetpp.s-874B	0.24751	0.9954	1.0353	0.4020	0.9924	0.4621	sms@0.2487
623.xalancbmk.s-700B	0.38237	1.0045	1.0044	0.0507	0.9972	0.0035	spp@0.3806

5.5 llc_1024set_16way_1024KiB (LLC, half, 1024 sets × 16 ways, 1024 KiB)**Normal prefetcher baselines:**

trace	pref	IPC	spd/no_pref	L2 miss rate	sel.acc	timeli.	cov. vs no_pref
602.gcc.s-734B	ampm	0.42992	1.1806	0.0422	0.1312	0.9983	1.8365
602.gcc.s-734B	no_pref	0.36414	1.0000	0.5025	0.0000	0.0000	0.0000
602.gcc.s-734B	sandbox	0.43597	1.1973	0.0177	0.1018	0.9998	1.9204
602.gcc.s-734B	sms	0.42079	1.1556	0.1019	0.4588	0.9994	1.5733
602.gcc.s-734B	spp	0.42829	1.1762	0.1753	0.0591	0.9986	1.3051
619.lbm.s-4268B	ampm	0.37538	1.1653	0.1971	0.3274	0.9525	1.6864
619.lbm.s-4268B	no_pref	0.32212	1.0000	0.9987	0.0000	0.0000	0.0000
619.lbm.s-4268B	sandbox	0.37509	1.1644	0.2818	0.2099	0.8669	1.6637
619.lbm.s-4268B	sms	0.38156	1.1845	0.1288	0.9433	0.9547	1.8186
619.lbm.s-4268B	spp	0.37845	1.1749	0.7648	0.0825	0.7439	0.6477
605.mcf.s-994B	ampm	0.18388	1.0383	0.6974	0.1777	0.9849	0.1221
605.mcf.s-994B	no_pref	0.17710	1.0000	0.7513	0.0000	0.0000	0.0000
605.mcf.s-994B	sandbox	0.16159	0.9124	0.7356	0.0331	0.7432	0.5905
605.mcf.s-994B	sms	0.18307	1.0337	0.6971	0.1947	0.7908	0.3386
605.mcf.s-994B	spp	0.18316	1.0342	0.7463	0.0201	0.9075	0.0195
620.omnetpp.s-874B	ampm	0.23484	1.0093	0.6649	0.1556	0.9590	0.1425
620.omnetpp.s-874B	no_pref	0.23267	1.0000	0.7012	0.0000	0.0000	0.0000
620.omnetpp.s-874B	sandbox	0.23837	1.0245	0.5293	0.0558	0.9673	1.1756
620.omnetpp.s-874B	sms	0.24225	1.0412	0.5459	0.2443	0.9801	0.6237
620.omnetpp.s-874B	spp	0.23374	1.0046	0.7008	0.0005	0.6304	0.0003
623.xalancbmk.s-700B	ampm	0.34908	0.9888	0.4151	0.0630	0.9416	0.1174
623.xalancbmk.s-700B	no_pref	0.35303	1.0000	0.3696	0.0000	0.0000	0.0000
623.xalancbmk.s-700B	sandbox	0.33621	0.9524	0.4980	0.0493	0.9457	1.1772
623.xalancbmk.s-700B	sms	0.34154	0.9675	0.5387	0.1509	0.9634	0.6474
623.xalancbmk.s-700B	spp	0.35359	1.0016	0.3609	0.0049	0.8102	0.0172

Standalone v3.1 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42785	0.9814	1.1750	0.9737	0.9985	0.0501	sandbox@0.4360
619.lbm.s-4268B	0.38564	1.0107	1.1972	0.9793	0.9852	0.0243	sms@0.3816
605.mcf.s-994B	0.18320	0.9963	1.0344	0.5026	0.7660	0.7000	ampm@0.1839
620.omnetpp.s-874B	0.24026	0.9918	1.0326	0.4547	0.9906	0.5497	sms@0.2422
623.xalancbmk.s-700B	0.36387	1.0291	1.0307	0.6826	0.9871	0.2199	spp@0.3536

Standalone v3.3 replay:

trace	IPC	spd/best_norm	spd/no_pref	sel.acc	timeli.	L2 miss rate	best normal
602.gcc.s-734B	0.42813	0.9820	1.1757	0.9809	0.9990	0.0494	sandbox@0.4360
619.lbm.s-4268B	0.38488	1.0087	1.1948	0.9861	0.9934	0.0178	sms@0.3816
605.mcf.s-994B	0.18195	0.9895	1.0274	0.3992	0.6850	0.7048	ampm@0.1839
620.omnetpp.s-874B	0.24096	0.9947	1.0356	0.4444	0.9927	0.5381	sms@0.2422
623.xalancbmk.s-700B	0.37868	1.0710	1.0727	0.7504	0.9970	0.0461	spp@0.3536

5.6 llc_4096set_16way_4096KiB (LLC, double, 4096 MiB-class)

Only `normal/RUN_INF0.txt` is present in the bundle (no `summary.csv`); the largest-LLC baseline run is incomplete / not exported. `RUN_KIND=normal.prefetcher.baseline`, prefetchers = ampm no_pref sandbox sms spp, `WARMUP=SIM=25M`.

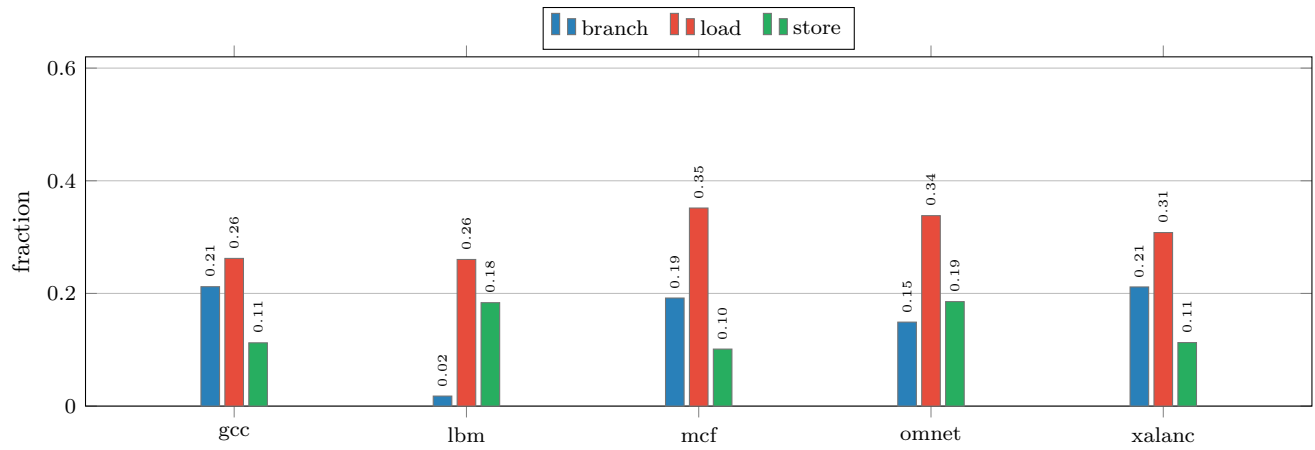
6 Visual comparison diagrams

Design logic: for each headline metric we put the *best normal prefetcher* (gray) side by side with the two standalone NN replays **v3.1** (blue) and **v3.3** (orange), across all five benchmarks. “best normal” is the per-benchmark winning prefetcher by IPC (gcc=sandbox, lbm=sms, mcf=ampm, omnet=sms, xalanc=spp). All values come from the same ROI-25M event-evidence run as the tables above.

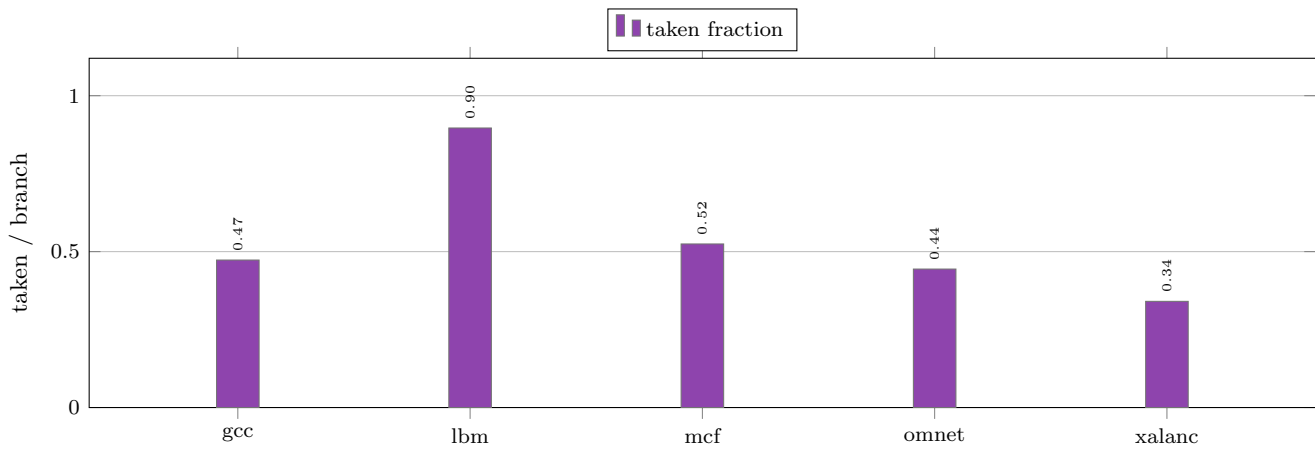
6.1 Workload characterization

Memory fraction $\approx$ load+store, so only the distinct categories (branch / load / store) are charted; the redundant “memory” series is dropped. Taken fraction is the share of *branches* that are taken.

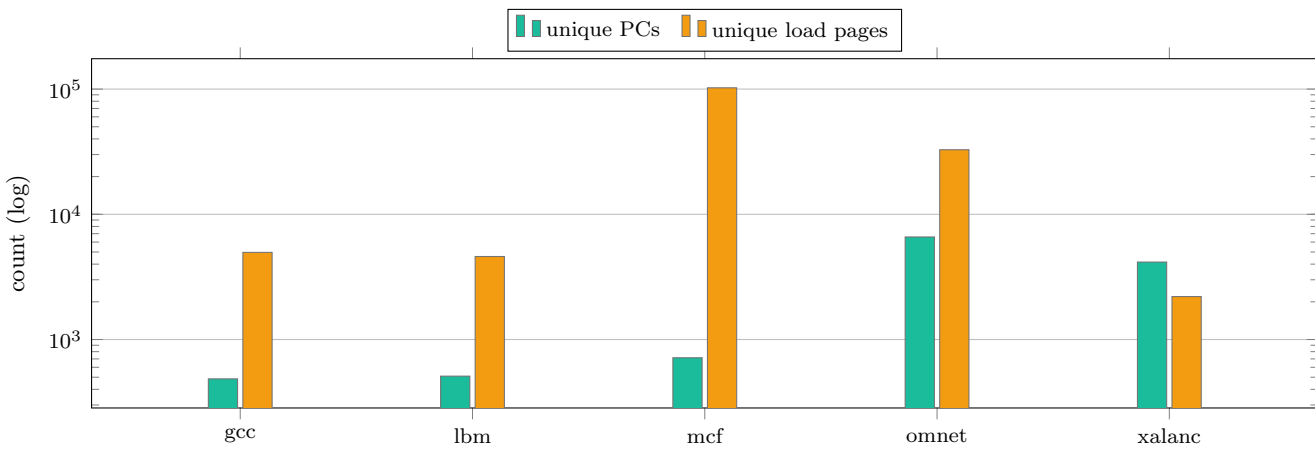
Instruction mix — fraction of dynamic instructions



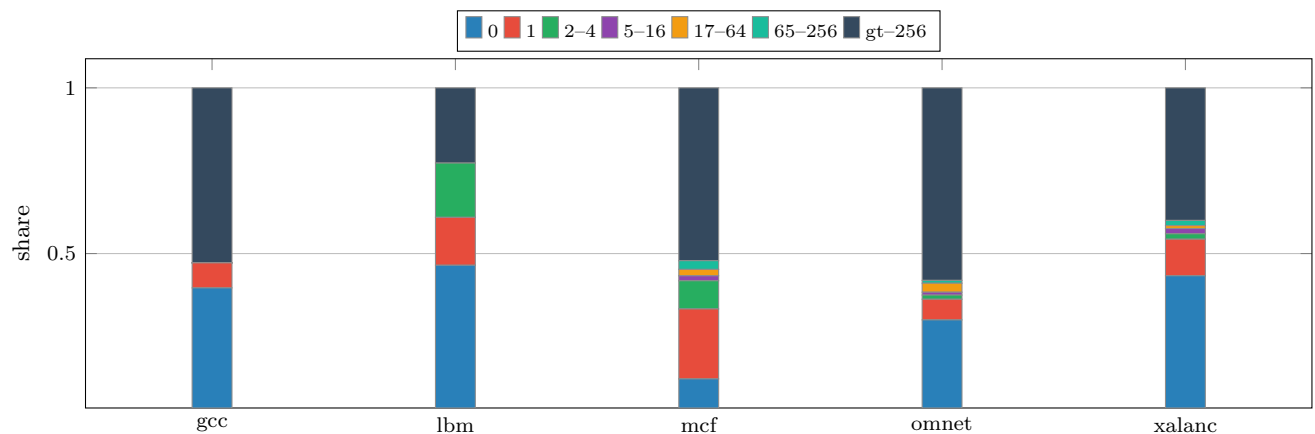
Taken-branch fraction (of branches)



Working-set spread — unique PCs and unique load pages (log)

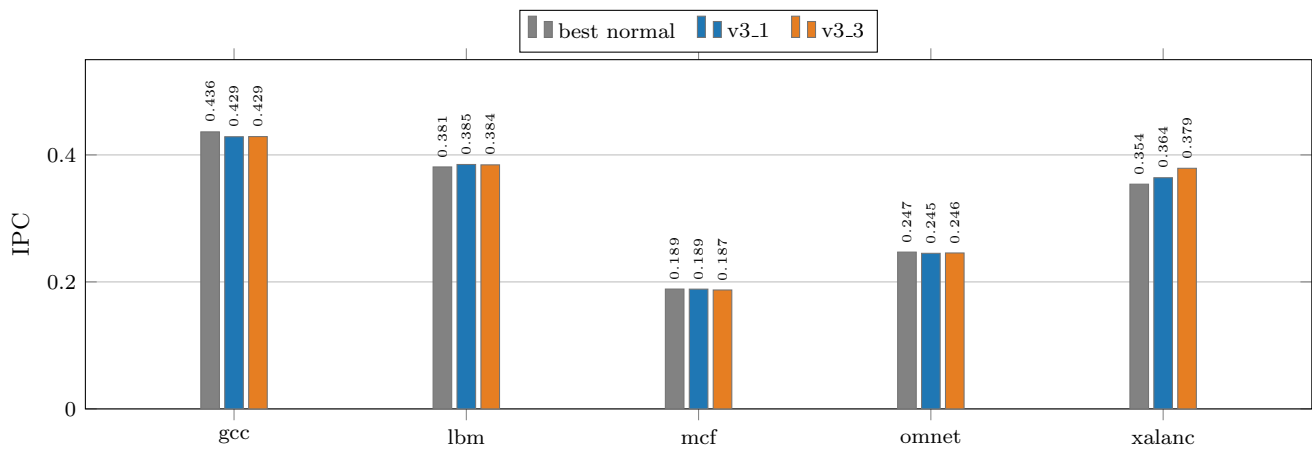


Absolute load line-delta bucket composition (share of observed deltas)

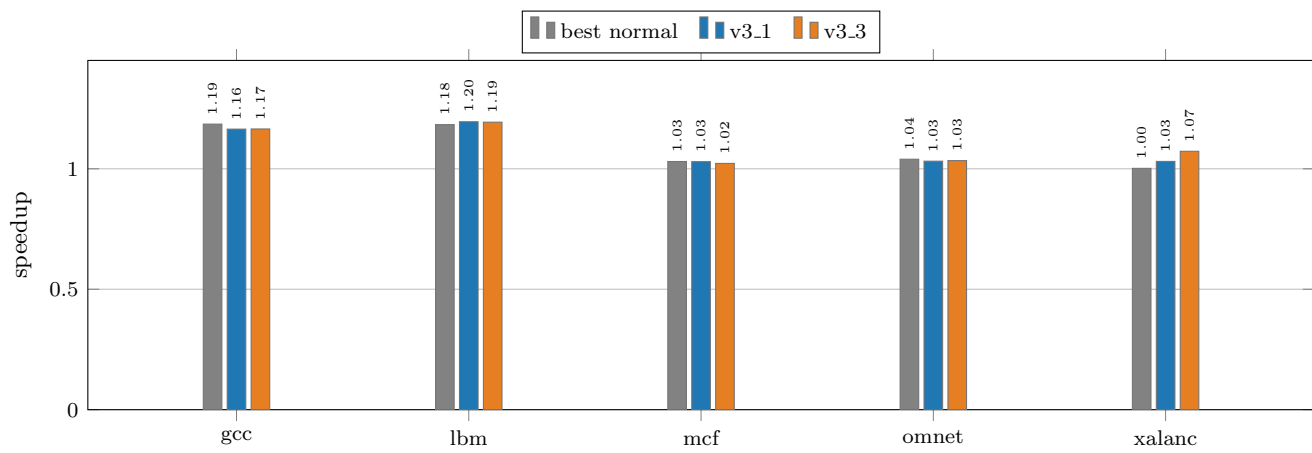


6.2 Best normal vs. NN replay — metric by metric

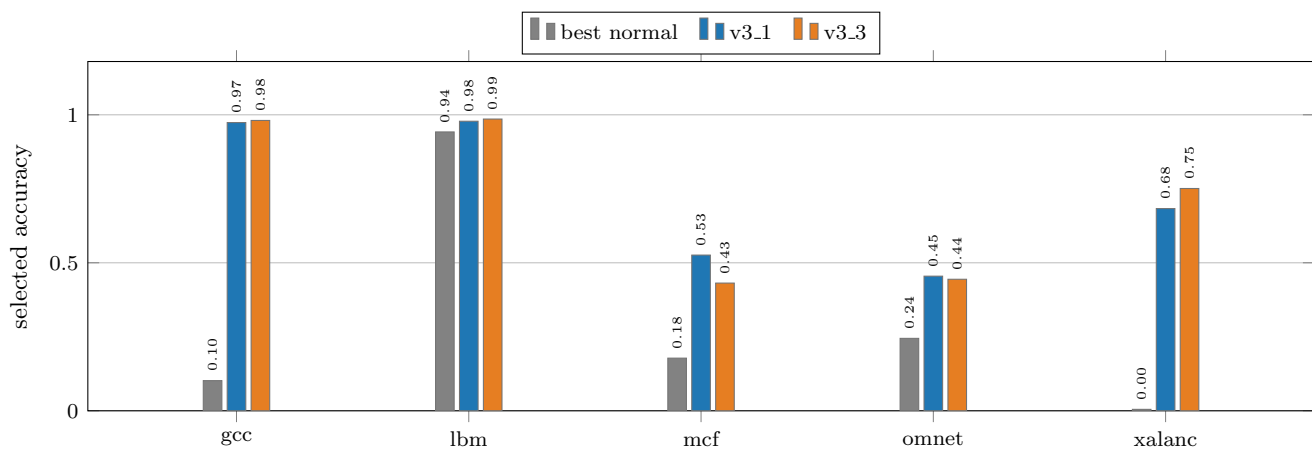
IPC: best normal vs. v3.1 vs. v3.3



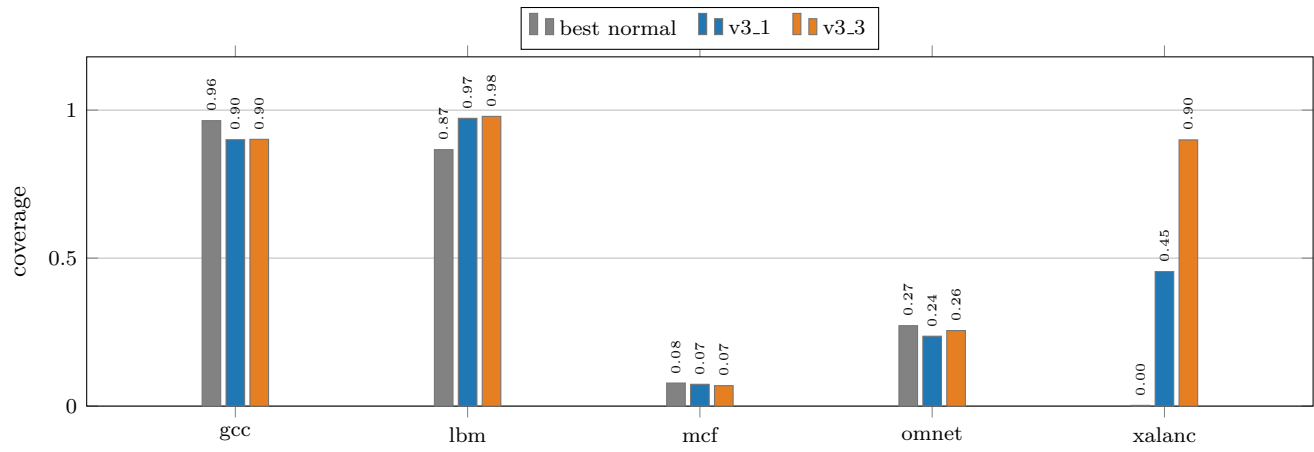
Speedup vs. no_pref



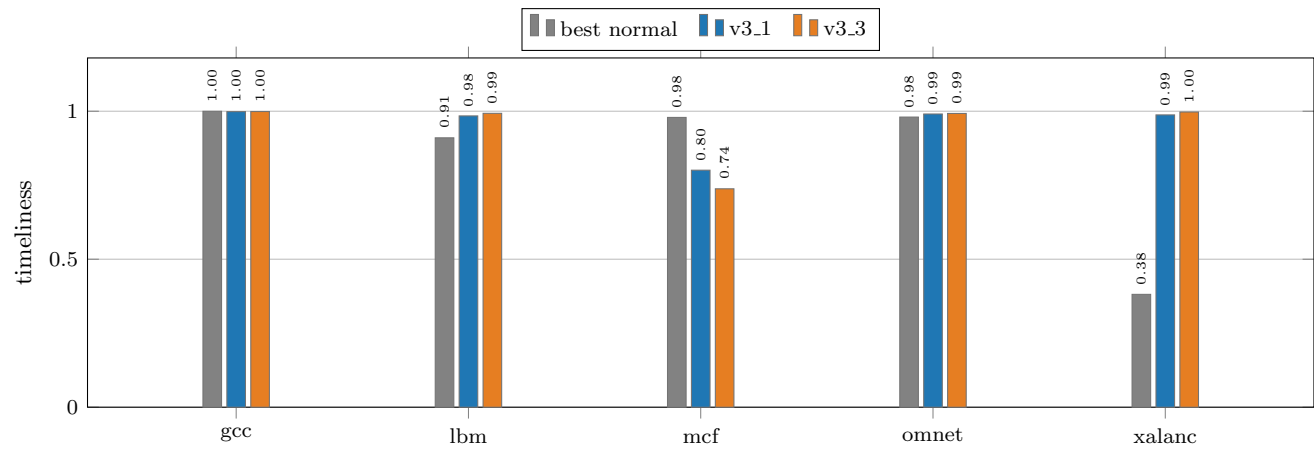
Selected accuracy (useful / selected)



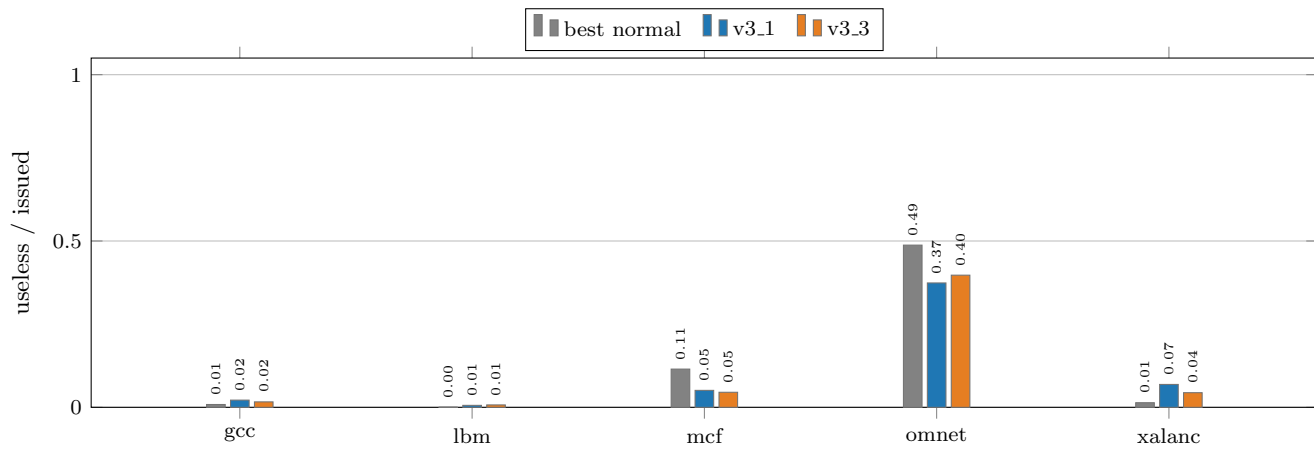
Unique demand-event coverage



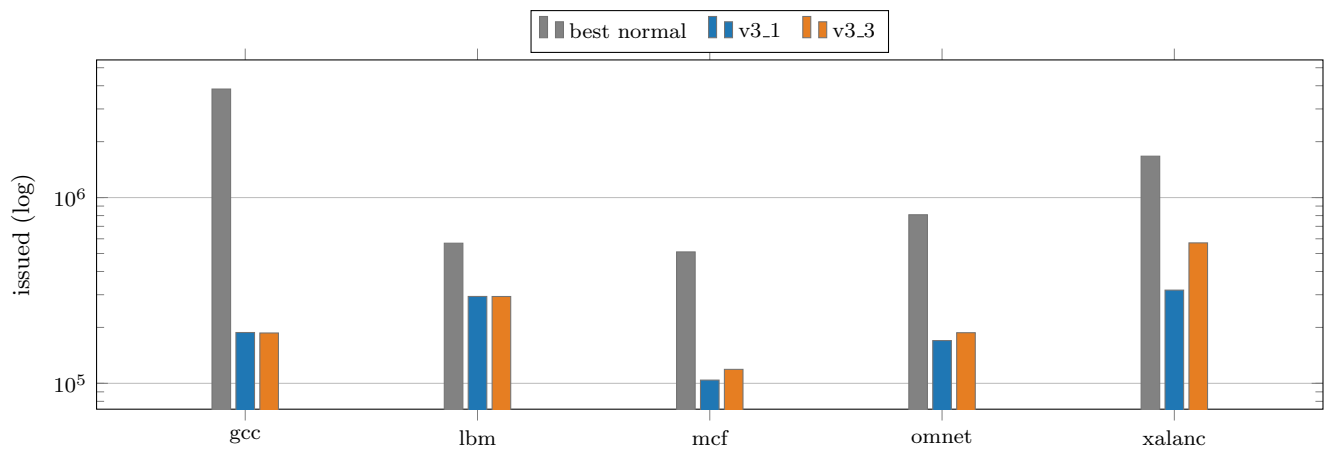
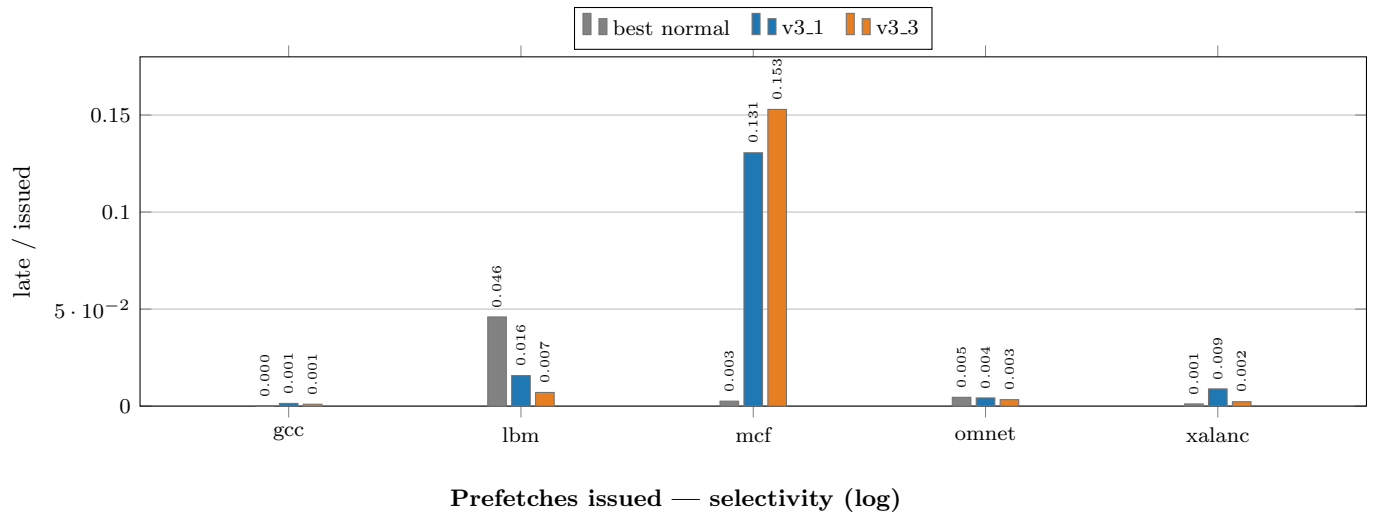
Event timeliness (over covered events)



Useless rate (useless / issued)



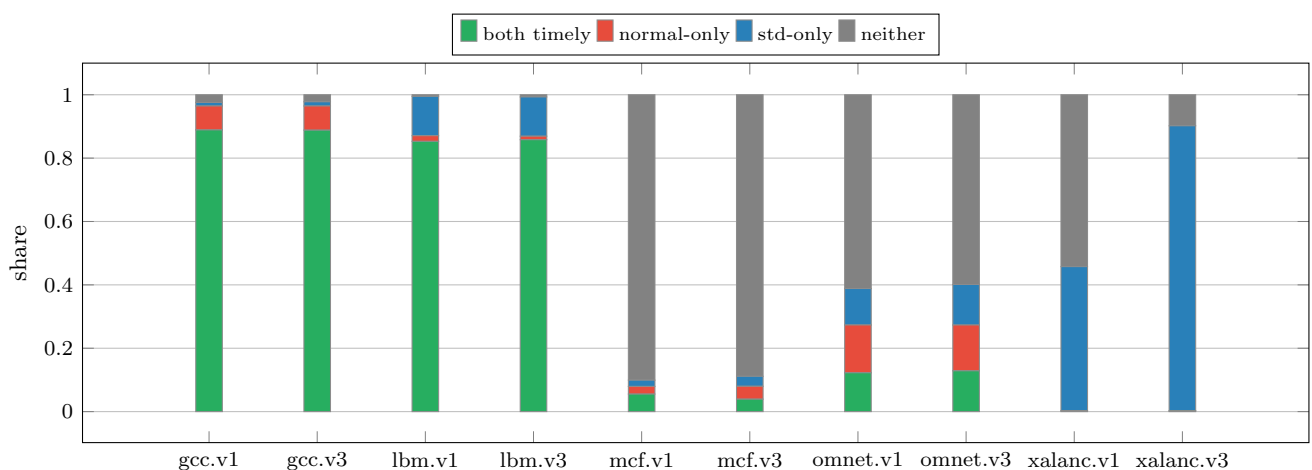
Late rate (late / issued)



6.3 Demand-event outcome composition (vs. best normal)

For each *trace.variant*, the share of demand events that are: both timely, normal-only timely, standalone-only timely, or neither. Shows where the NN wins (large std-only) or loses (large normal-only / neither).

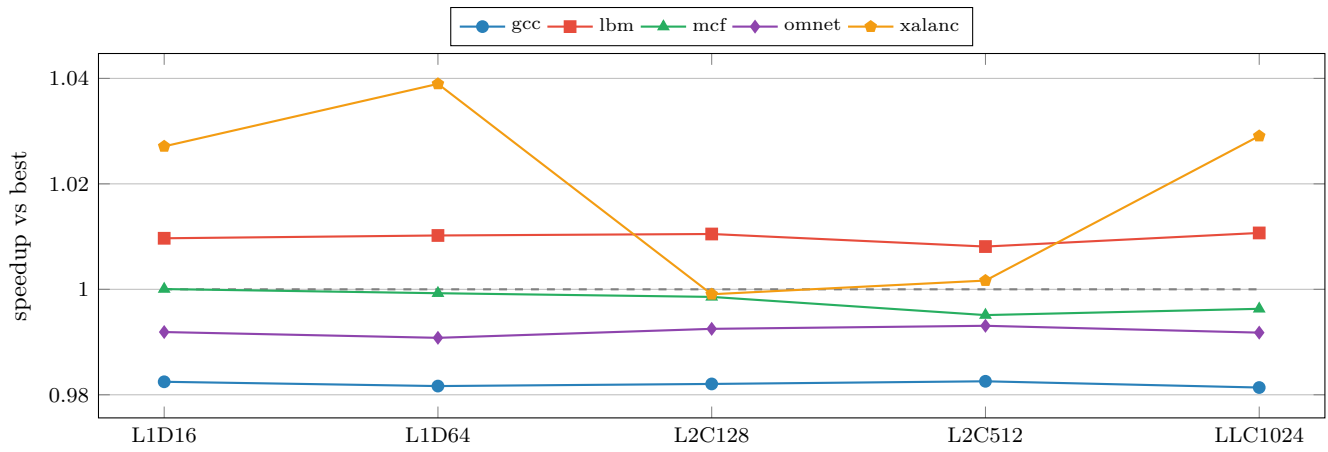
Demand-event outcome share (per trace.variant)



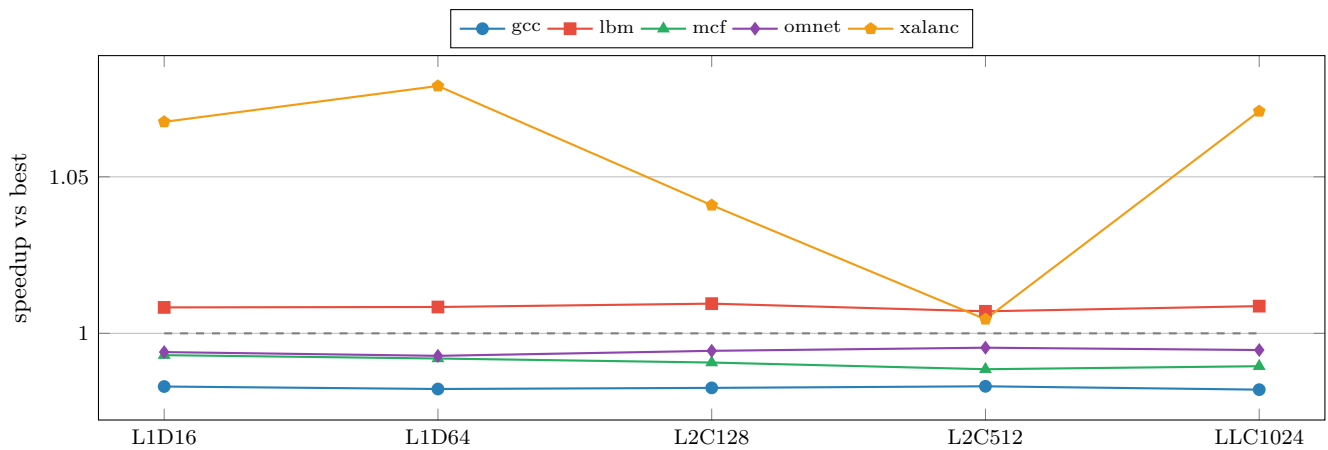
6.4 Cache-capacity sweep trends (frozen lists)

Dashed gray line = parity (speedup 1.0). Frozen lists use the baseline-capacity oracle (not capacity-trained), so these isolate capacity sensitivity.

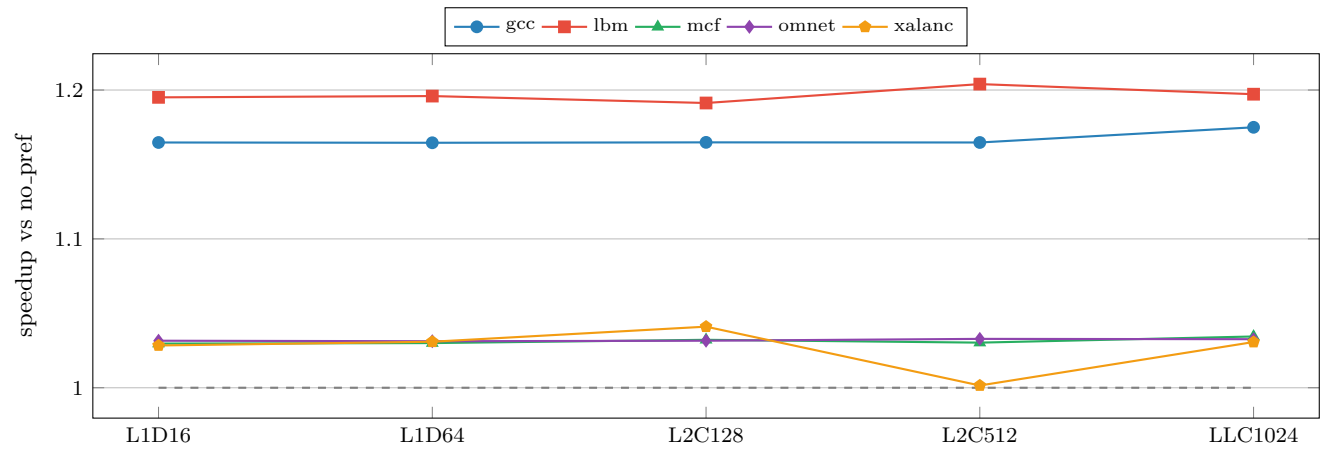
Speedup vs. best normal across capacity — v3_1



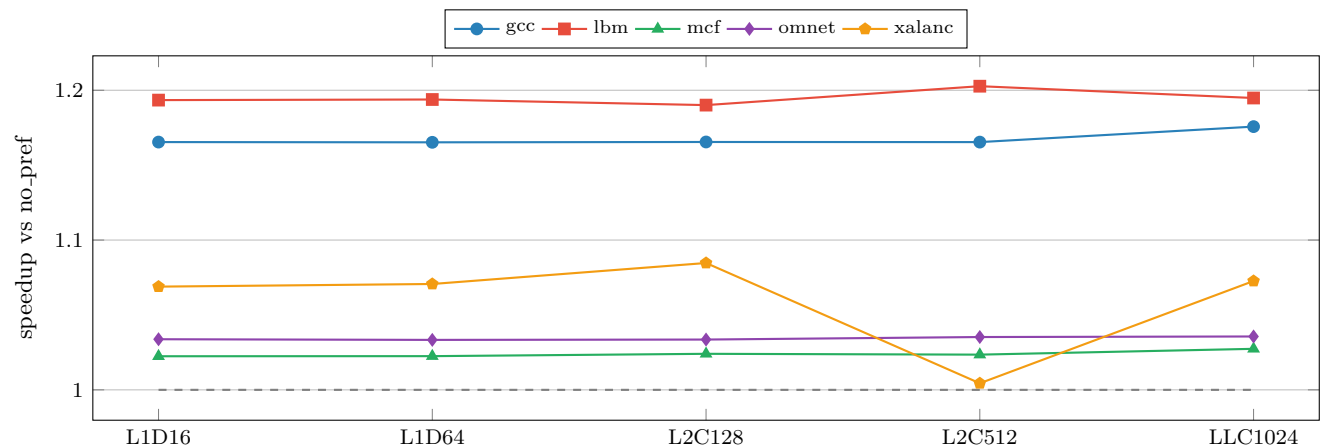
Speedup vs. best normal across capacity — v3.3



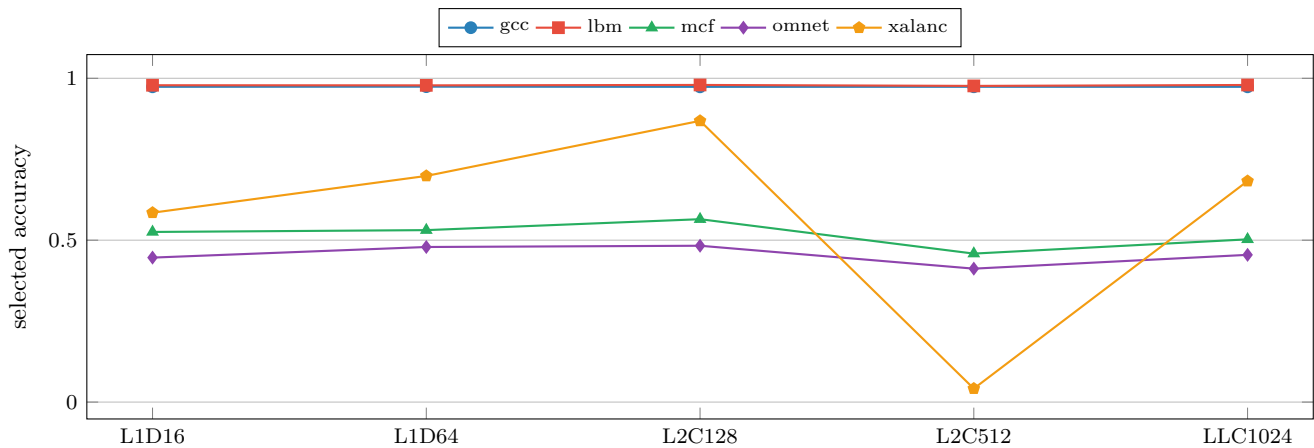
Speedup vs. no_pref across capacity — v3.1



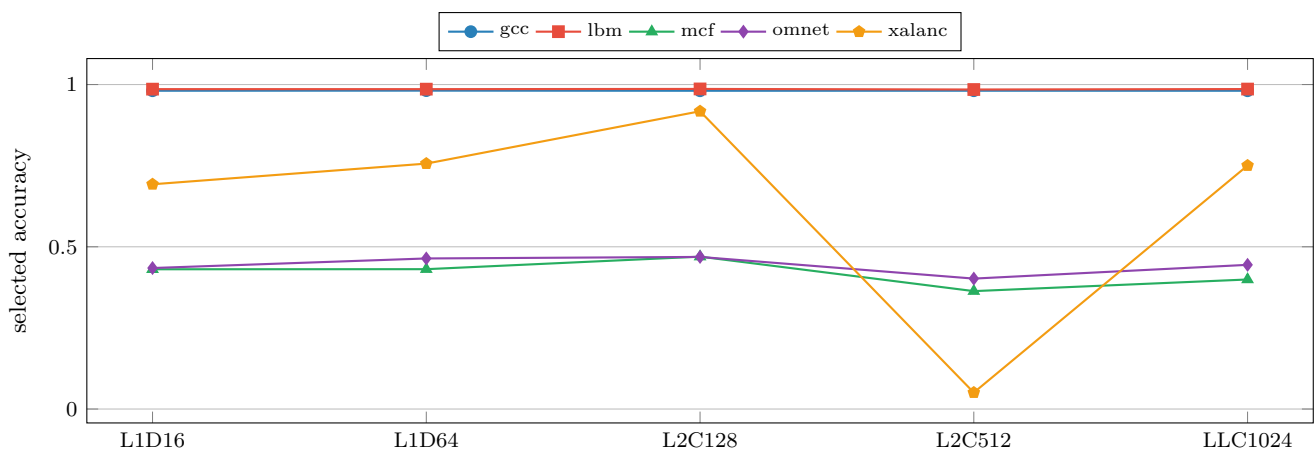
Speedup vs. no_pref across capacity — v3.3



Selected accuracy across capacity — v3.1



Selected accuracy across capacity — v3.3



7 Consolidated conclusions (every stated / directly-measured finding)

- 602.gcc:** best normal = sandbox (IPC 0.43628). Both NN variants are slightly below (v3.1 -0.00765 , v3.3 -0.00741) but reach very high selected accuracy (~ 0.97 – 0.98) and event timeliness (>0.998). NN covers $\sim 90\%$ of unique demand events; normal-only-timely events ($\sim 15k$) are dominated by “no earlier selected standalone export.”
- 619.lbm:** both NN variants *beat* the best normal (sms): v3.1 $+0.00387$, v3.3 $+0.00329$ IPC; high coverage (~ 0.97 – 0.98) and timeliness (~ 0.98 – 0.99). Standalone-only-timely events ($\sim 36k$) far exceed normal-only (~ 3 – $5k$): the NN is timely on events normal prefetchers miss.
- 605.mcf:** hardest workload. Best normal = ampm (IPC 0.18874); both NN variants essentially tie or lose slightly (v3.1 -0.00012 , v3.3 -0.00145). Very low unique-event coverage (~ 0.07) and lower timeliness (~ 0.74 – 0.80); huge “neither timely” bucket (~ 640 – $654k$). Tagged low/very-low unique-event coverage. 102,356 unique load pages (by far the most scattered footprint).
- 620.omnetpp:** best normal = sms (IPC 0.24695); NN slightly below (v3.1 -0.00192 , v3.3 -0.00136). Highest unique-PC count in ROI (6591); high counter timeliness (~ 0.99) but modest coverage (~ 0.24 – 0.26).
- 623.xalancbmk:** largest NN win. Best normal = spp (IPC 0.35391); v3.1 $+0.01016$, v3.3 $+0.02502$. v3.3 reaches coverage 0.899 and timeliness 0.997 with 570k issued. Standalone-only-timely events are enormous (v3.1 203,732; v3.3 404,054) vs. tiny normal-only (~ 120 – 252): NN dominates the demand-event timeliness here.
- Capacity sweep (frozen, not capacity-trained):** across L1D 16/64 KiB, L2C 128/512 KiB and LLC 1024 KiB, the v3.1/v3.3 frozen replays keep `speedup_vs_no_pref` >1 on every trace and remain close to or above the best normal on gcc, lbm, omnetpp, xalancbmk; mcf stays the weakest. Notable: at L2C-512 KiB and LLC-1024, xalancbmk’s frozen NN `speedup_vs_best_normal` collapses toward ~ 1.0 (e.g. v3.1 sel.acc drops to ~ 0.04 at L2C-512), illustrating frozen-list capacity sensitivity. xalancbmk v3.3 recovers strongly at L2C-128 (`spd/no_pref` 1.085) and L2C-256 (1.085).

7. **Methodological limitations (must carry forward):** (a) no assembly mapping — trace PCs cannot be tied to disassembly; (b) frozen standalone exports record *selections only*, so “no earlier selected standalone export” is not proof the candidate bank lacked the line; (c) a full decision ledger (every candidate, score/probabilities, candidate-source, reject_reason) is required next to separate candidate absence from policy rejection; (d) the evidence report asserts measured association only, not causation.

All numbers transcribed directly from the bundle’s CSV/JSON/MD artifacts; no values were imputed. Where a file was absent (llc_4096 summary), that absence is noted rather than estimated.

Appendix A — Full per-benchmark detail tables

Complete (untruncated) top-N listings for every benchmark.

pc	dyn instr	instr share	load	store	branch	taken
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602.gcc.s-734B

Top PCs (all 200 rows).

pc	dyn instr	instr share	load	store	branch	taken
0x40f700	923,760	0.036950	0	0	0	0
0x40f703	923,760	0.036950	0	0	923,760	307,920
0x40f705	615,840	0.024634	0	0	0	0
0x40f708	615,840	0.024634	0	0	0	0
0x40f70a	615,840	0.024634	615,840	0	0	0
0x40f70f	615,840	0.024634	615,840	0	0	0
0x40f714	615,840	0.024634	615,840	0	0	0
0x40f719	615,840	0.024634	0	0	615,840	615,839
0x40f511	308,754	0.012350	0	0	0	0
0x40f514	308,754	0.012350	0	0	308,754	308,754
0x40f52a	308,754	0.012350	0	0	0	0
0x40f52d	308,754	0.012350	0	0	308,754	1
0x40f762	308,754	0.012350	0	0	0	0
0x40f766	308,754	0.012350	308,754	0	0	0
0x40f767	308,754	0.012350	308,754	0	0	0
0x40f769	308,754	0.012350	308,754	0	0	0
0x40f76b	308,754	0.012350	308,754	0	0	0
0x40f76d	308,754	0.012350	308,754	0	0	0
0x40f76f	308,754	0.012350	308,754	0	0	0
0x40f770	308,754	0.012350	308,754	0	0	0
0x40f7eb	308,754	0.012350	0	0	0	0
0x40f7ee	308,754	0.012350	0	0	308,754	308,754
0x40f886	308,754	0.012350	308,754	0	0	0
0x40f88a	308,754	0.012350	308,754	0	0	0
0x40f88e	308,754	0.012350	0	0	0	0
0x40f891	308,754	0.012350	0	0	0	0
0x40f894	308,754	0.012350	0	0	308,754	308,398
0x40f7b0	308,754	0.012350	308,754	0	0	0
0x40f7b4	308,754	0.012350	0	0	0	0
0x40f7b7	308,754	0.012350	0	0	308,754	1
0x40f7d6	308,754	0.012350	0	0	0	0
0x40f7da	308,754	0.012350	0	0	0	0
0x40f7dd	308,754	0.012350	0	0	0	0
0x40f7e0	308,754	0.012350	0	0	0	0
0x40f7e3	308,754	0.012350	0	0	0	0
0x40f7e6	308,754	0.012350	0	308,754	0	0
0x40f4f0	308,754	0.012350	0	308,754	0	0
0x40f4f1	308,754	0.012350	0	308,754	0	0
0x40f4f3	308,754	0.012350	0	308,754	0	0
0x40f52f	308,753	0.012350	0	0	0	0
0x40f532	308,753	0.012350	0	0	308,753	0
0x40f534	308,753	0.012350	308,753	0	0	0
0x40f538	308,753	0.012350	308,753	0	0	0
0x40f53d	308,753	0.012350	0	0	308,753	0
0x40f53f	308,753	0.012350	0	0	0	0
0x40f542	308,753	0.012350	0	0	308,753	0
0x40f548	308,753	0.012350	0	0	0	0
0x40f54b	308,753	0.012350	0	0	308,753	833
0x40f7b9	308,753	0.012350	308,753	0	0	0
0x40f7be	308,753	0.012350	0	0	0	0
0x40f7c0	308,753	0.012350	308,753	0	0	0
0x40f7c3	308,753	0.012350	0	0	308,753	308,753
0x40f4f5	308,753	0.012350	0	308,753	0	0
0x40f4f7	308,753	0.012350	0	308,753	0	0
0x40f4f9	308,753	0.012350	0	308,753	0	0
0x40f4fa	308,753	0.012350	0	308,753	0	0
0x40f4fb	308,753	0.012350	0	0	0	0
0x40f4fe	308,753	0.012350	0	0	0	0
0x40f501	308,753	0.012350	0	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x40f504	308,753	0.012350	0	308,753	0	0
0x40f508	308,753	0.012350	0	0	0	0
0x40f50b	308,753	0.012350	0	0	0	0
0x40f50e	308,753	0.012350	0	0	0	0
0x40f6ef	307,921	0.012317	0	0	0	0
0x40f6f1	307,921	0.012317	0	0	0	0
0x40f551	307,920	0.012317	307,920	0	0	0
0x40f555	307,920	0.012317	0	0	307,920	0
0x40f55b	307,920	0.012317	0	0	0	0
0x40f560	307,920	0.012317	0	0	0	0
0x40f562	307,920	0.012317	0	0	307,920	307,920
0xa521f0	14,028	0.000561	0	0	0	0
0xa521f3	14,028	0.000561	0	0	0	0
0xa521f6	14,028	0.000561	0	0	0	0
0xa521f8	14,028	0.000561	0	0	0	0
0xa521fd	14,028	0.000561	0	0	0	0
0xa52200	14,028	0.000561	0	0	14,028	13,249
0xa4c410	2,742	0.000110	0	2,742	0	0
0xa4c412	2,742	0.000110	0	2,742	0	0
0xa4c413	2,742	0.000110	0	2,742	0	0
0xa4c414	2,742	0.000110	0	0	0	0
0xa4c416	2,742	0.000110	2,742	0	0	0
0xa4c41d	2,742	0.000110	2,742	0	0	0
0xa4c41f	2,742	0.000110	0	0	2,742	2,742
0xa4c43c	2,742	0.000110	0	0	0	0
0xa4c43f	2,742	0.000110	2,742	0	0	0
0xa4c443	2,742	0.000110	2,742	0	0	0
0xa4c447	2,742	0.000110	0	0	0	0
0xa4c449	2,742	0.000110	0	0	2,742	1,370
0xa4c463	2,742	0.000110	0	0	0	0
0xa4c467	2,742	0.000110	2,742	0	0	0
0xa4c468	2,742	0.000110	2,742	0	0	0
0xa4c46a	2,742	0.000110	2,742	0	0	0
0xa5217d	2,499	0.000100	0	0	0	0
0xa52180	2,499	0.000100	0	0	2,499	2,082
0xa52170	2,498	0.000100	2,498	0	0	0
0xa52175	2,498	0.000100	0	0	0	0
0xa52178	2,498	0.000100	0	0	0	0
0xa5217a	2,498	0.000100	0	0	0	0
0xa521dd	2,498	0.000100	0	0	0	0
0xa521df	2,498	0.000100	0	2,498	0	0
0xa521e4	2,498	0.000100	0	0	2,498	1,719
0xa52213	2,498	0.000100	2,498	0	0	0
0xa5221a	2,498	0.000100	0	0	0	0
0xa5221c	2,498	0.000100	2,498	0	0	0
0xa52221	2,498	0.000100	2,498	0	0	0
0xa52225	2,498	0.000100	2,498	0	0	0
0xa5222a	2,498	0.000100	0	0	0	0
0xa5222e	2,498	0.000100	2,498	0	0	0
0xa52232	2,498	0.000100	0	0	0	0
0xa52235	2,498	0.000100	0	0	0	0
0xa52238	2,498	0.000100	0	0	2,498	0
0xa5223a	2,498	0.000100	0	0	0	0
0xa5223c	2,498	0.000100	0	0	2,498	1,356
0xa52210	1,719	0.000069	0	0	0	0
0xa4c44b	1,372	0.000055	0	0	0	0
0xa4c44d	1,372	0.000055	0	0	1,372	1,372
0xa52368	1,370	0.000055	1,370	0	0	0
0xa5236c	1,370	0.000055	0	0	1,370	1,362
0xa522a4	1,370	0.000055	1,370	0	0	0
0xa522a8	1,370	0.000055	0	1,370	0	0
0xa4c44f	1,370	0.000055	0	1,370	0	0
0xa4c454	1,370	0.000055	1,370	0	0	0
0xa4c45b	1,370	0.000055	1,370	0	0	0
0xa4c45f	1,370	0.000055	0	1,370	0	0

pc	dyn instr	instr share	load	store	branch	taken
0xa522ad	1,370	0.000055	1,370	0	0	0
0xa522b4	1,370	0.000055	0	0	0	0
0xa522b6	1,370	0.000055	1,370	0	0	0
0xa522bb	1,370	0.000055	1,370	0	0	0
0xa522bf	1,370	0.000055	0	0	1,370	0
0xa522c5	1,370	0.000055	1,370	0	0	0
0xa522ca	1,370	0.000055	0	0	1,370	1,370
0xa522e0	1,370	0.000055	1,370	0	0	0
0xa522e5	1,370	0.000055	0	0	0	0
0xa522e7	1,370	0.000055	0	0	1,370	0
0xa522e9	1,370	0.000055	1,370	0	0	0
0xa522ee	1,370	0.000055	1,370	0	0	0
0xa522f2	1,370	0.000055	1,370	0	0	0
0xa522f6	1,370	0.000055	0	0	0	0
0xa522f9	1,370	0.000055	0	0	1,370	1,370
0xa5231a	1,370	0.000055	1,370	0	0	0
0xa5231f	1,370	0.000055	0	1,370	0	0
0x40d3f0	1,370	0.000055	0	1,370	0	0
0x40d3f1	1,370	0.000055	0	1,370	0	0
0x40d3f3	1,370	0.000055	0	1,370	0	0
0x40d3f5	1,370	0.000055	0	1,370	0	0
0x40d3f6	1,370	0.000055	0	1,370	0	0
0x40d3f7	1,370	0.000055	0	0	0	0
0x40d3fa	1,370	0.000055	0	0	0	0
0x40d3fc	1,370	0.000055	0	0	0	0
0x40d3ff	1,370	0.000055	1,370	0	0	0
0x40d403	1,370	0.000055	0	0	0	0
0x40d406	1,370	0.000055	0	0	1,370	0
0x40d40c	1,370	0.000055	1,370	0	0	0
0x40d410	1,370	0.000055	0	0	0	0
0x40d412	1,370	0.000055	0	0	1,370	1,180
0x40d482	1,370	0.000055	0	0	0	0
0x40d484	1,370	0.000055	0	0	0	0
0x40d487	1,370	0.000055	0	0	0	0
0x40d48a	1,370	0.000055	0	0	0	0
0x40d48f	1,370	0.000055	0	0	0	0
0x40d491	1,370	0.000055	0	0	0	0
0x40d494	1,370	0.000055	1,370	0	0	0
0x40d499	1,370	0.000055	0	0	0	0
0x40d49c	1,370	0.000055	0	0	0	0
0x40d49f	1,370	0.000055	0	0	0	0
0x40d4a2	1,370	0.000055	0	0	0	0
0x40d4a5	1,370	0.000055	0	0	1,370	1,014
0x40d56b	1,370	0.000055	0	0	0	0
0x40d56f	1,370	0.000055	1,370	0	0	0
0x40d570	1,370	0.000055	1,370	0	0	0
0x40d572	1,370	0.000055	1,370	0	0	0
0x40d574	1,370	0.000055	1,370	0	0	0
0x40d575	1,370	0.000055	1,370	0	0	0
0xa52324	1,370	0.000055	0	0	0	0
0xa52326	1,370	0.000055	0	0	1,370	1,014
0xa52270	1,356	0.000054	0	0	0	0
0xa52272	1,356	0.000054	0	0	0	0
0xa52276	1,356	0.000054	0	0	0	0
0xa52279	1,356	0.000054	0	0	1,356	1,356
0xa5223e	1,142	0.000046	1,142	0	0	0
0xa52242	1,142	0.000046	0	0	1,142	1,128
0xa521b0	992	0.000040	0	0	0	0
0xa521b4	992	0.000040	0	0	992	661
0x40f75c	834	0.000033	0	834	0	0
0x40f760	834	0.000033	0	0	0	0
0x40f61a	833	0.000033	0	0	0	0
0x40f61d	833	0.000033	0	0	833	0
0x40f61f	833	0.000033	0	833	0	0
0x40f623	833	0.000033	0	0	833	833

pc	dyn instr	instr share	load	store	branch	taken
0x40f746	833	0.000033	833	0	0	0
0x40f74b	833	0.000033	833	0	0	0
0x40f74f	833	0.000033	0	833	0	0
0x40f753	833	0.000033	833	0	0	0
0x40f758	833	0.000033	833	0	0	0
0xa521e6	779	0.000031	0	0	0	0
0xa52202	779	0.000031	0	0	779	779
0xa52193	662	0.000026	0	0	0	0
0xa52196	662	0.000026	0	0	662	662
0xa521d0	661	0.000026	0	0	0	0
0xa521d3	661	0.000026	661	0	0	0

Top load line deltas (all 100) and top load pages (all 100).

line_delta	count	share	page	load operands	share
0	2,598,631	0.396437	34,358,534,190	2,189,035	0.333951
1	494,542	0.075445	652,008	174,383	0.026603
5,083,415	6,603	0.001007	652,009	174,001	0.026545
5,631,043	6,380	0.000973	652,005	173,660	0.026493
5,627,328	6,092	0.000929	652,007	173,555	0.026477
5,684,124	5,372	0.000820	652,003	173,470	0.026464
5,082,884	5,337	0.000814	652,006	173,470	0.026464
5,631,574	5,314	0.000811	652,004	173,090	0.026406
5,630,512	5,126	0.000782	652,002	168,721	0.025739
5,080,230	4,734	0.000722	652,010	159,160	0.024281
5,684,655	4,699	0.000717	652,011	151,254	0.023075
-5,631,043	4,449	0.000679	652,012	151,088	0.023049
5,629,981	4,296	0.000655	652,013	150,874	0.023017
6,018,321	4,247	0.000648	652,015	150,505	0.022960
-5,627,328	4,193	0.000640	652,016	150,341	0.022935
-5,083,415	4,169	0.000636	652,017	150,176	0.022910
13,644,122	4,112	0.000627	652,014	150,012	0.022885
6,018,320	4,069	0.000621	652,018	92,543	0.014118
-5,684,124	4,017	0.000613	1,789,864	83,137	0.012683
-5,082,884	3,888	0.000593	4,309	8,701	0.001327
-5,630,512	3,843	0.000586	217,499	8,239	0.001257
-5,631,574	3,825	0.000584	1,833,957	2,485	0.000379
5,629,982	3,719	0.000567	260,274	1,916	0.000292
5,070,675	3,669	0.000560	253,898	1,426	0.000218
-5,629,981	3,651	0.000557	258,778	1,373	0.000209
3,383,150	3,637	0.000555	563,185	1,016	0.000155
-18,041,570	3,623	0.000553	563,190	1,016	0.000155
-18,141,850	3,577	0.000546	563,195	1,016	0.000155
12,844,165	3,514	0.000536	563,200	1,016	0.000155
21,610,738	3,470	0.000529	563,995	1,016	0.000155
13,848,230	3,454	0.000527	564,010	1,016	0.000155
2,206,700	3,439	0.000525	564,015	1,016	0.000155
-15,371,980	3,439	0.000525	564,020	1,016	0.000155
6,993,130	3,424	0.000522	564,025	1,016	0.000155
2,843,515	3,401	0.000519	564,030	1,016	0.000155
12,948,198	3,335	0.000509	564,035	1,016	0.000155
10,957,478	3,334	0.000509	564,040	1,016	0.000155
-16,355,240	3,301	0.000504	564,045	1,016	0.000155
9,098,900	3,301	0.000504	564,060	1,016	0.000155
-15,376,757	3,300	0.000503	564,065	1,016	0.000155
12,945,013	3,299	0.000503	564,080	1,016	0.000155
-6,018,320	3,266	0.000498	564,085	1,016	0.000155
-15,365,610	3,256	0.000497	564,090	1,016	0.000155
9,189,070	3,243	0.000495	572,575	1,016	0.000155
8,650,998	3,239	0.000494	572,580	1,016	0.000155
2,205,108	3,236	0.000494	572,585	1,016	0.000155
-13,192,557	3,236	0.000494	572,590	1,016	0.000155
-13,422,320	3,210	0.000490	572,595	1,016	0.000155
5,721,805	3,209	0.000490	572,625	1,016	0.000155
-15,386,312	3,185	0.000486	572,630	1,016	0.000155

line_delta	count	share	page	load operands	share
9,974,828	3,173	0.000484	563,186	1,014	0.000155
2,367,470	3,164	0.000483	563,187	1,014	0.000155
-17,308,840	3,164	0.000483	563,188	1,014	0.000155
-15,869,205	3,163	0.000483	563,191	1,014	0.000155
3,845,337	3,135	0.000478	563,192	1,014	0.000155
8,174,472	3,135	0.000478	563,193	1,014	0.000155
23,779,438	3,128	0.000477	563,196	1,014	0.000155
4,492,207	3,123	0.000476	563,197	1,014	0.000155
5,748,870	3,121	0.000476	563,198	1,014	0.000155
4,469,387	3,120	0.000476	563,201	1,014	0.000155
-16,308,013	3,120	0.000476	563,202	1,014	0.000155
9,105,270	3,117	0.000476	563,203	1,014	0.000155
24,081,862	3,116	0.000475	563,993	1,014	0.000155
2,182,298	3,109	0.000474	563,996	1,014	0.000155
-18,797,717	3,109	0.000474	563,997	1,014	0.000155
-2,198,945,912,354	3,097	0.000472	563,998	1,014	0.000155
5,720,743	3,094	0.000472	564,011	1,014	0.000155
4,491,145	3,092	0.000472	564,012	1,014	0.000155
12,871,767	3,091	0.000472	564,013	1,014	0.000155
2,185,482	3,088	0.000471	564,016	1,014	0.000155
15,799,997	3,077	0.000469	564,017	1,014	0.000155
-16,262,938	3,075	0.000469	564,018	1,014	0.000155
6,014,605	3,071	0.000469	564,026	1,014	0.000155
5,632,105	3,071	0.000469	564,027	1,014	0.000155
20,379,743	3,055	0.000466	564,028	1,014	0.000155
20,374,435	3,049	0.000465	564,031	1,014	0.000155
22,056,480	3,035	0.000463	564,032	1,014	0.000155
19,641,745	3,035	0.000463	564,033	1,014	0.000155
18,909,281	3,035	0.000463	564,036	1,014	0.000155
17,351,908	3,015	0.000460	564,037	1,014	0.000155
3,666,006	3,009	0.000459	564,038	1,014	0.000155
20,241,265	3,007	0.000459	564,041	1,014	0.000155
5,082,353	2,994	0.000457	564,042	1,014	0.000155
4,535,200	2,994	0.000457	564,043	1,014	0.000155
-16,314,378	2,994	0.000457	564,046	1,014	0.000155
22,064,435	2,991	0.000456	564,047	1,014	0.000155
20,376,558	2,988	0.000456	564,048	1,014	0.000155
12,807,020	2,988	0.000456	564,061	1,014	0.000155
-5,080,230	2,986	0.000456	564,062	1,014	0.000155
-6,018,321	2,983	0.000455	564,063	1,014	0.000155
-5,684,655	2,981	0.000455	564,076	1,014	0.000155
-14,750,617	2,981	0.000455	564,077	1,014	0.000155
3,669,191	2,973	0.000454	564,078	1,014	0.000155
-13,810,294	2,973	0.000454	564,081	1,014	0.000155
-14,438,104	2,973	0.000454	564,082	1,014	0.000155
2,184,420	2,964	0.000452	564,083	1,014	0.000155
5,704,825	2,954	0.000451	564,086	1,014	0.000155
-250,968	2,947	0.000450	564,087	1,014	0.000155
2,901,370	2,934	0.000448	564,088	1,014	0.000155
3,669,722	2,930	0.000447	572,576	1,014	0.000155

Register usage (all 62 rows).

direction	register_id	dynamic uses	share within direction
source	26	5,621,295	0.153906
source	6	5,305,937	0.145272
source	25	4,681,019	0.128163
source	15	3,090,929	0.084627
source	4	2,470,151	0.067631
source	8	2,169,291	0.059393
source	7	2,165,769	0.059297
source	17	1,899,894	0.052018
source	9	1,857,065	0.050845
source	16	1,247,581	0.034158
source	10	991,340	0.027142

direction	register_id	dynamic uses	share within direction
source	55	925,211	0.025332
source	82	620,864	0.016999
source	5	316,678	0.008670
source	56	314,125	0.008600
source	49	313,791	0.008591
source	3	313,580	0.008586
source	27	312,979	0.008569
source	18	310,480	0.008501
source	29	309,220	0.008466
source	80	308,754	0.008453
source	11	308,754	0.008453
source	50	308,753	0.008453
source	64	308,753	0.008453
source	53	30,389	0.000832
source	30	8,438	0.000231
source	45	8,224	0.000225
source	47	4,113	0.000113
source	54	357	0.000010
source	33	356	0.000010
source	61	2	0.000000
destination	25	6,580,552	0.228758
destination	26	5,925,823	0.205998
destination	6	4,984,138	0.173262
destination	4	1,544,902	0.053705
destination	15	930,829	0.032358
destination	7	930,734	0.032355
destination	9	930,444	0.032345
destination	16	929,619	0.032316
destination	17	655,975	0.022803
destination	10	655,760	0.022796
destination	49	622,987	0.021657
destination	55	620,615	0.021574
destination	54	615,842	0.021408
destination	8	316,386	0.010998
destination	56	312,045	0.010848
destination	5	311,854	0.010841
destination	3	310,484	0.010793
destination	18	310,480	0.010793
destination	82	310,102	0.010780
destination	11	308,755	0.010733
destination	80	308,754	0.010733
destination	64	308,754	0.010733
destination	53	21,365	0.000743
destination	45	8,224	0.000286
destination	30	5,709	0.000198
destination	79	2,742	0.000095
destination	27	2,205	0.000077
destination	33	356	0.000012
destination	47	2	0.000000
destination	61	2	0.000000
destination	36	1	0.000000

619.lbm_s-4268B

Top PCs (all 200 rows).

pc	dyn instr	instr share	load	store	branch	taken
0x4027cb	117,755	0.004710	0	0	0	0
0x4027cf	117,755	0.004710	0	0	0	0
0x4027d5	117,755	0.004710	0	0	117,755	117,755
0x401cb0	117,755	0.004710	117,755	0	0	0
0x401cb7	117,755	0.004710	0	0	0	0
0x401cba	117,755	0.004710	117,755	0	0	0
0x401cbf	117,755	0.004710	0	0	117,755	71,616
0x401df0	71,616	0.002865	0	71,616	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x401df5	71,616	0.002865	71,616	0	0	0
0x401dfa	71,616	0.002865	0	71,616	0	0
0x401e02	71,616	0.002865	71,616	0	0	0
0x401e07	71,616	0.002865	0	71,616	0	0
0x401e0f	71,616	0.002865	71,616	0	0	0
0x401e14	71,616	0.002865	0	71,616	0	0
0x401e19	71,616	0.002865	71,616	0	0	0
0x401e1e	71,616	0.002865	0	71,616	0	0
0x401e26	71,616	0.002865	71,616	0	0	0
0x401e2b	71,616	0.002865	0	71,616	0	0
0x401e33	71,616	0.002865	71,616	0	0	0
0x401e38	71,616	0.002865	0	71,616	0	0
0x401e40	71,616	0.002865	71,616	0	0	0
0x401e45	71,616	0.002865	0	71,616	0	0
0x401e4d	71,616	0.002865	71,616	0	0	0
0x401e52	71,616	0.002865	0	71,616	0	0
0x401e5a	71,616	0.002865	71,616	0	0	0
0x401e5f	71,616	0.002865	0	71,616	0	0
0x401e67	71,616	0.002865	71,616	0	0	0
0x401e6c	71,616	0.002865	0	71,616	0	0
0x401e74	71,616	0.002865	71,616	0	0	0
0x401e79	71,616	0.002865	0	71,616	0	0
0x401e81	71,616	0.002865	71,616	0	0	0
0x401e86	71,616	0.002865	0	71,616	0	0
0x401e8e	71,616	0.002865	71,616	0	0	0
0x401e93	71,616	0.002865	0	71,616	0	0
0x401e9b	71,616	0.002865	71,616	0	0	0
0x401ea0	71,616	0.002865	0	71,616	0	0
0x401ea8	71,616	0.002865	71,616	0	0	0
0x401ead	71,616	0.002865	0	71,616	0	0
0x401eb5	71,616	0.002865	71,616	0	0	0
0x401ebd	71,616	0.002865	0	71,616	0	0
0x401ec5	71,616	0.002865	71,616	0	0	0
0x401ecd	71,616	0.002865	0	71,616	0	0
0x401ed5	71,616	0.002865	71,616	0	0	0
0x401edd	71,616	0.002865	0	71,616	0	0
0x401ee5	71,616	0.002865	0	0	71,616	71,616
0x402773	46,139	0.001846	0	0	0	0
0x402777	46,139	0.001846	0	46,139	0	0
0x402780	46,139	0.001846	46,139	0	0	0
0x402789	46,139	0.001846	0	0	0	0
0x40278d	46,139	0.001846	0	0	0	0
0x402792	46,139	0.001846	0	0	0	0
0x402797	46,139	0.001846	0	0	0	0
0x40279c	46,139	0.001846	0	46,139	0	0
0x4027a5	46,139	0.001846	46,139	0	0	0
0x4027ae	46,139	0.001846	0	0	0	0
0x4027b2	46,139	0.001846	0	0	0	0
0x4027b6	46,139	0.001846	0	0	0	0
0x4027ba	46,139	0.001846	0	0	0	0
0x4027be	46,139	0.001846	0	0	0	0
0x4027c2	46,139	0.001846	0	46,139	0	0
0x401cc5	46,139	0.001846	46,139	0	0	0
0x401ccb	46,139	0.001846	0	0	0	0
0x401cd0	46,139	0.001846	0	46,139	0	0
0x401cd6	46,139	0.001846	0	0	0	0
0x401cdb	46,139	0.001846	46,139	0	0	0
0x401ce1	46,139	0.001846	0	0	0	0
0x401ce6	46,139	0.001846	46,139	0	0	0
0x401cec	46,139	0.001846	0	46,139	0	0
0x401cf2	46,139	0.001846	0	0	0	0
0x401cf7	46,139	0.001846	46,139	0	0	0
0x401cfd	46,139	0.001846	0	46,139	0	0
0x401d03	46,139	0.001846	0	0	0	0
0x401d08	46,139	0.001846	46,139	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x401d0f	46,139	0.001846	0	0	0	0
0x401d14	46,139	0.001846	46,139	0	0	0
0x401d1a	46,139	0.001846	0	0	0	0
0x401d1f	46,139	0.001846	46,139	0	0	0
0x401d26	46,139	0.001846	0	0	0	0
0x401d2b	46,139	0.001846	46,139	0	0	0
0x401d32	46,139	0.001846	0	0	0	0
0x401d37	46,139	0.001846	46,139	0	0	0
0x401d3e	46,139	0.001846	0	0	0	0
0x401d43	46,139	0.001846	46,139	0	0	0
0x401d4a	46,139	0.001846	0	0	0	0
0x401d4f	46,139	0.001846	46,139	0	0	0
0x401d55	46,139	0.001846	0	46,139	0	0
0x401d5b	46,139	0.001846	0	0	0	0
0x401d60	46,139	0.001846	46,139	0	0	0
0x401d66	46,139	0.001846	0	46,139	0	0
0x401d6c	46,139	0.001846	0	0	0	0
0x401d71	46,139	0.001846	46,139	0	0	0
0x401d77	46,139	0.001846	0	0	0	0
0x401d7c	46,139	0.001846	46,139	0	0	0
0x401d82	46,139	0.001846	0	0	0	0
0x401d87	46,139	0.001846	46,139	0	0	0
0x401d8d	46,139	0.001846	0	46,139	0	0
0x401d93	46,139	0.001846	0	0	0	0
0x401d98	46,139	0.001846	46,139	0	0	0
0x401da1	46,139	0.001846	0	46,139	0	0
0x401da7	46,139	0.001846	0	0	0	0
0x401dac	46,139	0.001846	46,139	0	0	0
0x401db5	46,139	0.001846	0	0	0	0
0x401dba	46,139	0.001846	46,139	0	0	0
0x401dc4	46,139	0.001846	0	0	0	0
0x401dc9	46,139	0.001846	0	0	0	0
0x401dce	46,139	0.001846	0	0	0	0
0x401dd2	46,139	0.001846	0	0	0	0
0x401dd5	46,139	0.001846	0	46,139	0	0
0x401ddb	46,139	0.001846	0	0	46,139	46,139
0x401ef0	46,139	0.001846	0	0	0	0
0x401ef4	46,139	0.001846	46,139	0	0	0
0x401efa	46,139	0.001846	46,139	0	0	0
0x401f00	46,139	0.001846	0	0	0	0
0x401f04	46,139	0.001846	0	0	0	0
0x401f08	46,139	0.001846	46,139	0	0	0
0x401f0e	46,139	0.001846	46,139	0	0	0
0x401f14	46,139	0.001846	0	0	0	0
0x401f18	46,139	0.001846	0	0	0	0
0x401f1d	46,139	0.001846	0	0	0	0
0x401f22	46,139	0.001846	46,139	0	0	0
0x401f29	46,139	0.001846	0	0	0	0
0x401f2e	46,139	0.001846	0	46,139	0	0
0x401f37	46,139	0.001846	0	46,139	0	0
0x401f3e	46,139	0.001846	0	46,139	0	0
0x401f44	46,139	0.001846	0	0	46,139	46,139
0x401f60	46,139	0.001846	0	0	0	0
0x401f65	46,139	0.001846	0	0	0	0
0x401f6a	46,139	0.001846	0	0	0	0
0x401f6f	46,139	0.001846	0	0	0	0
0x401f74	46,139	0.001846	46,139	0	0	0
0x401f7b	46,139	0.001846	46,139	0	0	0
0x401f82	46,139	0.001846	0	0	0	0
0x401f87	46,139	0.001846	0	0	0	0
0x401f8c	46,139	0.001846	0	0	0	0
0x401f91	46,139	0.001846	46,139	0	0	0
0x401f97	46,139	0.001846	46,139	0	0	0
0x401f9d	46,139	0.001846	0	0	0	0
0x401f9f	46,139	0.001846	0	46,139	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x401fa9	46,139	0.001846	0	46,139	0	0
0x401fb2	46,139	0.001846	0	46,139	0	0
0x401fbb	46,139	0.001846	0	46,139	0	0
0x401fc1	46,139	0.001846	0	46,139	0	0
0x401fc7	46,139	0.001846	0	46,139	0	0
0x401fce	46,139	0.001846	0	46,139	0	0
0x401fd5	46,139	0.001846	0	46,139	0	0
0x401fdb	46,139	0.001846	0	46,139	0	0
0x401fe2	46,139	0.001846	0	46,139	0	0
0x401fe8	46,139	0.001846	0	46,139	0	0
0x401fef	46,139	0.001846	0	0	46,139	46,139
0x402000	46,139	0.001846	0	0	0	0
0x402005	46,139	0.001846	0	0	0	0
0x40200a	46,139	0.001846	0	0	0	0
0x40200f	46,139	0.001846	0	0	0	0
0x402014	46,139	0.001846	0	0	0	0
0x402019	46,139	0.001846	46,139	0	0	0
0x402020	46,139	0.001846	46,139	0	0	0
0x402027	46,139	0.001846	0	0	0	0
0x40202c	46,139	0.001846	0	0	0	0
0x402031	46,139	0.001846	0	0	0	0
0x402036	46,139	0.001846	0	0	0	0
0x40203b	46,139	0.001846	0	0	0	0
0x40203f	46,139	0.001846	0	0	0	0
0x402044	46,139	0.001846	0	0	0	0
0x402048	46,139	0.001846	0	0	0	0
0x40204c	46,139	0.001846	0	0	0	0
0x402050	46,139	0.001846	0	0	0	0
0x402054	46,139	0.001846	0	0	0	0
0x402058	46,139	0.001846	46,139	0	0	0
0x402060	46,139	0.001846	0	0	0	0
0x402065	46,139	0.001846	46,139	0	0	0
0x40206d	46,139	0.001846	46,139	0	0	0
0x402075	46,139	0.001846	0	0	0	0
0x402079	46,139	0.001846	0	0	0	0
0x40207d	46,139	0.001846	0	0	0	0
0x402081	46,139	0.001846	0	46,139	0	0
0x402087	46,139	0.001846	0	0	0	0
0x40208c	46,139	0.001846	46,139	0	0	0
0x402094	46,139	0.001846	0	0	0	0
0x402099	46,139	0.001846	0	0	0	0
0x40209e	46,139	0.001846	0	0	0	0
0x4020a3	46,139	0.001846	0	0	0	0
0x4020a7	46,139	0.001846	0	0	0	0
0x4020ab	46,138	0.001846	0	0	0	0
0x4020b0	46,138	0.001846	0	0	0	0
0x4020b5	46,138	0.001846	0	0	0	0
0x4020ba	46,138	0.001846	0	0	0	0
0x4020be	46,138	0.001846	0	0	0	0
0x4020c2	46,138	0.001846	0	46,138	0	0
0x4020c8	46,138	0.001846	0	0	0	0
0x4020cc	46,138	0.001846	0	0	0	0
0x4020d1	46,138	0.001846	0	0	0	0
0x4020d5	46,138	0.001846	0	0	0	0
0x4020d9	46,138	0.001846	0	0	0	0
0x4020dd	46,138	0.001846	0	46,138	0	0
0x4020e3	46,138	0.001846	0	0	0	0
0x4020e8	46,138	0.001846	0	0	0	0
0x4020ed	46,138	0.001846	0	46,138	0	0
0x4020f3	46,138	0.001846	0	0	0	0
0x4020f7	46,138	0.001846	0	0	0	0
0x4020fc	46,138	0.001846	0	0	0	0

Top load line deltas (all 100) and top load pages (all 100).

	line_delta	count	share	page	load operands	share
	0	3,024,797	0.464814	34,358,891,164	2,860,567	0.439578
	1	571,220	0.087778	1,179	461,384	0.070900
	-1	369,107	0.056720	34,190,983,063	974	0.000150
	-2	289,568	0.044497	34,190,983,073	974	0.000150
	3	289,567	0.044497	34,190,983,075	974	0.000150
	2	197,292	0.030317	34,190,983,078	974	0.000150
	-3	151,153	0.023227	34,190,983,080	974	0.000150
-2,198,968,959,038		92,277	0.014180	34,190,983,083	974	0.000150
	4	92,276	0.014180	34,190,983,103	974	0.000150
2,198,968,959,037		46,138	0.007090	34,190,983,105	974	0.000150
-2,198,968,959,037		46,138	0.007090	34,190,983,108	974	0.000150
2,198,968,959,038		46,138	0.007090	34,190,983,110	974	0.000150
	-4	46,138	0.007090	34,190,984,033	974	0.000150
2,198,968,959,039		46,138	0.007090	34,190,984,048	974	0.000150
-10,746,162,659		8	0.000001	34,190,984,465	974	0.000150
-10,746,162,654		8	0.000001	34,190,984,620	974	0.000150
-10,746,162,649		8	0.000001	34,190,984,628	974	0.000150
-10,746,162,614		8	0.000001	34,190,984,630	974	0.000150
-10,746,162,544		8	0.000001	34,190,984,633	974	0.000150
-10,746,162,519		8	0.000001	34,190,984,635	974	0.000150
-10,746,162,514		8	0.000001	34,190,984,638	974	0.000150
-10,746,162,509		8	0.000001	34,190,984,643	974	0.000150
-10,746,162,504		8	0.000001	34,190,984,645	974	0.000150
-10,746,162,499		8	0.000001	34,190,984,665	974	0.000150
-10,746,162,454		8	0.000001	34,190,984,668	974	0.000150
-10,746,162,449		8	0.000001	34,190,984,670	974	0.000150
-10,746,162,444		8	0.000001	34,190,984,673	974	0.000150
-10,746,162,424		8	0.000001	34,190,985,595	974	0.000150
-10,746,162,339		8	0.000001	34,190,985,630	974	0.000150
-10,746,162,334		8	0.000001	34,190,985,638	974	0.000150
-10,746,162,314		8	0.000001	34,190,986,188	974	0.000150
-10,746,162,309		8	0.000001	34,190,986,190	974	0.000150
-10,746,162,304		8	0.000001	34,190,986,198	974	0.000150
-10,746,162,269		8	0.000001	34,190,986,200	974	0.000150
-10,746,162,264		8	0.000001	34,190,986,203	974	0.000150
-10,746,162,259		8	0.000001	34,190,986,205	974	0.000150
-10,746,162,229		8	0.000001	34,190,986,208	974	0.000150
-10,746,162,224		8	0.000001	34,190,986,228	974	0.000150
-10,746,162,204		8	0.000001	34,190,986,230	974	0.000150
-10,746,162,199		8	0.000001	34,190,986,233	974	0.000150
-10,746,162,194		8	0.000001	34,190,986,240	974	0.000150
-10,746,162,189		8	0.000001	34,190,982,634	972	0.000149
-10,746,162,154		8	0.000001	34,190,983,044	972	0.000149
-10,746,162,149		8	0.000001	34,190,983,052	972	0.000149
-10,746,162,144		8	0.000001	34,190,983,059	972	0.000149
-10,746,162,119		8	0.000001	34,190,983,066	972	0.000149
-10,746,162,114		8	0.000001	34,190,983,067	972	0.000149
-10,746,162,044		8	0.000001	34,190,983,071	972	0.000149
-10,746,162,039		8	0.000001	34,190,983,074	972	0.000149
-10,746,162,024		8	0.000001	34,190,983,079	972	0.000149
-10,746,162,019		8	0.000001	34,190,983,081	972	0.000149
-10,746,162,014		8	0.000001	34,190,983,082	972	0.000149
-10,746,162,009		8	0.000001	34,190,983,084	972	0.000149
-10,746,162,004		8	0.000001	34,190,983,102	972	0.000149
-10,746,161,999		8	0.000001	34,190,983,104	972	0.000149
-10,746,161,994		8	0.000001	34,190,983,106	972	0.000149
-10,746,161,959		8	0.000001	34,190,983,107	972	0.000149
-10,746,161,954		8	0.000001	34,190,983,114	972	0.000149
-10,746,161,949		8	0.000001	34,190,983,122	972	0.000149
-10,746,161,844		8	0.000001	34,190,983,782	972	0.000149
-10,746,161,839		8	0.000001	34,190,983,819	972	0.000149
-10,746,161,834		8	0.000001	34,190,983,919	972	0.000149
-10,746,161,829		8	0.000001	34,190,984,052	972	0.000149
-10,746,161,824		8	0.000001	34,190,984,272	972	0.000149
-10,746,161,819		8	0.000001	34,190,984,591	972	0.000149

line.delta	count	share	page	load operands	share
-10,746,161,814	8	0.000001	34,190,984,606	972	0.000149
-10,746,161,809	8	0.000001	34,190,984,614	972	0.000149
-10,746,161,764	8	0.000001	34,190,984,622	972	0.000149
-10,746,161,759	8	0.000001	34,190,984,629	972	0.000149
-10,746,161,744	8	0.000001	34,190,984,634	972	0.000149
-10,746,161,739	8	0.000001	34,190,984,636	972	0.000149
-10,746,161,734	8	0.000001	34,190,984,637	972	0.000149
-10,746,161,729	8	0.000001	34,190,984,641	972	0.000149
-10,746,161,724	8	0.000001	34,190,984,642	972	0.000149
-10,746,161,719	8	0.000001	34,190,984,644	972	0.000149
-10,746,161,704	8	0.000001	34,190,984,646	972	0.000149
-10,746,161,699	8	0.000001	34,190,984,647	972	0.000149
-10,746,161,694	8	0.000001	34,190,984,664	972	0.000149
-10,746,161,689	8	0.000001	34,190,984,666	972	0.000149
-10,746,161,654	8	0.000001	34,190,984,667	972	0.000149
-10,746,161,649	8	0.000001	34,190,984,669	972	0.000149
-10,746,161,644	8	0.000001	34,190,984,677	972	0.000149
-10,746,161,639	8	0.000001	34,190,984,867	972	0.000149
-10,746,161,634	8	0.000001	34,190,985,117	972	0.000149
-10,746,161,629	8	0.000001	34,190,985,381	972	0.000149
-10,746,161,624	8	0.000001	34,190,985,389	972	0.000149
-10,746,161,619	8	0.000001	34,190,985,611	972	0.000149
-10,746,161,614	8	0.000001	34,190,985,619	972	0.000149
-10,746,161,544	8	0.000001	34,190,985,752	972	0.000149
-10,746,161,524	8	0.000001	34,190,986,161	972	0.000149
-10,746,161,519	8	0.000001	34,190,986,192	972	0.000149
-10,746,161,514	8	0.000001	34,190,986,196	972	0.000149
-10,746,161,509	8	0.000001	34,190,986,197	972	0.000149
-10,746,161,504	8	0.000001	34,190,986,199	972	0.000149
-10,746,161,499	8	0.000001	34,190,986,204	972	0.000149
-10,746,161,494	8	0.000001	34,190,986,206	972	0.000149
-10,746,161,489	8	0.000001	34,190,986,207	972	0.000149
-10,746,161,484	8	0.000001	34,190,986,209	972	0.000149
-10,746,161,469	8	0.000001	34,190,986,227	972	0.000149
-10,746,161,464	8	0.000001	34,190,986,229	972	0.000149

Register usage (all 46 rows).

direction	register_id	dynamic uses	share within direction
source	10	5,658,425	0.121308
source	6	5,213,628	0.111773
source	101	4,152,435	0.089022
source	100	3,368,099	0.072207
source	3	3,185,586	0.068294
source	103	3,091,262	0.066272
source	106	2,353,050	0.050446
source	4	2,237,329	0.047965
source	102	2,030,078	0.043522
source	115	1,891,683	0.040555
source	105	1,753,248	0.037587
source	110	1,614,834	0.034620
source	107	1,338,011	0.028685
source	9	1,289,088	0.027636
source	114	1,107,325	0.023739
source	112	1,061,188	0.022750
source	108	1,061,179	0.022750
source	26	906,927	0.019443
source	113	784,350	0.016815
source	109	784,350	0.016815
source	104	625,276	0.013405
source	111	553,660	0.011870
source	25	373,927	0.008016
source	30	117,755	0.002524
source	55	92,278	0.001978
destination	101	3,229,670	0.156982

direction	register_id	dynamic uses	share within direction
destination	103	2,583,738	0.125586
destination	100	2,399,193	0.116616
destination	9	1,289,088	0.062658
destination	105	1,245,727	0.060550
destination	102	1,199,590	0.058308
destination	115	1,107,333	0.053823
destination	107	1,107,318	0.053823
destination	106	1,061,184	0.051580
destination	114	645,943	0.031397
destination	112	645,943	0.031397
destination	110	599,795	0.029154
destination	104	532,998	0.025907
destination	113	507,519	0.024669
destination	109	507,519	0.024669
destination	25	445,543	0.021656
destination	26	445,543	0.021656
destination	108	369,106	0.017941
destination	111	369,105	0.017941
destination	55	163,894	0.007966
destination	10	117,755	0.005724

605.mcf_s-994B

Top PCs (all 200 rows).

pc	dyn instr	instr share	load	store	branch	taken
0x402440	811,790	0.032472	811,790	0	0	0
0x402443	811,790	0.032472	811,790	0	0	0
0x402446	811,790	0.032472	811,790	0	0	0
0x40244a	811,790	0.032472	0	0	0	0
0x40244f	811,789	0.032472	811,789	0	0	0
0x402453	811,789	0.032472	0	0	811,789	52,952
0x40246d	811,789	0.032472	811,789	0	0	0
0x402455	758,837	0.030353	0	0	0	0
0x40245a	758,837	0.030353	0	0	758,837	124,567
0x40245c	634,270	0.025371	634,270	0	0	0
0x40245e	634,270	0.025371	634,270	0	0	0
0x402460	634,270	0.025371	0	0	0	0
0x402465	634,270	0.025371	0	0	0	0
0x40246a	634,270	0.025371	0	0	0	0
0x406383	396,095	0.015844	396,095	0	0	0
0x406388	396,095	0.015844	396,095	396,095	0	0
0x40638c	396,095	0.015844	0	0	0	0
0x40638e	396,095	0.015844	0	0	396,095	171,014
0x406380	396,094	0.015844	0	0	0	0
0x406fa0	342,303	0.013692	0	0	0	0
0x406fa3	342,303	0.013692	342,303	0	0	0
0x406fa8	342,303	0.013692	342,303	342,303	0	0
0x406fac	342,302	0.013692	0	0	0	0
0x406fae	342,302	0.013692	0	0	342,302	167,170
0x406394	225,081	0.009003	0	0	225,081	225,081
0x4063b2	225,081	0.009003	0	0	0	0
0x4063b5	225,081	0.009003	0	0	0	0
0x4063b8	225,081	0.009003	0	0	225,081	222,876
0x406fb0	175,132	0.007005	0	0	175,132	0
0x406fb2	175,132	0.007005	0	0	0	0
0x406fb5	175,132	0.007005	0	0	175,132	175,132
0x406fdd	175,132	0.007005	0	0	0	0
0x406fe0	175,132	0.007005	0	0	0	0
0x406fe3	175,132	0.007005	0	0	175,132	171,289
0x406990	174,693	0.006988	0	174,693	0	0
0x406998	174,693	0.006988	0	0	0	0
0x40699b	174,693	0.006988	0	174,693	0	0
0x4069a0	174,693	0.006988	0	0	174,693	3,679
0x40632f	174,692	0.006988	0	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x406332	174,692	0.006988	0	174,692	0	0
0x406337	174,692	0.006988	0	0	174,692	173,218
0x406350	173,218	0.006929	173,218	0	0	0
0x406358	173,218	0.006929	0	0	173,218	173,218
0x40636e	173,218	0.006929	0	0	0	0
0x406371	173,218	0.006929	0	0	0	0
0x4069a6	171,014	0.006841	171,014	0	0	0
0x4069ae	171,014	0.006841	171,014	0	0	0
0x4069b6	171,014	0.006841	0	0	171,014	171,014
0x406f90	171,014	0.006841	0	0	0	0
0x406f93	171,014	0.006841	0	0	0	0
0x407000	167,170	0.006687	167,170	0	0	0
0x407008	167,170	0.006687	167,170	0	0	0
0x40700d	167,170	0.006687	0	0	0	0
0x407010	167,170	0.006687	0	0	167,170	167,170
0x407045	167,170	0.006687	167,170	0	0	0
0x407049	167,170	0.006687	167,170	0	0	0
0x40704c	167,170	0.006687	0	167,170	0	0
0x407050	167,170	0.006687	0	167,170	0	0
0x407053	167,170	0.006687	0	0	167,170	167,170
0x407061	167,170	0.006687	0	0	0	0
0x407064	167,170	0.006687	0	0	167,170	167,170
0x406310	167,170	0.006687	0	0	0	0
0x406313	167,170	0.006687	0	0	0	0
0x406316	167,170	0.006687	0	167,170	0	0
0x406321	167,170	0.006687	0	0	0	0
0x406324	167,170	0.006687	0	0	0	0
0x406327	167,170	0.006687	167,170	0	0	0
0x4083f0	32,904	0.001316	0	0	0	0
0x4083f4	32,904	0.001316	32,904	0	0	0
0x4083f9	32,904	0.001316	0	0	0	0
0x4083fd	32,904	0.001316	32,904	32,904	0	0
0x408401	32,904	0.001316	32,904	0	0	0
0x408406	32,904	0.001316	0	0	0	0
0x408408	32,904	0.001316	0	0	32,904	11,819
0x40840a	21,085	0.000843	21,085	0	0	0
0x40840e	21,085	0.000843	21,085	0	0	0
0x408413	21,085	0.000843	21,085	0	0	0
0x408417	21,085	0.000843	0	21,085	0	0
0x40841b	21,085	0.000843	0	21,085	0	0
0x40841f	21,085	0.000843	21,085	0	0	0
0x408424	21,085	0.000843	0	0	0	0
0x408428	21,085	0.000843	0	0	0	0
0x40842b	21,085	0.000843	0	0	21,085	14,803
0x407e70	18,101	0.000724	0	0	0	0
0x407e73	18,101	0.000724	0	0	18,101	0
0x407e79	18,101	0.000724	18,101	0	0	0
0x407e81	18,101	0.000724	0	0	18,101	18,101
0x4083ed	18,101	0.000724	0	0	0	0
0x4083ef	18,101	0.000724	0	0	0	0
0x408480	18,101	0.000724	18,101	0	0	0
0x408485	18,101	0.000724	0	0	0	0
0x408488	18,101	0.000724	18,101	18,101	0	0
0x408490	18,101	0.000724	18,101	18,101	0	0
0x408495	18,101	0.000724	18,101	18,101	0	0
0x40849a	18,101	0.000724	18,101	0	0	0
0x40849f	18,101	0.000724	0	0	0	0
0x4084a2	18,101	0.000724	0	0	18,101	12,161
0x407ba0	11,706	0.000468	0	0	0	0
0x407ba4	11,706	0.000468	11,706	0	0	0
0x407ba9	11,706	0.000468	0	0	0	0
0x407bad	11,706	0.000468	11,706	11,706	0	0
0x407bb1	11,706	0.000468	0	0	0	0
0x407bb3	11,706	0.000468	0	0	11,706	3,098
0x407bb5	8,608	0.000344	8,608	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x407bb9	8,608	0.000344	8,608	0	0	0
0x407bbe	8,608	0.000344	8,608	0	0	0
0x407bc2	8,608	0.000344	0	8,608	0	0
0x407bc6	8,608	0.000344	0	8,608	0	0
0x407bca	8,608	0.000344	8,608	0	0	0
0x407bcf	8,608	0.000344	0	0	0	0
0x407bd3	8,608	0.000344	8,608	0	0	0
0x407bd8	8,608	0.000344	0	0	8,608	7,616
0x407480	7,522	0.000301	7,522	0	0	0
0x407485	7,522	0.000301	7,522	0	0	0
0x40748a	7,522	0.000301	0	0	0	0
0x40748e	7,522	0.000301	7,522	0	0	0
0x407493	7,522	0.000301	0	7,522	0	0
0x407498	7,522	0.000301	0	0	0	0
0x40749c	7,522	0.000301	7,522	0	0	0
0x4074a4	7,522	0.000301	0	7,522	0	0
0x4074a9	7,522	0.000301	0	0	7,522	7,022
0x405f30	7,522	0.000301	0	0	0	0
0x405f33	7,522	0.000301	0	0	0	0
0x405f36	7,522	0.000301	0	0	0	0
0x405f3a	7,522	0.000301	0	0	0	0
0x405f3d	7,522	0.000301	0	0	0	0
0x405f41	7,522	0.000301	0	7,522	0	0
0x405f46	7,522	0.000301	0	7,522	0	0
0x405f4b	7,522	0.000301	0	0	7,522	812
0x406150	7,522	0.000301	7,522	0	0	0
0x406158	7,522	0.000301	0	0	7,522	7,522
0x4061e0	7,522	0.000301	7,522	0	0	0
0x4061e3	7,522	0.000301	7,522	0	0	0
0x4061e7	7,522	0.000301	0	7,522	0	0
0x4061ea	7,522	0.000301	0	7,522	0	0
0x4061ee	7,522	0.000301	0	0	7,522	7,522
0x4062e0	7,522	0.000301	0	0	0	0
0x4062e4	7,522	0.000301	0	7,522	0	0
0x4062ec	7,522	0.000301	0	0	0	0
0x4062f0	7,522	0.000301	0	0	0	0
0x4062f4	7,522	0.000301	0	0	0	0
0x4062f7	7,522	0.000301	0	7,522	0	0
0x406302	7,522	0.000301	0	0	0	0
0x406305	7,522	0.000301	0	0	0	0
0x406308	7,522	0.000301	0	0	0	0
0x40630b	7,522	0.000301	0	0	7,522	7,522
0x404fb0	7,022	0.000281	0	7,022	0	0
0x404fb5	7,022	0.000281	7,022	0	0	0
0x404fbd	7,022	0.000281	0	0	0	0
0x404fc0	7,022	0.000281	7,022	0	0	0
0x404fc8	7,022	0.000281	0	0	0	0
0x404fcb	7,022	0.000281	0	0	0	0
0x404fce	7,022	0.000281	0	0	0	0
0x404fd1	7,022	0.000281	0	0	0	0
0x404fd5	7,022	0.000281	0	0	0	0
0x404fd8	7,022	0.000281	0	0	7,022	0
0x404fde	7,022	0.000281	7,022	0	0	0
0x404fe3	7,022	0.000281	0	0	0	0
0x404fe6	7,022	0.000281	0	0	0	0
0x404fe9	7,022	0.000281	0	0	0	0
0x404fec	7,022	0.000281	7,022	0	0	0
0x404ff1	7,022	0.000281	0	0	7,022	0
0x404ff7	7,022	0.000281	0	0	0	0
0x404ffb	7,022	0.000281	0	0	0	0
0x405001	7,022	0.000281	0	0	0	0
0x405005	7,022	0.000281	0	0	0	0
0x405009	7,022	0.000281	0	0	7,022	7,022
0x405080	7,022	0.000281	0	0	0	0
0x405083	7,022	0.000281	7,022	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x405088	7,022	0.000281	0	0	0	0
0x40508b	7,022	0.000281	0	0	0	0
0x405090	7,022	0.000281	7,022	0	0	0
0x405093	7,022	0.000281	7,022	0	0	0
0x405096	7,022	0.000281	0	7,022	0	0
0x405099	7,022	0.000281	0	0	0	0
0x40509d	7,022	0.000281	0	7,022	0	0
0x4050a0	7,022	0.000281	0	0	0	0
0x4050a4	7,022	0.000281	0	0	0	0
0x4050a7	7,022	0.000281	0	0	0	0
0x4050ab	7,022	0.000281	0	0	7,022	0
0x4050ad	7,022	0.000281	0	0	7,022	7,022
0x4056a0	7,022	0.000281	0	0	0	0
0x4056a3	7,022	0.000281	0	0	0	0
0x4056a6	7,022	0.000281	7,022	0	0	0
0x4056ab	7,022	0.000281	0	0	0	0
0x4056ae	7,022	0.000281	0	0	0	0
0x4056b1	7,022	0.000281	0	0	0	0
0x4056b4	7,022	0.000281	0	0	0	0
0x4056b7	7,022	0.000281	0	0	0	0
0x4056ba	7,022	0.000281	0	0	0	0
0x4056be	7,022	0.000281	0	0	0	0
0x4056c1	7,022	0.000281	7,022	0	0	0
0x4056c6	7,022	0.000281	0	0	7,022	7,022
0x405cb0	7,022	0.000281	0	0	0	0
0x405cb3	7,022	0.000281	0	0	7,022	358
0x405cd2	7,021	0.000281	0	0	0	0
0x405cd5	7,021	0.000281	0	0	7,021	223
0x405cdb	6,798	0.000272	0	0	0	0
0x405cde	6,798	0.000272	0	0	0	0
0x405ce0	6,798	0.000272	0	0	0	0

Top load line deltas (all 100) and top load pages (all 100).

line_delta	count	share	page	load operands	share
-1	1,416,790	0.161160	34,357,359,865	3,592,630	0.408661
0	1,071,402	0.121872	34,122,636,279	285,881	0.032519
1	438,372	0.049865	34,357,359,864	260,071	0.029583
2	399,747	0.045471	34,122,166,985	200,107	0.022762
-30,034,854	163,325	0.018578	34,122,169,806	162,697	0.018507
-2	163,076	0.018550	34,122,636,209	61,720	0.007021
3	125,970	0.014329	34,122,636,242	48,291	0.005493
-29,854,322	94,228	0.010718	34,122,636,256	45,149	0.005136
15,052,163,820	68,452	0.007786	34,122,319,480	34,464	0.003920
15,052,344,359	36,778	0.004183	34,122,636,224	33,792	0.003844
-3	33,922	0.003859	34,122,369,975	31,586	0.003593
-17,041,110	19,034	0.002165	34,122,346,873	30,502	0.003470
-18,517,483	18,852	0.002144	34,122,636,233	24,711	0.002811
-20,270,638	17,197	0.001956	34,122,636,244	23,578	0.002682
15,042,584,657	17,197	0.001956	34,122,163,818	23,550	0.002679
-4	15,403	0.001752	34,122,636,236	23,327	0.002653
5	13,467	0.001532	34,122,636,216	23,099	0.002628
-5	13,308	0.001514	34,122,636,264	22,134	0.002518
-15,875,072	13,207	0.001502	34,122,636,270	21,939	0.002496
15,039,352,987	12,544	0.001427	34,122,636,251	21,714	0.002470
-6	12,238	0.001392	34,122,636,273	21,298	0.002423
15,052,547,026	11,771	0.001339	34,122,636,276	21,064	0.002396
-30,236,057	11,767	0.001338	34,122,636,228	20,209	0.002299
15,040,831,510	11,602	0.001320	34,122,636,262	19,681	0.002239
-7	11,393	0.001296	34,122,636,212	19,560	0.002225
15,041,651,344	9,589	0.001091	34,122,636,249	19,358	0.002202
-19,338,276	9,587	0.001091	34,122,334,063	19,186	0.002182
4	9,095	0.001035	34,122,636,268	18,730	0.002131
-9	8,340	0.000949	34,122,636,245	18,679	0.002125
-8	8,155	0.000928	34,122,636,248	18,532	0.002108

line.delta	count	share	page	load operands	share
15,043,224,878	7,609	0.000866	34,122,636,230	18,204	0.002071
-20,910,866	7,607	0.000865	34,122,636,261	18,168	0.002067
-14,712,556	7,265	0.000826	34,122,636,240	18,071	0.002056
15,039,360,494	7,138	0.000812	34,122,636,211	18,037	0.002052
-17,049,533	7,134	0.000811	34,122,388,231	18,029	0.002051
-10	6,487	0.000738	34,122,636,238	17,876	0.002033
15,037,024,425	6,429	0.000731	34,122,636,259	17,703	0.002014
-11	5,982	0.000680	34,122,636,210	17,539	0.001995
-13	5,486	0.000624	34,122,636,253	17,514	0.001992
-12	5,290	0.000602	34,122,636,257	17,433	0.001983
-14	5,182	0.000589	34,122,636,220	17,431	0.001983
6	5,156	0.000586	34,122,636,260	17,377	0.001977
15,041,288,442	4,963	0.000565	34,122,636,267	17,274	0.001965
-18,975,953	4,960	0.000564	34,122,636,254	17,197	0.001956
15,038,184,562	4,785	0.000544	34,122,636,213	17,197	0.001956
-18	4,690	0.000533	34,122,636,225	17,189	0.001955
15,045,434,238	4,634	0.000527	34,122,636,258	17,179	0.001954
-23,124,161	4,632	0.000527	34,122,636,241	17,142	0.001950
-19,824,903	4,626	0.000526	34,122,636,226	17,079	0.001943
15,042,137,964	4,626	0.000526	34,122,636,234	17,072	0.001942
15,040,106,840	4,378	0.000498	34,122,636,247	17,058	0.001940
-17,794,979	4,375	0.000498	34,122,636,239	17,041	0.001938
-15	4,222	0.000480	34,122,636,237	16,972	0.001931
15,040,361,607	4,107	0.000467	34,122,636,222	16,952	0.001928
-18,050,653	4,103	0.000467	34,122,636,250	16,897	0.001922
-16	4,085	0.000465	34,122,636,266	16,865	0.001918
15,042,924,149	3,931	0.000447	34,122,636,231	16,853	0.001917
-20,610,598	3,925	0.000446	34,122,636,246	16,794	0.001910
9	3,839	0.000437	34,122,636,255	16,721	0.001902
-22	3,829	0.000436	34,122,636,274	16,684	0.001898
15,043,527,919	3,685	0.000419	34,122,636,218	16,579	0.001886
-21,213,914	3,682	0.000419	34,122,636,235	16,540	0.001881
-17	3,585	0.000408	34,122,636,243	16,409	0.001867
15,038,618,820	3,573	0.000406	34,122,636,252	16,391	0.001864
-16,307,058	3,572	0.000406	34,122,636,277	16,312	0.001855
-23	3,560	0.000405	34,122,636,221	16,242	0.001848
-20	3,517	0.000400	34,122,636,275	16,195	0.001842
-21	3,466	0.000394	34,122,636,214	16,145	0.001836
-19	3,451	0.000393	34,122,636,263	16,130	0.001835
-27	3,373	0.000384	34,122,636,217	16,064	0.001827
7	3,288	0.000374	34,122,636,265	16,035	0.001824
15,041,757,513	3,221	0.000366	34,122,636,229	16,014	0.001822
-19,447,613	3,215	0.000366	34,122,636,227	16,003	0.001820
15,040,948,301	3,211	0.000365	34,122,636,215	15,944	0.001814
-18,636,031	3,207	0.000365	34,122,636,272	15,920	0.001811
15,038,539,892	3,032	0.000345	34,122,636,219	15,812	0.001799
-16,229,444	3,031	0.000345	34,122,636,271	15,734	0.001790
-25	2,956	0.000336	34,122,636,269	15,647	0.001780
-54	2,943	0.000335	34,122,636,232	15,589	0.001773
-24	2,926	0.000333	34,122,636,278	15,525	0.001766
-36	2,893	0.000329	34,122,309,476	15,286	0.001739
-15,866,630	2,889	0.000329	34,122,636,223	14,798	0.001683
-26	2,692	0.000306	34,122,369,857	14,282	0.001625
-45	2,681	0.000305	34,122,406,359	13,792	0.001569
-31	2,646	0.000301	34,122,339,733	9,970	0.001134
-32	2,644	0.000301	34,122,326,459	9,374	0.001066
-41	2,563	0.000292	34,122,274,955	9,279	0.001055
-28	2,522	0.000287	34,122,358,196	8,763	0.000997
-40	2,516	0.000286	34,122,354,215	8,617	0.000980
-29	2,500	0.000284	34,122,314,175	7,894	0.000898
-30	2,459	0.000280	34,122,304,741	7,692	0.000875
15,042,293,878	2,438	0.000277	34,122,381,446	7,159	0.000814
-19,980,825	2,437	0.000277	34,122,345,048	6,480	0.000737
-33	2,435	0.000277	34,122,332,404	6,476	0.000737
10	2,429	0.000276	34,122,382,679	6,471	0.000736

line.delta	count	share	page	load operands	share
15,041,966,313	2,418	0.000275	34,122,329,141	5,189	0.000590
-19,655,366	2,417	0.000275	34,122,388,335	4,956	0.000564
15,043,446,614	2,403	0.000273	34,122,324,023	4,954	0.000564
-21,132,769	2,401	0.000273	34,122,306,012	4,915	0.000559
-34	2,379	0.000271	34,122,331,052	4,464	0.000508

Register usage (all 60 rows).

direction	register_id	dynamic uses	share within direction
source	26	5,615,303	0.167917
source	6	5,117,445	0.153029
source	25	5,008,952	0.149785
source	9	2,294,129	0.068602
source	5	2,214,891	0.066233
source	3	1,812,779	0.054208
source	4	1,760,150	0.052635
source	8	1,719,250	0.051412
source	56	1,466,319	0.043848
source	16	1,079,603	0.032284
source	10	971,289	0.029045
source	15	913,789	0.027325
source	17	785,379	0.023486
source	11	739,189	0.022104
source	55	667,724	0.019967
source	7	395,077	0.011814
source	29	360,242	0.010772
source	18	157,254	0.004702
source	13	136,159	0.004072
source	14	69,979	0.002093
source	12	38,572	0.001153
source	54	36,791	0.001100
source	45	26,432	0.000790
source	53	22,191	0.000664
source	47	10,143	0.000303
source	73	9,643	0.000288
source	79	6,710	0.000201
source	67	3,703	0.000111
source	76	1,360	0.000041
source	49	500	0.000015
destination	26	6,433,755	0.247569
destination	25	5,152,493	0.198266
destination	56	3,493,473	0.134428
destination	6	1,730,195	0.066577
destination	4	1,722,074	0.066265
destination	9	1,406,710	0.054130
destination	3	1,241,450	0.047771
destination	8	985,704	0.037930
destination	55	667,858	0.025699
destination	5	656,625	0.025267
destination	15	539,369	0.020755
destination	10	476,181	0.018323
destination	17	415,454	0.015987
destination	11	356,587	0.013721
destination	16	243,809	0.009382
destination	7	111,163	0.004278
destination	13	78,298	0.003013
destination	18	64,949	0.002499
destination	14	47,374	0.001823
destination	54	46,076	0.001773
destination	53	22,191	0.000854
destination	47	20,286	0.000781
destination	45	19,544	0.000752
destination	12	19,286	0.000742
destination	67	14,428	0.000555
destination	79	6,710	0.000258

direction	register_id	dynamic uses	share within direction
destination	46	6,664	0.000256
destination	30	6,664	0.000256
destination	76	1,360	0.000052
destination	49	1,000	0.000038

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Top PCs (all 200 rows).

pc	dyn instr	instr share	load	store	branch	taken
0x471325	140,198	0.005608	0	0	0	0
0x471328	140,198	0.005608	0	0	140,198	7,215
0x47132e	132,983	0.005319	0	0	0	0
0x471331	132,983	0.005319	0	0	0	0
0x471334	132,983	0.005319	0	0	132,983	832
0x471386	132,983	0.005319	0	0	0	0
0x471389	132,983	0.005319	132,983	0	0	0
0x47138d	132,983	0.005319	0	0	0	0
0x471390	132,983	0.005319	132,983	0	0	0
0x471394	132,983	0.005319	132,983	0	0	0
0x471398	132,983	0.005319	132,983	0	0	0
0x47139c	132,983	0.005319	0	0	132,983	3,433
0x471336	132,151	0.005286	0	0	0	0
0x471339	132,151	0.005286	132,151	0	0	0
0x47133d	132,151	0.005286	132,151	0	0	0
0x471341	132,151	0.005286	0	0	0	0
0x471343	132,151	0.005286	0	0	0	0
0x471346	132,151	0.005286	0	0	0	0
0x471349	132,151	0.005286	132,151	0	0	0
0x47134d	132,151	0.005286	132,151	0	0	0
0x471351	132,151	0.005286	132,151	0	0	0
0x471355	132,151	0.005286	0	0	132,151	43,935
0x47139e	129,550	0.005182	0	0	129,550	126,834
0x471310	129,440	0.005178	0	129,440	0	0
0x471314	129,440	0.005178	0	129,440	0	0
0x471317	129,440	0.005178	129,440	0	0	0
0x47131b	129,440	0.005178	0	129,440	0	0
0x47131f	129,440	0.005178	0	129,440	0	0
0x471323	129,440	0.005178	0	0	0	0
0x471357	88,216	0.003529	0	0	88,216	56,286
0x400670	61,945	0.002478	61,945	0	61,945	61,945
0x63e5e0	61,945	0.002478	0	0	0	0
0x63e5e2	61,945	0.002478	0	0	0	0
0x63e5e4	61,945	0.002478	0	0	0	0
0x63e5e8	61,945	0.002478	0	0	0	0
0x63e5ec	61,945	0.002478	0	0	0	0
0x63e5ef	61,945	0.002478	0	0	61,945	0
0x63e5f1	61,945	0.002478	0	0	0	0
0x63e5f4	61,945	0.002478	0	0	61,945	30
0x63f810	61,945	0.002478	0	0	0	0
0x63f814	61,945	0.002478	61,945	0	0	0
0x63f818	61,945	0.002478	61,945	0	0	0
0x63f81c	61,945	0.002478	0	0	0	0
0x63f81e	61,945	0.002478	61,945	0	0	0
0x63e5f6	61,915	0.002477	61,915	0	0	0
0x63e5fa	61,915	0.002477	61,915	0	0	0
0x63e5fe	61,915	0.002477	61,915	0	0	0
0x63e603	61,915	0.002477	61,915	0	0	0
0x63e608	61,915	0.002477	0	0	0	0
0x63e60c	61,915	0.002477	0	0	0	0
0x63e610	61,915	0.002477	0	0	0	0
0x63e614	61,915	0.002477	0	0	0	0
0x63e618	61,915	0.002477	0	0	0	0
0x63e61c	61,915	0.002477	0	0	0	0
0x63e622	61,915	0.002477	0	0	61,915	61,915

pc	dyn instr	instr share	load	store	branch	taken
0x471384	61,090	0.002444	0	0	0	0
0x55b6e3	37,818	0.001513	0	0	0	0
0x55b6e7	37,818	0.001513	37,818	0	0	0
0x55b6eb	37,818	0.001513	37,818	0	0	0
0x55b6ef	37,818	0.001513	37,818	0	0	0
0x55b6f4	37,818	0.001513	37,818	0	0	0
0x55b6f9	37,818	0.001513	0	0	0	0
0x55b6fc	37,818	0.001513	0	0	37,818	6,923
0x55b6d0	37,817	0.001513	0	37,817	0	0
0x55b6d2	37,817	0.001513	0	0	0	0
0x55b6d5	37,817	0.001513	0	37,817	0	0
0x55b6d7	37,817	0.001513	0	0	0	0
0x55b6da	37,817	0.001513	0	37,817	0	0
0x55b6dc	37,817	0.001513	0	0	0	0
0x55b6df	37,817	0.001513	0	37,817	0	0
0x55b6e1	37,817	0.001513	0	37,817	0	0
0x55b6e2	37,817	0.001513	0	37,817	0	0
0x4712a0	37,276	0.001491	0	0	0	0
0x4712a2	37,276	0.001491	0	0	0	0
0x4712a4	37,276	0.001491	0	0	0	0
0x4712a7	37,276	0.001491	37,276	0	0	0
0x4712ab	37,276	0.001491	37,276	0	0	0
0x4712af	37,276	0.001491	37,276	0	0	0
0x4712b3	37,276	0.001491	0	0	37,276	4,337
0x4712b5	32,939	0.001318	0	0	32,939	26,930
0x471359	31,930	0.001277	31,930	0	0	0
0x47135d	31,930	0.001277	31,930	0	0	0
0x471362	31,930	0.001277	0	0	31,930	0
0x471364	31,930	0.001277	0	0	31,930	0
0x471366	31,930	0.001277	31,930	0	0	0
0x47136d	31,930	0.001277	31,930	0	0	0
0x471374	31,930	0.001277	0	0	31,930	16,323
0x55b6fe	30,895	0.001236	30,895	0	0	0
0x55b701	30,895	0.001236	0	0	0	0
0x55b704	30,895	0.001236	0	0	30,895	0
0x55b706	30,895	0.001236	0	0	0	0
0x55b709	30,895	0.001236	0	30,895	0	0
0x55b70e	30,895	0.001236	0	30,895	0	0
0x55b713	30,895	0.001236	0	30,895	0	0
0x55b717	30,895	0.001236	0	30,895	0	0
0x55b71b	30,895	0.001236	0	30,895	0	0
0x55b720	30,895	0.001236	0	0	0	0
0x55b722	30,895	0.001236	30,895	0	0	0
0x55b726	30,895	0.001236	30,895	0	0	0
0x55b72a	30,895	0.001236	30,895	0	0	0
0x55b72f	30,895	0.001236	30,895	0	0	0
0x55b734	30,895	0.001236	0	0	30,895	0
0x55b736	30,895	0.001236	0	0	0	0
0x55b739	30,895	0.001236	0	0	30,895	30,895
0x55b7b0	30,895	0.001236	30,895	0	0	0
0x55b7b4	30,895	0.001236	0	0	0	0
0x55b7b7	30,895	0.001236	0	0	30,895	8,559
0x4712d4	26,930	0.001077	0	0	0	0
0x4712d7	26,930	0.001077	0	26,930	0	0
0x4712db	26,930	0.001077	0	26,930	0	0
0x4712de	26,930	0.001077	0	0	0	0
0x4712e1	26,930	0.001077	0	0	26,930	26,602
0x55b7b9	22,336	0.000893	0	0	0	0
0x55b7bc	22,336	0.000893	0	0	22,336	0
0x55b7c2	22,336	0.000893	0	0	0	0
0x55b7c5	22,336	0.000893	0	22,336	0	0
0x55b7ca	22,336	0.000893	0	22,336	0	0
0x55b7cf	22,336	0.000893	0	22,336	0	0
0x55b7d3	22,336	0.000893	0	22,336	0	0
0x55b7d8	22,336	0.000893	0	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x55b7da	22,336	0.000893	22,336	0	0	0
0x55b7de	22,336	0.000893	22,336	0	0	0
0x55b7e3	22,336	0.000893	22,336	0	0	0
0x55b7e8	22,336	0.000893	0	0	22,336	22,336
0x55b73b	22,336	0.000893	22,336	0	0	0
0x55b73f	22,336	0.000893	0	0	0	0
0x55b742	22,336	0.000893	0	0	0	0
0x55b745	22,336	0.000893	22,336	0	0	0
0x55b748	22,336	0.000893	0	22,336	0	0
0x55b74d	22,336	0.000893	0	22,336	0	0
0x55b752	22,336	0.000893	22,336	0	0	0
0x55b756	22,336	0.000893	0	0	0	0
0x55b75a	22,336	0.000893	22,336	0	0	0
0x55b75b	22,336	0.000893	22,336	0	0	0
0x55b75c	22,336	0.000893	22,336	0	0	0
0x55b75e	22,336	0.000893	22,336	0	0	0
0x55b760	22,336	0.000893	22,336	0	0	0
0x55b762	22,336	0.000893	22,336	0	0	0
0x55b764	22,336	0.000893	0	0	22,336	22,336
0x4c2770	21,474	0.000859	0	21,474	0	0
0x4c2777	21,474	0.000859	21,474	0	0	0
0x550fd0	20,569	0.000823	0	20,569	0	0
0x550fd2	20,569	0.000823	0	20,569	0	0
0x550fd4	20,569	0.000823	0	20,569	0	0
0x550fd5	20,569	0.000823	0	0	0	0
0x550fd8	20,569	0.000823	20,569	0	0	0
0x550fdf	20,569	0.000823	20,569	0	0	0
0x550fe3	20,569	0.000823	0	0	20,569	20,569
0x550ff8	20,569	0.000823	20,569	0	0	0
0x550fff	20,569	0.000823	0	0	0	0
0x551001	20,569	0.000823	0	0	0	0
0x551004	20,569	0.000823	0	0	20,569	0
0x55100a	20,569	0.000823	20,569	0	20,569	20,569
0x5510cd	20,569	0.000823	20,569	0	0	0
0x5510d4	20,569	0.000823	20,569	0	0	0
0x5510d8	20,569	0.000823	0	0	20,569	20,569
0x5510ed	20,569	0.000823	20,569	0	0	0
0x5510f4	20,569	0.000823	0	0	0	0
0x5510f7	20,569	0.000823	0	0	20,569	78
0x55115d	20,569	0.000823	20,569	0	0	0
0x551164	20,569	0.000823	20,569	0	0	0
0x551168	20,569	0.000823	0	0	20,569	20,569
0x5511ce	20,569	0.000823	20,569	0	0	0
0x5511cf	20,569	0.000823	20,569	0	0	0
0x5511d1	20,569	0.000823	20,569	0	0	0
0x5511d3	20,569	0.000823	20,569	0	0	0
0x5510f9	20,491	0.000820	0	0	0	0
0x5510fc	20,491	0.000820	0	0	20,491	5,162
0x5124d0	16,725	0.000669	0	16,725	0	0
0x5124d1	16,725	0.000669	0	16,725	0	0
0x5124d3	16,725	0.000669	0	16,725	0	0
0x5124d5	16,725	0.000669	0	16,725	0	0
0x5124d7	16,725	0.000669	0	16,725	0	0
0x5124d9	16,725	0.000669	0	16,725	0	0
0x5124da	16,725	0.000669	0	16,725	0	0
0x5124db	16,725	0.000669	0	0	0	0
0x5124de	16,725	0.000669	0	0	0	0
0x5124e1	16,725	0.000669	16,725	0	0	0
0x5124e5	16,725	0.000669	16,725	0	0	0
0x5124e9	16,725	0.000669	0	0	0	0
0x5124eb	16,725	0.000669	0	0	16,725	0
0x5124ed	16,725	0.000669	0	0	0	0
0x5124f1	16,725	0.000669	0	0	0	0
0x5124f3	16,725	0.000669	0	0	16,725	16,725
0x51257f	16,725	0.000669	0	16,725	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x512583	16,725	0.000669	0	0	0	0
0x512586	16,725	0.000669	0	16,725	0	0
0x51258a	16,725	0.000669	0	16,725	0	0
0x51258e	16,725	0.000669	0	0	0	0
0x512590	16,725	0.000669	16,725	0	0	0
0x512594	16,725	0.000669	0	16,725	0	0
0x512598	16,725	0.000669	0	0	0	0
0x51259c	16,725	0.000669	16,725	0	0	0
0x51259d	16,725	0.000669	16,725	0	0	0
0x51259f	16,725	0.000669	16,725	0	0	0
0x5125a1	16,725	0.000669	16,725	0	0	0
0x5125a3	16,725	0.000669	16,725	0	0	0
0x5125a5	16,725	0.000669	16,725	0	0	0
0x5125a6	16,725	0.000669	16,725	0	0	0
0x471376	15,607	0.000624	0	0	15,607	15,607

Top load line deltas (all 100) and top load pages (all 100).

line_delta	count	share	page	load operands	share
0	2,534,417	0.299840	34,356,209,220	2,847,970	0.336936
1	460,970	0.054536	6,442	942,446	0.111498
244,559	221,852	0.026247	1,793	439,723	0.052022
244,031	84,903	0.010045	33,619	348,748	0.041259
-1	66,199	0.007832	2,621	341,260	0.040374
2	47,178	0.005582	1,794	134,210	0.015878
-244,559	41,151	0.004868	2,629	121,499	0.014374
-297,746	41,074	0.004859	1,723	110,111	0.013027
-21	37,856	0.004479	2,622	96,806	0.011453
21	35,190	0.004163	1,790	89,706	0.010613
1,739,318	33,676	0.003984	33,620	78,930	0.009338
-2,198,796,977,812	32,211	0.003811	1,724	71,542	0.008464
-52,971	30,741	0.003637	2,656	51,796	0.006128
-2,241	28,771	0.003404	2,616	37,817	0.004474
297,753	27,967	0.003309	33,622	37,212	0.004402
-22	26,851	0.003177	33,632	34,620	0.004096
-2	25,929	0.003068	33,625	34,030	0.004026
-2,198,797,222,383	25,220	0.002984	1,792	33,072	0.003913
2,198,796,977,811	23,727	0.002807	2,657	28,778	0.003405
-4,455	23,299	0.002756	6,388	28,166	0.003332
-2,198,797,222,381	22,340	0.002643	1,774	27,518	0.003256
-2,198,796,977,814	22,222	0.002629	6,671	23,774	0.002813
580	22,193	0.002626	1,764	23,274	0.002753
-57,972	22,189	0.002625	1,733	22,746	0.002691
2,198,796,977,823	21,759	0.002574	2,630	22,192	0.002625
-2,198,797,221,845	21,474	0.002541	33,645	22,188	0.002625
-300,486	21,432	0.002536	6,432	21,474	0.002541
52,913	20,569	0.002433	1,747	21,432	0.002536
52,928	19,989	0.002365	2,624	16,967	0.002007
2,198,797,275,350	19,503	0.002307	2,623	16,595	0.001963
-4,476	18,906	0.002237	6,668	14,841	0.001756
4,476	18,906	0.002237	1,791	13,049	0.001544
-2,198,797,275,372	18,775	0.002221	1,765	12,838	0.001519
2,198,796,977,813	18,279	0.002163	2,632	12,783	0.001512
-241,004	18,194	0.002152	2,634	10,737	0.001270
-52,992	17,989	0.002128	1,777	10,737	0.001270
-2,198,797,275,371	17,844	0.002111	33,621	9,789	0.001158
-2,198,797,222,382	17,306	0.002047	1,797	8,857	0.001048
11	16,122	0.001907	1,937	6,013	0.000711
57,447	15,637	0.001850	6,553	5,377	0.000636
2,198,797,275,371	15,458	0.001829	1,795	4,960	0.000587
52,685	15,371	0.001819	33,623	4,453	0.000527
-52,685	15,371	0.001819	63,654	4,415	0.000522
52,971	15,370	0.001818	33,624	4,118	0.000487
52,707	15,370	0.001818	33,631	3,420	0.000405
52,992	15,279	0.001808	33,626	3,138	0.000371

line_delta	count	share	page	load operands	share
2,198,796,977,814	15,266	0.001806	33,644	2,785	0.000329
2,198,797,222,664	14,613	0.001729	6,672	2,741	0.000324
1,739,319	14,319	0.001694	1,732	2,433	0.000288
4,455	13,870	0.001641	33,627	2,154	0.000255
2,198,797,275,329	13,720	0.001623	33,630	2,124	0.000251
25	13,492	0.001596	33,629	1,870	0.000221
-3	12,670	0.001499	33,628	1,638	0.000194
-2,198,796,977,811	12,622	0.001493	33,643	1,281	0.000152
2,198,796,977,815	12,525	0.001482	48,206	1,148	0.000136
42	12,157	0.001438	33,633	1,074	0.000127
-42	12,082	0.001429	46,424	1,065	0.000126
-297,743	12,015	0.001421	25,294	1,056	0.000125
55,332	11,636	0.001377	57,441	1,046	0.000124
-297,741	11,636	0.001377	45,479	1,044	0.000124
-297,725	11,636	0.001377	33,634	1,029	0.000122
3	11,419	0.001351	20,061	1,029	0.000122
-6	11,334	0.001341	12,053	1,024	0.000121
1,739,320	10,772	0.001274	39,504	905	0.000107
-328	10,767	0.001274	20,166	902	0.000107
632	10,758	0.001273	21,432	901	0.000107
-631	10,747	0.001271	10,324	899	0.000106
297,956	10,739	0.001271	60,968	883	0.000104
-2,198,797,222,346	10,737	0.001270	10,638	879	0.000104
301,917	10,737	0.001270	9,412	860	0.000102
-244,025	10,737	0.001270	39,770	856	0.000101
2,198,796,977,820	10,737	0.001270	25,296	832	0.000098
-2,198,797,221,517	10,737	0.001270	19,285	829	0.000098
-297,956	10,737	0.001270	63,614	823	0.000097
2,198,795,238,494	10,737	0.001270	7,411	813	0.000096
2,198,797,221,843	10,737	0.001270	24,314	813	0.000096
2,198,797,222,330	10,737	0.001270	56,918	808	0.000096
2,198,796,977,816	10,737	0.001270	15,902	801	0.000095
-244,031	10,674	0.001263	15,890	792	0.000094
244,032	10,674	0.001263	17,313	788	0.000093
297,755	10,674	0.001263	25,447	778	0.000092
2,115	10,622	0.001257	33,635	771	0.000091
13	10,549	0.001248	33,642	759	0.000090
-2,198,797,279,826	10,470	0.001239	10,022	755	0.000089
1,739,321	10,346	0.001224	51,413	751	0.000089
2,198,797,275,567	10,291	0.001218	6,769	737	0.000087
-2,198,797,275,351	10,251	0.001213	45,057	735	0.000087
-2,198,797,275,349	10,251	0.001213	11,080	716	0.000085
2,150	9,970	0.001180	33,641	714	0.000084
241,005	9,968	0.001179	22,555	711	0.000084
-238,762	9,968	0.001179	21,229	710	0.000084
18	9,926	0.001174	24,531	709	0.000084
2,198,796,977,812	9,820	0.001162	31,348	709	0.000084
-2,198,797,220,024	9,410	0.001113	16,873	707	0.000084
-2,198,797,222,379	9,288	0.001099	35,484	702	0.000083
-2,198,797,222,380	9,168	0.001085	12,059	698	0.000083
2,198,797,275,564	9,163	0.001084	30,803	687	0.000081
2,198,797,275,515	9,161	0.001084	24,652	687	0.000081
-297,745	8,923	0.001056	30,825	683	0.000081
297,752	8,923	0.001056	31,183	679	0.000080

Register usage (all 110 rows).

direction	register_id	dynamic uses	share within direction
source	6	6,528,022	0.183276
source	26	5,041,935	0.141554
source	25	3,445,394	0.096731
source	10	3,057,723	0.085847
source	3	2,147,579	0.060294
source	7	1,841,693	0.051706
source	9	1,261,394	0.035414

direction	register_id	dynamic uses	share within direction
source	55	1,116,175	0.031337
source	17	1,110,100	0.031166
source	8	931,499	0.026152
source	56	927,256	0.026033
source	4	924,840	0.025965
source	18	840,338	0.023593
source	12	770,110	0.021621
source	47	654,753	0.018382
source	14	605,302	0.016994
source	15	561,676	0.015769
source	5	525,320	0.014749
source	100	375,151	0.010532
source	54	371,088	0.010418
source	101	312,692	0.008779
source	61	299,166	0.008399
source	13	262,474	0.007369
source	67	262,423	0.007368
source	16	262,141	0.007360
source	27	230,861	0.006481
source	11	140,137	0.003934
source	49	126,279	0.003545
source	53	79,162	0.002222
source	45	68,222	0.001915
source	30	65,034	0.001826
source	102	61,981	0.001740
source	29	60,424	0.001696
source	79	56,298	0.001581
source	35	34,656	0.000973
source	23	28,166	0.000791
source	108	22,388	0.000629
source	33	22,292	0.000626
source	112	22,180	0.000623
source	111	19,462	0.000546
source	109	19,454	0.000546
source	110	19,454	0.000546
source	32	16,981	0.000477
source	28	15,603	0.000438
source	36	14,944	0.000420
source	59	12,406	0.000348
source	73	11,673	0.000328
source	48	10,620	0.000298
source	39	10,193	0.000286
source	64	7,452	0.000209
source	82	4,438	0.000125
source	76	1,359	0.000038
source	68	29	0.000001
source	74	23	0.000001
source	103	20	0.000001
source	104	16	0.000000
source	40	11	0.000000
source	50	2	0.000000
source	46	2	0.000000
destination	6	5,472,771	0.194047
destination	25	5,231,149	0.185480
destination	26	5,211,965	0.184800
destination	10	2,144,747	0.076046
destination	3	1,038,184	0.036811
destination	7	914,548	0.032427
destination	56	795,103	0.028192
destination	9	754,691	0.026759
destination	8	717,798	0.025451
destination	17	657,666	0.023319
destination	4	549,404	0.019480
destination	18	474,297	0.016817
destination	55	469,015	0.016630

direction	register_id	dynamic uses	share within direction
destination	54	417,967	0.014820
destination	47	413,254	0.014653
destination	5	342,728	0.012152
destination	15	334,657	0.011866
destination	101	312,660	0.011086
destination	14	271,132	0.009613
destination	100	257,830	0.009142
destination	12	220,822	0.007830
destination	16	217,451	0.007710
destination	13	133,019	0.004716
destination	67	132,983	0.004715
destination	102	123,896	0.004393
destination	11	93,601	0.003319
destination	27	85,012	0.003014
destination	49	82,767	0.002935
destination	79	55,434	0.001966
destination	45	44,083	0.001563
destination	61	35,093	0.001244
destination	53	24,226	0.000859
destination	112	22,180	0.000786
destination	82	15,802	0.000560
destination	109	15,300	0.000542
destination	110	15,300	0.000542
destination	111	15,300	0.000542
destination	33	13,914	0.000493
destination	36	13,906	0.000493
destination	29	13,042	0.000462
destination	73	11,680	0.000414
destination	108	11,242	0.000399
destination	64	8,436	0.000299
destination	59	6,205	0.000220
destination	32	4,150	0.000147
destination	76	3,992	0.000142
destination	50	2,602	0.000092
destination	30	170	0.000006
destination	68	58	0.000002
destination	104	16	0.000001
destination	103	12	0.000000

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Top PCs (all 200 rows).

pc	dyn instr	instr share	load	store	branch	taken
0x638eb4	443,132	0.017725	0	0	0	0
0x638eb7	443,132	0.017725	0	0	443,132	0
0x638ebd	443,132	0.017725	443,132	0	0	0
0x638ec0	443,132	0.017725	443,132	0	0	0
0x638ec3	443,132	0.017725	0	0	0	0
0x638ec5	443,132	0.017725	0	0	0	0
0x638ec8	443,132	0.017725	0	0	0	0
0x638ecb	443,132	0.017725	0	0	443,132	253
0x638ed1	442,879	0.017715	442,879	0	0	0
0x638ed5	442,879	0.017715	0	0	0	0
0x638ed8	442,879	0.017715	0	0	442,879	401,933
0x638eb0	441,991	0.017680	441,991	0	0	0
0x64cc60	369,969	0.014799	369,969	0	0	0
0x64cc64	369,969	0.014799	0	0	369,969	0
0x64cc66	369,969	0.014799	369,969	0	0	0
0x64cc6a	369,969	0.014799	0	0	369,969	0
0x64cc6c	369,969	0.014799	369,969	0	0	0
0x64cc6f	369,969	0.014799	0	0	369,969	0
0x64cc71	369,969	0.014799	369,969	0	0	0
0x64cc75	369,969	0.014799	0	0	369,969	570
0x64cc77	369,399	0.014776	0	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x64cc7b	369,399	0.014776	0	0	0	0
0x64cc7e	369,399	0.014776	0	0	0	0
0x64cc82	369,399	0.014776	0	0	0	0
0x64cc86	369,399	0.014776	0	0	369,399	368,828
0x64a790	119,754	0.004790	0	0	0	0
0x64a793	119,754	0.004790	0	0	119,754	10,265
0x64a798	109,490	0.004380	0	0	0	0
0x64a79c	109,490	0.004380	0	0	0	0
0x64a7a0	109,490	0.004380	109,490	0	0	0
0x64a7a3	109,490	0.004380	0	0	0	0
0x64a7a7	109,490	0.004380	0	0	109,490	109,490
0x64a795	109,489	0.004380	109,489	0	0	0
0x678c10	53,600	0.002144	0	0	0	0
0x678c13	53,600	0.002144	0	0	53,600	6,844
0x678c30	49,040	0.001962	0	0	0	0
0x678c33	49,040	0.001962	0	0	49,040	4,563
0x678c15	46,756	0.001870	46,756	0	0	0
0x678c18	46,756	0.001870	0	0	0	0
0x678c1b	46,756	0.001870	0	46,756	0	0
0x678c1e	46,756	0.001870	0	0	0	0
0x678c20	46,756	0.001870	0	0	46,756	10,261
0x678c35	44,477	0.001779	44,477	44,477	0	0
0x678c38	44,477	0.001779	0	0	44,477	7,981
0x678c22	43,339	0.001734	43,339	0	0	0
0x678c3a	41,059	0.001642	41,059	0	0	0
0x638eda	40,946	0.001638	40,946	0	0	0
0x638ede	40,946	0.001638	0	0	0	0
0x638ee2	40,946	0.001638	0	0	0	0
0x638ee6	40,946	0.001638	0	0	0	0
0x638ee9	40,946	0.001638	0	0	0	0
0x638eec	40,946	0.001638	0	0	40,946	40,058
0x7b358c	37,292	0.001492	0	0	0	0
0x7b358e	37,292	0.001492	0	0	37,292	33,903
0x7b3580	33,903	0.001356	0	0	0	0
0x7b3584	33,903	0.001356	0	0	0	0
0x7b3586	33,903	0.001356	0	0	33,903	0
0x64a740	27,371	0.001095	27,371	0	0	0
0x64a744	27,371	0.001095	27,371	0	0	0
0x64a748	27,371	0.001095	0	0	27,371	6,842
0x678230	23,952	0.000958	0	0	0	0
0x678234	23,952	0.000958	23,952	0	0	0
0x406bb0	21,589	0.000864	21,589	0	0	0
0x406bb4	21,589	0.000864	0	21,589	0	0
0x406bb9	21,589	0.000864	21,589	0	0	0
0x406bbe	21,589	0.000864	0	21,589	0	0
0x406bc3	21,589	0.000864	21,589	0	0	0
0x406bc8	21,589	0.000864	0	21,589	0	0
0x406bcd	21,589	0.000864	21,589	0	0	0
0x406bd2	21,589	0.000864	0	21,589	0	0
0x406bd7	21,589	0.000864	21,589	0	0	0
0x406bdc	21,589	0.000864	0	21,589	0	0
0x406be1	21,589	0.000864	21,589	0	0	0
0x406be6	21,589	0.000864	0	21,589	0	0
0x406beb	21,589	0.000864	21,589	0	0	0
0x406bf0	21,589	0.000864	0	21,589	0	0
0x406bf5	21,589	0.000864	21,589	0	0	0
0x406bfa	21,589	0.000864	0	21,589	0	0
0x406bfe	21,589	0.000864	0	0	0	0
0x406c02	21,589	0.000864	0	0	0	0
0x406c06	21,589	0.000864	0	0	21,589	19,315
0x64a7af	20,530	0.000821	20,530	0	0	0
0x64a74a	20,529	0.000821	0	0	0	0
0x64a74f	20,529	0.000821	20,529	0	0	0
0x64a754	20,529	0.000821	0	0	20,529	10,265
0x64a75a	20,529	0.000821	0	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x64a75f	20,529	0.000821	20,529	0	0	0
0x64a764	20,529	0.000821	0	0	20,529	10,265
0x64a76a	20,529	0.000821	0	0	0	0
0x64a76c	20,529	0.000821	0	0	0	0
0x64a76f	20,529	0.000821	0	0	20,529	10,265
0x5a13f0	15,960	0.000638	15,960	0	0	0
0x5a13f5	15,960	0.000638	15,960	0	0	0
0x5a13f9	15,960	0.000638	0	0	0	0
0x5a13fb	15,960	0.000638	0	0	0	0
0x5a13fe	15,960	0.000638	0	0	0	0
0x5a1401	15,960	0.000638	0	0	15,960	5,700
0x678240	14,829	0.000593	0	0	0	0
0x678244	14,829	0.000593	14,829	0	0	0
0x7b13d0	14,170	0.000567	0	0	0	0
0x7b13d2	14,170	0.000567	14,170	14,170	0	0
0x7b13d5	14,170	0.000567	0	0	0	0
0x7b13d8	14,170	0.000567	0	0	14,170	1,837
0x7b1510	14,170	0.000567	0	0	0	0
0x7b1514	14,170	0.000567	0	0	0	0
0x7b1518	14,170	0.000567	0	0	0	0
0x7b151b	14,170	0.000567	0	0	14,170	12,753
0x498410	13,687	0.000547	0	0	0	0
0x498413	13,687	0.000547	0	0	13,687	9,124
0x5e378e	13,682	0.000547	0	0	0	0
0x5e3791	13,682	0.000547	0	0	0	0
0x5e3795	13,682	0.000547	13,682	0	0	0
0x5e3796	13,682	0.000547	13,682	0	0	0
0x5e3798	13,682	0.000547	13,682	0	0	0
0x5e379a	13,682	0.000547	13,682	0	0	0
0x5e379c	13,682	0.000547	13,682	0	0	0
0x5e379e	13,682	0.000547	13,682	0	0	0
0x5e379f	13,682	0.000547	13,682	0	0	0
0x5e2b60	13,681	0.000547	0	13,681	0	0
0x5e2b61	13,681	0.000547	0	13,681	0	0
0x5e2b63	13,681	0.000547	0	13,681	0	0
0x5e2b65	13,681	0.000547	0	13,681	0	0
0x5e2b67	13,681	0.000547	0	13,681	0	0
0x5e2b69	13,681	0.000547	0	13,681	0	0
0x5e2b6a	13,681	0.000547	0	0	0	0
0x5e2b6e	13,681	0.000547	0	0	0	0
0x5e2b71	13,681	0.000547	0	0	0	0
0x5e2b74	13,681	0.000547	0	0	0	0
0x5e2b77	13,681	0.000547	0	0	0	0
0x5e2b7a	13,681	0.000547	0	0	0	0
0x5e2b7d	13,681	0.000547	13,681	0	0	0
0x5e2b7f	13,681	0.000547	0	0	0	0
0x5e2b82	13,681	0.000547	0	0	0	0
0x5e2b85	13,681	0.000547	0	0	13,681	0
0x5e2b8b	13,681	0.000547	13,681	0	13,681	13,681
0x54a690	12,545	0.000502	12,545	0	0	0
0x54a697	12,545	0.000502	12,545	0	0	0
0x54a69a	12,545	0.000502	12,545	0	12,545	12,545
0x616a40	12,545	0.000502	12,545	0	0	0
0x616a47	12,545	0.000502	12,545	0	0	0
0x406cd0	12,541	0.000502	12,541	0	0	0
0x406cd3	12,541	0.000502	0	12,541	0	0
0x406cd6	12,541	0.000502	0	0	0	0
0x406cda	12,541	0.000502	0	0	0	0
0x406cde	12,541	0.000502	0	0	0	0
0x406ce1	12,541	0.000502	0	0	12,541	5,982
0x7b13de	12,333	0.000493	0	0	12,333	12,333
0x5a12e0	11,410	0.000456	11,410	0	0	0
0x5a12e5	11,410	0.000456	11,410	0	0	0
0x5a12ea	11,410	0.000456	0	0	0	0
0x5a12ec	11,410	0.000456	0	0	0	0

pc	dyn instr	instr share	load	store	branch	taken
0x5a12ef	11,410	0.000456	0	0	0	0
0x5a12f2	11,410	0.000456	0	0	11,410	9,128
0x6441a0	11,402	0.000456	11,402	0	0	0
0x6441a4	11,402	0.000456	11,402	0	0	0
0x5a1548	10,265	0.000411	0	0	0	0
0x5a154b	10,265	0.000411	0	0	0	0
0x5a154f	10,265	0.000411	10,265	0	0	0
0x5a1550	10,265	0.000411	10,265	0	0	0
0x5a1552	10,265	0.000411	10,265	0	0	0
0x5a1554	10,265	0.000411	10,265	0	0	0
0x5a1556	10,265	0.000411	10,265	0	0	0
0x5a1558	10,265	0.000411	10,265	0	0	0
0x5a1559	10,265	0.000411	10,265	0	0	0
0x5a0ff1	10,265	0.000411	10,265	0	0	0
0x5a0ff5	10,265	0.000411	0	0	10,265	10,265
0x5a1006	10,265	0.000411	0	10,265	0	0
0x5a100a	10,265	0.000411	10,265	0	0	0
0x5a100e	10,265	0.000411	0	0	0	0
0x5a1012	10,265	0.000411	10,265	0	0	0
0x5a1017	10,265	0.000411	0	0	0	0
0x5a101a	10,265	0.000411	0	0	10,265	0
0x5a101c	10,265	0.000411	0	10,265	0	0
0x5a101f	10,265	0.000411	0	10,265	0	0
0x5a1024	10,265	0.000411	0	0	0	0
0x5a1027	10,265	0.000411	0	0	0	0
0x5a102b	10,265	0.000411	10,265	0	0	0
0x5a102c	10,265	0.000411	10,265	0	0	0
0x5a102e	10,265	0.000411	10,265	0	0	0
0x5a1030	10,265	0.000411	10,265	0	0	0
0x5a1032	10,265	0.000411	10,265	0	0	0
0x5a1034	10,265	0.000411	10,265	0	0	0
0x5a1035	10,265	0.000411	10,265	0	0	0
0x5a0fc0	10,265	0.000411	0	10,265	0	0
0x5a0fc1	10,265	0.000411	0	10,265	0	0
0x5a0fc3	10,265	0.000411	0	10,265	0	0
0x5a0fc5	10,265	0.000411	0	10,265	0	0
0x5a0fc7	10,265	0.000411	0	10,265	0	0
0x5a0fc9	10,265	0.000411	0	10,265	0	0
0x5a0fca	10,265	0.000411	0	0	0	0
0x5a0fce	10,265	0.000411	0	0	0	0
0x5a0fd1	10,265	0.000411	0	0	0	0
0x5a0fd4	10,265	0.000411	0	0	0	0
0x5a0fd7	10,265	0.000411	10,265	0	0	0
0x5a0fdc	10,265	0.000411	0	0	0	0
0x5a0fe0	10,265	0.000411	0	0	0	0
0x5a0fe4	10,265	0.000411	0	0	0	0
0x5a0fe7	10,265	0.000411	0	0	0	0
0x5a0fea	10,265	0.000411	0	0	0	0
0x5a0fec	10,264	0.000411	0	10,264	0	0

Top load line deltas (all 100) and top load pages (all 100).

	line.delta	count	share	page	load operands	share
	0	3,335,073	0.432993	34,356,208,070	1,078,809	0.140062
	1	705,989	0.091659	34,356,208,071	1,033,557	0.134187
	-1	144,400	0.018747	125,425	497,408	0.064579
	2	53,250	0.006913	125,426	366,336	0.047561
	2,242	50,160	0.006512	125,424	274,401	0.035626
	-2,242	45,600	0.005920	5,867	255,806	0.033211
	-6	32,299	0.004193	125,427	235,264	0.030544
	-229	29,666	0.003852	94,325	180,166	0.023391
	-5	29,162	0.003786	94,417	124,327	0.016141
	-233,209	28,511	0.003702	3,333	112,760	0.014640
	6	28,343	0.003680	125,428	104,192	0.013527
	229	27,384	0.003555	94,550	94,662	0.012290

line_delta	count	share	page	load operands	share
2,135	25,102	0.003259	94,513	79,198	0.010282
-2,135	22,820	0.002963	94,549	74,106	0.009621
-2,198,791,279,792	22,800	0.002960	94,548	74,100	0.009620
-2	20,287	0.002634	94,547	66,173	0.008591
3	19,535	0.002536	2,278	54,742	0.007107
5	18,406	0.002390	2,223	51,322	0.006663
-91	14,825	0.001925	2,254	41,046	0.005329
-4	14,481	0.001880	94,546	38,794	0.005037
-5,896,885	13,687	0.001777	2,262	38,762	0.005032
91	13,686	0.001777	94,370	31,026	0.004028
-1,741	13,686	0.001777	2,133	30,803	0.003999
-5,896,883	13,684	0.001777	3,358	28,792	0.003738
2,198,791,279,792	13,680	0.001776	94,533	28,504	0.003701
-5,892,016	13,680	0.001776	94,324	23,940	0.003108
-161,376	12,569	0.001632	94,543	21,793	0.002829
-2,198,791,265,797	12,545	0.001629	2,113	21,665	0.002813
-233,195	12,545	0.001629	5,854	19,412	0.002520
233,202	12,545	0.001629	2,134	18,247	0.002369
2,198,796,941,052	12,545	0.001629	3,336	17,100	0.002220
-2,198,797,171,987	12,545	0.001629	94,512	15,974	0.002074
2,198,797,171,991	12,545	0.001629	2,261	15,968	0.002073
2,198,797,170,724	11,410	0.001481	94,416	15,964	0.002073
2,198,797,170,725	11,410	0.001481	94,366	15,178	0.001971
-2,198,796,941,030	11,409	0.001481	94,368	14,745	0.001914
-2,198,796,941,038	11,403	0.001480	94,361	14,139	0.001836
-2,198,797,172,315	11,402	0.001480	3,360	13,686	0.001777
4	11,055	0.001435	2,259	12,545	0.001629
2,198,796,941,074	10,262	0.001332	94,351	11,824	0.001535
8,380	10,260	0.001332	94,402	11,693	0.001518
-2,198,791,267,784	9,128	0.001185	5,856	11,406	0.001481
233,213	9,124	0.001185	2,212	11,406	0.001481
1,658	8,260	0.001072	3,337	11,402	0.001480
-2,198,796,941,041	7,985	0.001037	94,563	10,620	0.001379
2,198,797,170,720	7,981	0.001036	2,131	10,263	0.001332
-32	6,920	0.000898	94,568	9,954	0.001292
2,198,796,941,030	6,846	0.000889	94,608	9,881	0.001283
-5,912,232	6,846	0.000889	94,365	9,696	0.001259
-5,896,879	6,846	0.000889	94,603	9,624	0.001249
-2,198,796,941,039	6,844	0.000889	94,562	9,580	0.001244
5,666,235	6,843	0.000888	94,346	9,174	0.001191
71,690	6,843	0.000888	2,253	9,120	0.001184
161,377	6,843	0.000888	94,569	8,933	0.001160
-159,634	6,843	0.000888	94,356	8,516	0.001106
-2,198,797,172,316	6,841	0.000888	94,606	8,351	0.001084
2,198,797,170,705	6,840	0.000888	94,390	8,328	0.001081
-2,198,797,170,713	6,840	0.000888	94,326	8,141	0.001057
2,198,797,170,713	6,840	0.000888	94,552	7,688	0.000998
-3	6,307	0.000819	94,567	7,669	0.000996
-1,658	5,982	0.000777	94,341	7,200	0.000935
159,718	5,980	0.000776	94,580	7,143	0.000927
-23	5,869	0.000762	94,572	6,906	0.000897
-2,198,796,941,027	5,705	0.000741	95,522	6,903	0.000896
-2,198,791,273,839	5,705	0.000741	94,584	6,844	0.000889
-233,201	5,703	0.000740	2,255	6,841	0.000888
2,198,797,170,712	5,701	0.000740	2,156	6,840	0.000888
-2,198,797,170,720	5,701	0.000740	107,957	6,678	0.000867
-5,829,140	5,701	0.000740	125,995	6,674	0.000866
-2,198,796,941,017	5,701	0.000740	94,570	6,521	0.000847
-8,381	5,700	0.000740	94,357	6,488	0.000842
-2,198,791,265,392	5,700	0.000740	94,829	6,468	0.000840
-78,251	5,700	0.000740	99,382	6,259	0.000813
47	4,876	0.000633	94,362	6,132	0.000796
2,198,796,941,029	4,565	0.000593	94,579	5,966	0.000775
-2,198,796,941,029	4,565	0.000593	94,566	5,953	0.000773
-2,198,796,941,032	4,564	0.000593	94,605	5,865	0.000761

line.delta	count	share	page	load operands	share
2,198,796,941,028	4,564	0.000593	2,130	5,700	0.000740
2,198,797,180,016	4,564	0.000593	125,429	5,699	0.000740
-2,198,791,265,556	4,564	0.000593	95,398	5,675	0.000737
238,990	4,564	0.000593	94,578	5,620	0.000730
-2,198,791,273,840	4,564	0.000593	94,327	5,460	0.000709
-2,198,791,265,544	4,564	0.000593	94,415	5,403	0.000701
-5,896,881	4,564	0.000593	5,868	5,377	0.000698
1,625	4,563	0.000592	94,574	5,369	0.000697
-2,198,791,265,322	4,563	0.000592	95,427	5,355	0.000695
-2,198,791,273,827	4,562	0.000592	95,118	5,313	0.000690
-233,212	4,562	0.000592	94,364	5,151	0.000669
5,665,474	4,562	0.000592	95,362	5,066	0.000658
2,198,797,174,247	4,561	0.000592	95,454	5,060	0.000657
2,198,796,941,010	4,561	0.000592	97,397	5,060	0.000657
-2,198,791,273,896	4,561	0.000592	97,567	5,060	0.000657
-5,906,264	4,561	0.000592	97,523	5,060	0.000657
8,379	4,560	0.000592	94,843	4,940	0.000641
-2,221	4,560	0.000592	95,022	4,940	0.000641
-5,829,139	4,560	0.000592	94,367	4,888	0.000635
5,829,139	4,560	0.000592	94,340	4,841	0.000629
-5,892,552	4,560	0.000592	94,363	4,821	0.000626
2,198,797,172,322	4,560	0.000592	94,561	4,789	0.000622
-5,892,015	4,560	0.000592	94,588	4,789	0.000622

Register usage (all 88 rows).

direction	register_id	dynamic uses	share within direction
source	26	5,902,607	0.160487
source	25	5,003,764	0.136048
source	6	4,345,871	0.118160
source	10	4,157,115	0.113028
source	7	3,153,578	0.085743
source	4	2,521,477	0.068557
source	9	1,560,021	0.042416
source	8	1,539,701	0.041863
source	18	1,096,368	0.029809
source	3	1,085,120	0.029503
source	15	1,011,410	0.027499
source	17	737,508	0.020052
source	55	655,031	0.017810
source	54	579,199	0.015748
source	32	505,840	0.013753
source	35	471,637	0.012823
source	5	464,729	0.012636
source	56	429,903	0.011689
source	100	302,702	0.008230
source	16	251,464	0.006837
source	41	119,752	0.003256
source	47	114,919	0.003125
source	29	82,069	0.002231
source	27	76,665	0.002084
source	101	76,386	0.002077
source	61	69,322	0.001885
source	49	63,738	0.001733
source	11	61,089	0.001661
source	13	53,916	0.001466
source	40	41,050	0.001116
source	53	40,396	0.001098
source	12	39,571	0.001076
source	14	27,755	0.000755
source	45	24,942	0.000678
source	23	19,412	0.000528
source	59	13,509	0.000367
source	64	12,975	0.000353
source	33	12,547	0.000341

direction	register_id	dynamic uses	share within direction
source	62	10,265	0.000279
source	30	10,219	0.000278
source	82	8,489	0.000231
source	102	5,703	0.000155
source	48	5,703	0.000155
source	74	5,701	0.000155
source	79	4,664	0.000127
source	68	3,389	0.000092
source	76	253	0.000007
destination	25	8,129,609	0.275773
destination	26	6,509,291	0.220808
destination	6	3,784,844	0.128390
destination	10	1,641,630	0.055687
destination	7	1,501,940	0.050949
destination	54	1,032,623	0.035029
destination	9	948,697	0.032182
destination	4	843,629	0.028618
destination	8	816,803	0.027708
destination	55	708,721	0.024041
destination	3	614,533	0.020846
destination	17	497,535	0.016877
destination	56	425,975	0.014450
destination	18	339,471	0.011516
destination	15	313,848	0.010646
destination	5	287,242	0.009744
destination	100	273,059	0.009263
destination	16	170,757	0.005792
destination	45	150,152	0.005093
destination	47	106,948	0.003628
destination	61	51,331	0.001741
destination	49	43,333	0.001470
destination	11	42,270	0.001434
destination	101	38,764	0.001315
destination	12	35,151	0.001192
destination	27	34,213	0.001161
destination	53	25,573	0.000867
destination	13	22,431	0.000761
destination	64	19,899	0.000675
destination	14	9,402	0.000319
destination	30	9,343	0.000317
destination	50	7,981	0.000271
destination	79	6,841	0.000232
destination	68	6,778	0.000230
destination	102	5,703	0.000193
destination	74	5,448	0.000185
destination	76	4,710	0.000160
destination	59	4,563	0.000155
destination	33	4,411	0.000150
destination	82	2,788	0.000095
destination	48	1,140	0.000039

Appendix B — Evidence CSVs (full)

normal_prefetcher_evidence.csv (all 45 rows)

trace	pref	IPC	spd/np	miss	acc	sel.acc	timeli.	cov	tags
602.gcc_s-734B	no_pref	0.36800	1.0000	0.5025	0.0000	0.0000	0.0000	0.0000	low unique event coverage
602.gcc_s-734B	stride	0.37006	1.0056	0.4163	0.2642	0.2642	0.9976	0.1716	
602.gcc_s-734B	streamer	0.43161	1.1729	0.0383	0.1037	0.1037	0.9999	0.9234	high useless fraction
602.gcc_s-734B	ampm	0.43069	1.1704	0.0421	0.1311	0.1312	0.9984	0.9157	
602.gcc_s-734B	spp	0.42836	1.1640	0.1753	0.0590	0.0591	0.9985	0.6508	
602.gcc_s-734B	ipcp	0.36796	0.9999	0.5025	0.2447	0.2447	0.9200	0.0000	
602.gcc_s-734B	sms	0.42213	1.1471	0.1018	0.4587	0.4587	0.9994	0.7971	

trace	pref	IPC	spd/np	miss	acc	sel.acc	timeli.	cov	tags
602.gcc_s-734B	sandbox	0.43628	1.1855	0.0177	0.1016	0.1018	0.9998	0.9644	
602.gcc_s-734B	power7	0.43075	1.1705	0.0428	0.1128	0.1178	0.9997	0.9144	
619.lbm_s-4268B	no_pref	0.32200	1.0000	0.9987	0.0000	0.0000	0.0000	0.0000	
619.lbm_s-4268B	stride	0.32465	1.0082	0.7702	0.4304	0.4304	0.9057	0.2282	low unique event coverage
619.lbm_s-4268B	streamer	0.36368	1.1294	0.5163	0.7656	0.7658	0.7811	0.4825	
619.lbm_s-4268B	ampm	0.37480	1.1640	0.1989	0.2471	0.3271	0.9516	0.7991	
619.lbm_s-4268B	spp	0.37933	1.1780	0.7655	0.0501	0.0824	0.7451	0.2332	low unique event coverage
619.lbm_s-4268B	ipcp	0.32200	1.0000	0.9987	0.0864	0.0864	0.8750	0.0000	
619.lbm_s-4268B	sms	0.38105	1.1834	0.1312	0.9421	0.9421	0.9535	0.8664	
619.lbm_s-4268B	sandbox	0.37440	1.1627	0.2891	0.1456	0.2090	0.8632	0.7094	
619.lbm_s-4268B	power7	0.36128	1.1220	0.4545	0.5992	0.6150	0.8274	0.5441	
605.mcf_s-994B	no_pref	0.18318	1.0000	0.7513	0.0000	0.0000	0.0000	0.0000	
605.mcf_s-994B	stride	0.18359	1.0022	0.7464	0.3023	0.3023	0.9999	0.0066	low unique event coverage;very low unique event coverage
605.mcf_s-994B	streamer	0.18350	1.0017	0.7120	0.0871	0.0917	0.8947	0.1159	low unique event coverage;high useless fraction
605.mcf_s-994B	ampm	0.18874	1.0304	0.6974	0.1775	0.1779	0.9858	0.0779	low unique event coverage;very low unique event coverage
605.mcf_s-994B	spp	0.18874	1.0304	0.7464	0.0202	0.0204	0.9196	0.0070	low unique event coverage;very low unique event coverage
605.mcf_s-994B	ipcp	0.18318	1.0000	0.7513	0.2955	0.2955	0.6190	0.0000	
605.mcf_s-994B	sms	0.18777	1.0251	0.6924	0.1991	0.1991	0.8074	0.1238	low unique event coverage;high useless fraction
605.mcf_s-994B	sandbox	0.16905	0.9229	0.7249	0.0290	0.0339	0.7631	0.2132	low unique event coverage;high useless fraction
605.mcf_s-994B	power7	0.18163	0.9915	0.7266	0.0885	0.0936	0.8796	0.0959	low unique event coverage;very low unique event coverage;high useless fraction
620.omnetpp_s-874B	no_pref	0.23750	1.0000	0.7013	0.0000	0.0000	0.0000	0.0000	
620.omnetpp_s-874B	stride	0.23833	1.0035	0.6949	0.3028	0.3028	1.0000	0.0094	low unique event coverage;very low unique event coverage
620.omnetpp_s-874B	streamer	0.24228	1.0201	0.6165	0.1463	0.1681	0.9776	0.1624	low unique event coverage;high useless fraction
620.omnetpp_s-874B	ampm	0.23976	1.0095	0.6648	0.1454	0.1558	0.9603	0.0691	low unique event coverage;very low unique event coverage;high useless fraction
620.omnetpp_s-874B	spp	0.23869	1.0050	0.7008	0.0004	0.0004	0.6852	0.0002	low unique event coverage;very low unique event coverage
620.omnetpp_s-874B	ipcp	0.23771	1.0009	0.6981	0.1852	0.1852	0.9349	0.0068	low unique event coverage;very low unique event coverage;high useless fraction
620.omnetpp_s-874B	sms	0.24695	1.0398	0.5456	0.2447	0.2447	0.9819	0.2717	high useless fraction
620.omnetpp_s-874B	sandbox	0.24433	1.0288	0.5265	0.0490	0.0561	0.9718	0.4235	high useless fraction
620.omnetpp_s-874B	power7	0.24232	1.0203	0.6180	0.1444	0.1665	0.9746	0.1621	low unique event coverage;high useless fraction
623.xalancbmk_s-700B	no_pref	0.35321	1.0000	0.3696	0.0000	0.0000	0.0000	0.0000	
623.xalancbmk_s-700B	stride	0.35340	1.0005	0.3679	0.0629	0.0701	0.7154	0.0047	low unique event coverage;very low unique event coverage

trace	pref	IPC	spd/np	miss	acc	sel.acc	timeli.	cov	tags
623.xalancbmk_s- streamer 700B		0.33644	0.9525	0.5994	0.0452	0.0520	0.9426	0.0569	low unique event coverage;very low unique event coverage;high useless fraction
623.xalancbmk_s- ampm 700B		0.34924	0.9888	0.4151	0.0572	0.0630	0.9417	0.0418	low unique event coverage;very low unique event coverage
623.xalancbmk_s- spp 700B		0.35391	1.0020	0.3608	0.0047	0.0048	0.8069	0.0025	low unique event coverage;very low unique event coverage
623.xalancbmk_s- ipcp 700B		0.35241	0.9977	0.3785	0.0673	0.0673	0.9803	0.0063	low unique event coverage;very low unique event coverage
623.xalancbmk_s- sms 700B		0.34167	0.9673	0.5387	0.1509	0.1509	0.9634	0.1523	low unique event coverage;high useless fraction
623.xalancbmk_s- sandbox 700B		0.34524	0.9774	0.4924	0.0416	0.0499	0.9579	0.3658	high useless fraction
623.xalancbmk_s- power7 700B		0.34330	0.9719	0.5051	0.0466	0.0547	0.9169	0.0383	low unique event coverage;very low unique event coverage;high useless fraction

standalone_prefetcher_evidence.csv (all 10 rows)

trace	std	IPC	spd/best	spd/np	sel.acc	timeli.	evtcov	evttim	tags
602.gcc_s-734B	v3_1	0.42863	0.9825	1.1648	0.9738	0.9986	0.9000	0.9986	
619.lbm_s-4268B	v3_1	0.38492	1.0102	1.1954	0.9783	0.9842	0.9723	0.9842	
605.mcf_s-994B	v3_1	0.18862	0.9994	1.0297	0.5261	0.8012	0.0737	0.8008	low unique event coverage;very low unique event coverage
620.omnetpp_s-874B	v3_1	0.24503	0.9922	1.0317	0.4548	0.9908	0.2362	0.9907	low unique event coverage;high useless fraction
623.xalancbmk_s-700B	v3_1	0.36407	1.0287	1.0307	0.6832	0.9872	0.4543	0.9876	
602.gcc_s-734B	v3_3	0.42887	0.9830	1.1654	0.9809	0.9990	0.9014	0.9990	
619.lbm_s-4268B	v3_3	0.38434	1.0086	1.1936	0.9856	0.9929	0.9789	0.9929	
605.mcf_s-994B	v3_3	0.18729	0.9923	1.0224	0.4315	0.7383	0.0691	0.7379	low unique event coverage;very low unique event coverage
620.omnetpp_s-874B	v3_3	0.24559	0.9945	1.0341	0.4444	0.9926	0.2552	0.9925	high useless fraction
623.xalancbmk_s-700B	v3_3	0.37893	1.0707	1.0728	0.7510	0.9970	0.8994	0.9974	

normal_vs_standalone_evidence.csv (all 90 rows)

trace	std	normal	nIPC	sIPC	Δ IPC	both	n-only	s-only	neither
602.gcc_s-734B	v3_1	no_pref	0.36800	0.42863	0.06063			182,817	20,065
602.gcc_s-734B	v3_1	stride	0.37006	0.42863	0.05857	33,372	1,482	149,445	18,500
602.gcc_s-734B	v3_1	streamer	0.43161	0.42863	-0.00298	175,444	12,136	7,373	8,077
602.gcc_s-734B	v3_1	ampm	0.43069	0.42863	-0.00206	174,921	11,084	7,896	8,870
602.gcc_s-734B	v3_1	spp	0.42836	0.42863	0.00027	125,048	7,157	57,769	12,917
602.gcc_s-734B	v3_1	ipcp	0.36796	0.42863	0.06067			182,817	20,065
602.gcc_s-734B	v3_1	sms	0.42213	0.42863	0.00650	150,699	11,213	32,118	8,930
602.gcc_s-734B	v3_1	sandbox	0.43628	0.42863	-0.00765	180,554	15,340	2,263	4,933
602.gcc_s-734B	v3_1	power7	0.43075	0.42863	-0.00212	173,862	11,885	8,955	8,264
602.gcc_s-734B	v3_3	no_pref	0.36800	0.42887	0.06087			183,095	19,853
602.gcc_s-734B	v3_3	stride	0.37006	0.42887	0.05881	33,134	1,720	149,961	18,138
602.gcc_s-734B	v3_3	streamer	0.43161	0.42887	-0.00274	174,753	12,827	8,342	7,112
602.gcc_s-734B	v3_3	ampm	0.43069	0.42887	-0.00182	174,295	11,710	8,800	8,042
602.gcc_s-734B	v3_3	spp	0.42836	0.42887	0.00051	125,087	7,118	58,008	12,720
602.gcc_s-734B	v3_3	ipcp	0.36796	0.42887	0.06091			183,095	19,853
602.gcc_s-734B	v3_3	sms	0.42213	0.42887	0.00674	150,583	11,329	32,512	8,551
602.gcc_s-734B	v3_3	sandbox	0.43628	0.42887	-0.00741	180,405	15,489	2,690	4,508
602.gcc_s-734B	v3_3	power7	0.43075	0.42887	-0.00188	173,171	12,576	9,924	7,340

trace	std	normal	nIPC	sIPC	Δ IPC	both	n-only	s-only	neither
619.lbm_s-4268B	v3.1	no_pref	0.32200	0.38492	0.06292			286,489	3,548
619.lbm_s-4268B	v3.1	stride	0.32465	0.38492	0.06027	65,873	1,356	220,616	2,973
619.lbm_s-4268B	v3.1	streamer	0.36368	0.38492	0.02124	139,622	2,542	146,867	2,062
619.lbm_s-4268B	v3.1	ampm	0.37480	0.38492	0.01012	231,113	4,322	55,376	2,065
619.lbm_s-4268B	v3.1	spp	0.37933	0.38492	0.00559	67,616	1,086	218,873	2,608
619.lbm_s-4268B	v3.1	ipcp	0.32200	0.38492	0.06292			286,489	3,548
619.lbm_s-4268B	v3.1	sms	0.38105	0.38492	0.00387	250,052	5,218	36,437	1,457
619.lbm_s-4268B	v3.1	sandbox	0.37440	0.38492	0.01052	205,131	3,894	81,358	1,437
619.lbm_s-4268B	v3.1	power7	0.36128	0.38492	0.02364	157,195	3,132	129,294	2,104
619.lbm_s-4268B	v3.3	no_pref	0.32200	0.38434	0.06234			288,420	4,162
619.lbm_s-4268B	v3.3	stride	0.32465	0.38434	0.05969	66,397	832	222,023	3,587
619.lbm_s-4268B	v3.3	streamer	0.36368	0.38434	0.02066	140,507	1,657	147,913	2,649
619.lbm_s-4268B	v3.3	ampm	0.37480	0.38434	0.00954	232,867	2,568	55,553	2,667
619.lbm_s-4268B	v3.3	spp	0.37933	0.38434	0.00501	67,903	799	220,517	3,096
619.lbm_s-4268B	v3.3	ipcp	0.32200	0.38434	0.06234			288,420	4,162
619.lbm_s-4268B	v3.3	sms	0.38105	0.38434	0.00329	252,030	3,240	36,390	2,023
619.lbm_s-4268B	v3.3	sandbox	0.37440	0.38434	0.00994	206,484	2,541	81,936	2,022
619.lbm_s-4268B	v3.3	power7	0.36128	0.38434	0.02306	158,424	1,903	129,996	2,727
605.mcf_s-994B	v3.1	no_pref	0.18318	0.18862	0.00544			54,569	672,081
605.mcf_s-994B	v3.1	stride	0.18359	0.18862	0.00503	3,992	866	50,577	671,215
605.mcf_s-994B	v3.1	streamer	0.18350	0.18862	0.00512	41,015	44,793	13,554	613,373
605.mcf_s-994B	v3.1	ampm	0.18874	0.18862	-0.00012	40,159	17,513	14,410	653,507
605.mcf_s-994B	v3.1	spp	0.18874	0.18862	-0.00012	4,228	928	50,341	670,236
605.mcf_s-994B	v3.1	ipcp	0.18318	0.18862	0.00544			54,569	672,081
605.mcf_s-994B	v3.1	sms	0.18777	0.18862	0.00085	44,637	46,967	9,932	583,970
605.mcf_s-994B	v3.1	sandbox	0.16905	0.18862	0.01957	44,791	113,058	9,778	470,537
605.mcf_s-994B	v3.1	power7	0.18163	0.18862	0.00699	34,030	36,980	20,539	621,599
605.mcf_s-994B	v3.3	no_pref	0.18318	0.18729	0.00411			51,141	670,919
605.mcf_s-994B	v3.3	stride	0.18359	0.18729	0.00370	3,144	1,714	47,997	669,205
605.mcf_s-994B	v3.3	streamer	0.18350	0.18729	0.00379	31,459	54,349	19,682	606,654
605.mcf_s-994B	v3.3	ampm	0.18874	0.18729	-0.00145	28,390	29,282	22,751	640,924
605.mcf_s-994B	v3.3	spp	0.18874	0.18729	-0.00145	4,129	1,027	47,012	669,108
605.mcf_s-994B	v3.3	ipcp	0.18318	0.18729	0.00411			51,141	670,919
605.mcf_s-994B	v3.3	sms	0.18777	0.18729	-0.00048	36,872	54,732	14,269	575,501
605.mcf_s-994B	v3.3	sandbox	0.16905	0.18729	0.01824	35,772	122,077	15,369	469,130
605.mcf_s-994B	v3.3	power7	0.18163	0.18729	0.00566	26,642	44,368	24,499	617,055
620.omnetpp_s-874B	v3.1	no_pref	0.23750	0.24503	0.00753			74,759	241,090
620.omnetpp_s-874B	v3.1	stride	0.23833	0.24503	0.00670	1,627	1,363	73,132	239,729
620.omnetpp_s-874B	v3.1	streamer	0.24228	0.24503	0.00275	36,221	15,178	38,538	225,008
620.omnetpp_s-874B	v3.1	ampm	0.23976	0.24503	0.00527	16,308	5,580	58,451	235,202
620.omnetpp_s-874B	v3.1	spp	0.23869	0.24503	0.00634	29	20	74,730	241,070
620.omnetpp_s-874B	v3.1	ipcp	0.23771	0.24503	0.00732	1,551	588	73,208	240,427
620.omnetpp_s-874B	v3.1	sms	0.24695	0.24503	-0.00192	38,558	47,449	36,201	192,359
620.omnetpp_s-874B	v3.1	sandbox	0.24433	0.24503	0.00070	62,511	71,536	12,248	164,142
620.omnetpp_s-874B	v3.1	power7	0.24232	0.24503	0.00271	35,681	15,641	39,078	224,478
620.omnetpp_s-874B	v3.3	no_pref	0.23750	0.24559	0.00809			80,773	235,173
620.omnetpp_s-874B	v3.3	stride	0.23833	0.24559	0.00726	1,897	1,093	78,876	234,084
620.omnetpp_s-874B	v3.3	streamer	0.24228	0.24559	0.00331	37,399	14,000	43,374	220,227
620.omnetpp_s-874B	v3.3	ampm	0.23976	0.24559	0.00583	17,011	4,877	63,762	229,977

trace	std	normal	nIPC	sIPC	Δ IPC	both	n-only	s-only	neither
620.omnetpp-s-874B	v3.3	spp	0.23869	0.24559	0.00690	37	12	80,736	235,161
620.omnetpp-s-874B	v3.3	ipcp	0.23771	0.24559	0.00788	1,504	635	79,269	234,413
620.omnetpp-s-874B	v3.3	sms	0.24695	0.24559	-0.00136	40,520	45,487	40,253	188,378
620.omnetpp-s-874B	v3.3	sandbox	0.24433	0.24559	0.00126	65,651	68,396	15,122	161,300
620.omnetpp-s-874B	v3.3	power7	0.24232	0.24559	0.00327	36,965	14,357	43,808	219,803
623.xalancbmk_s-700B	v3.1	no_pref	0.35321	0.36407	0.01086			204,612	243,168
623.xalancbmk_s-700B	v3.1	stride	0.35340	0.36407	0.01067	1,875	240	202,737	242,595
623.xalancbmk_s-700B	v3.1	streamer	0.33644	0.36407	0.02763	18,783	6,827	185,829	235,547
623.xalancbmk_s-700B	v3.1	ampm	0.34924	0.36407	0.01483	15,972	2,870	188,640	239,715
623.xalancbmk_s-700B	v3.1	spp	0.35391	0.36407	0.01016	880	252	203,732	242,859
623.xalancbmk_s-700B	v3.1	ipcp	0.35241	0.36407	0.01166	2,493	324	202,119	242,803
623.xalancbmk_s-700B	v3.1	sms	0.34167	0.36407	0.02240	40,631	27,935	163,981	213,285
623.xalancbmk_s-700B	v3.1	sandbox	0.34524	0.36407	0.01883	82,491	82,247	122,121	157,284
623.xalancbmk_s-700B	v3.1	power7	0.34330	0.36407	0.02077	12,690	4,557	191,922	237,851
623.xalancbmk_s-700B	v3.3	no_pref	0.35321	0.37893	0.02572			405,066	44,210
623.xalancbmk_s-700B	v3.3	stride	0.35340	0.37893	0.02553	1,841	274	403,225	43,469
623.xalancbmk_s-700B	v3.3	streamer	0.33644	0.37893	0.04249	23,184	2,426	381,882	41,440
623.xalancbmk_s-700B	v3.3	ampm	0.34924	0.37893	0.02969	17,273	1,569	387,793	42,255
623.xalancbmk_s-700B	v3.3	spp	0.35391	0.37893	0.02502	1,012	120	404,054	43,997
623.xalancbmk_s-700B	v3.3	ipcp	0.35241	0.37893	0.02652	2,427	390	402,639	43,778
623.xalancbmk_s-700B	v3.3	sms	0.34167	0.37893	0.03726	62,407	6,159	342,659	37,714
623.xalancbmk_s-700B	v3.3	sandbox	0.34524	0.37893	0.03369	149,467	15,271	255,599	28,222
623.xalancbmk_s-700B	v3.3	power7	0.34330	0.37893	0.03563	15,579	1,668	389,487	42,125